Can Communicating Parents' Beliefs to Students Be a Pivotal Component of Parental Engagement? Evidence from School-Based Agricultural Education in Liberia

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Jimmy Lee, Northwestern University*

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Abstract

When school programs require students to introduce improved techniques on household farms, is informing students and parents about the program sufficient, or is there a case for further signaling parents' support to students? I study this question within a randomized evaluation of school-based agricultural education (SBAE) in Liberia, where students take school farm training into a home farming project. After parents view a promotional video of the program and students are encouraged to diffuse techniques to their parents, I randomize a nudge revealing to students their parents' optimism about their farm-management skills. Two years later, SBAE improves students' livelihoods only with the nudge: income rises 63%, on-time school fee payment 43–47%, and cost-effectiveness for attendance doubles. Analysis of mechanisms highlights a management aspect of agricultural technology adoption that can be addressed with nudges: even where parents become attentive to SBAE and adopt promoted techniques, third-party assurance can be pivotal in shifting students away from assisting in subsistence farming.

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1. Introduction

One of the greatest obstacles to rural development in developing countries is the challenge of logistics: a small number of trained staff must reach vast numbers of dispersed households. This constraint makes it difficult to promote practices that could raise productivity and living standards, such as improved farming techniques, better sanitation, or digital and financial literacy. Positioning schools as hubs for rural development offers a cost-effective solution. Drawing on existing infrastructure, teachers, students' initiative, and social ties, schools can reach households at scale. This in turn strengthens rural schools by encouraging attendance and enriching curricula with practical, experiential, and student-centered learning. It also offers an efficient platform to collect regular experimental data from rural households to inform the design of practices.

Projecting the broader impact of school-based programs hinges on whether they generate spillovers from schools into households. This is far from guaranteed: students depend on parents for most household decisions, yet parents live at a distance from schools and often hold inaccurate beliefs about what and how much students are learning (Dizon-Ross, 2019). My companion paper (Lee, 2025) highlights the pivotal role of engaging parents with program information in a three-year randomized evaluation of school-based agricultural extension (SBAE) across 197 junior high schools in Liberia. SBAE integrates experiential science instruction, extracurricular activities on school farms, and student home projects, with the aim of improving household technology adoption. While the program successfully trains teachers and students, it does not by itself increase technology adoption on any household parcels. By contrast, when parents are engaged through a promotional video and an annual farmer field day, the program achieves technology diffusion and becomes more cost-effective than most rigorously evaluated extension programs in sub-Saharan Africa.

Within the same experiment, this paper studies a sub-treatment aimed at another goal of SBAE—to inspire students to manage household farms and empower them to benefit from technology diffusion. Even when parents are engaged and adopt yield-improving techniques, they may overlook many potential changes in farm management—such as students' time spent assisting on subsistence parcels, management of sales-oriented parcels, and adoption of practices that generate more timely income to support school expenses. Standard household models (Chiappori & Mazzocco, 2017) predict that parental engagement suffices to increase delegation of farms by making parents more open to changes in farm management, as these models assume that students would learn parents' openness from natural interactions and propose changes accordingly. This assumption is reasonable here, as students can infer parents' types based on their (dramatically) increased visits to schools and adoption of technologies. Yet in practice, these actions might signal parents' beliefs about the technologies rather than openness to changes in farm management. Students can also learn about parents' openness from direct communication, but that could be subject to frictions, too. First, exchanging information is costly: parents and students experience SBAE differently, so sharing perspectives requires effort. Second, in high-risk farming environments, parents may hesitate

to voice optimism about SBAE's efficacy for fear of losing control over farming decisions—as monitoring students' behavior is costly. Third, behavioral dynamics may further dampen dialogue: parents might avoid the shame of learning from their children, while students might avoid the blame if their management leads to crop losses.

Aiming to overcome these communication frictions, I design a randomized nudge that signals to students parents' openness to delegating farm management, following parental engagement in a randomized evaluation of the program. The SBAE evaluation takes place in 100 SBAE schools and 97 control schools, sampling about 20 fourth to eighth grade students with prior farming experiences in each school. In a random subset of SBAE schools, parents are engaged through a promotional video, an annual farmer field day, or both (Lee, 2025). In 50 SBAE schools, sampled parents (one per household) view a video about SBAE during the first rainy season—6 months into the program—available in two randomized versions. The basic video describes SBAE as an innovative education model that integrates experiential science instruction, extracurricular activities on school farms, and student home projects to advance technology diffusion, educational outcomes, and youth empowerment. The persuasive video includes content from the basic video and adds testimonials from past participants and local authority figures, highlighting SBAE's effectiveness in enhancing students' farming skills and transforming lives through improved agricultural techniques.

To gauge parents' beliefs about students' improvements in farm management abilities under SBAE—a key factor determining their openness to delegation—we asked parents to state beliefs about students before and after viewing a video. These belief measures—elicited with incentives for accurate guesses—reflect students' improvements as depicted in the video. We define "anticipated improvements" as the difference between parents' pre-video beliefs about students' current abilities and their post-video beliefs about abilities one year later. Parents express overwhelming optimism: 97% predict improvement on at least one out of seven measures. 89% expect higher cassava yields, and 70%—up from 31% before viewing—believe students will even surpass typical adults in the community. The share of parents who express this optimism is similar for both videos, but anticipated harvest improvements are stronger among those who view the persuasive version. Beliefs about cassava yields remain higher at midline and endline compared to parents in schools without video sessions. At the end of each session, all parents consent to sharing anticipated improvements with students.

The nudge—the key treatment of interest—is a randomized add-on to an encouragement message delivered to students in a survey within a day after parents view the video. The survey first activates students' thinking about their potential growth under SBAE. To do this, students answer the same belief measures that their parents completed. Once this exercise is finished, a random half of students receive the nudge (henceforth, the *revelation nudge*). The nudge contains "revelation messages" for each belief measure—provided parents anticipate growth—two elements: (i) a qualitative statement, such as, "your father thinks that if you start growing cassava next year, you can harvest more tubers than if you start now", and (ii) a number indicating how much

parents expect students to improve in one year, such as the increase in the typical number of tubers students could harvest from a cassava plant. The nudge provides an informative signal of what parents think because students are aware of two features: (i) the incentivized elicitation of responses, and (ii) the careful elicitation procedure, which reminds subjects of the stated anticipated changes (including comparison of stated beliefs about students' harvests and typical adults' harvests) and allows revisions if needed. After the nudge (or not), all students receive an encouragement message reminding them about the SBAE program and urging them to share farming practices with their parents.

I next use students' survey responses immediately after the encouragement message to gauge the intent-to-treat effects of the revelation nudge. I find that students receiving the nudge are not more likely to express plans to look after farms or sell crops. Nor do they change farm locations. Instead, changes are more subtle. Policy reports, alongside a broad anthropology literature, suggest teenagers in rural Liberia aspire for a shift from assisting on traditional subsistence parcels—typically intercropping a wide variety of roots or vegetables with rice, relying on fallowing to recover soil fertility—towards managing sales-oriented parcels that provide timely income; while parents only accept these changes if teenagers gain sufficient farming skills. Further, SBAE emphasizes "learning by earning" in its home projects. Adoption of promoted methods (such as bed planting rather than flat planting) yields the greatest benefits when paired with practices that prioritize crop yields over traditional risk management. A typical example is planting a single cassava stake per hole, which enhances growth but reduces the fallback security of multiple stakes. Consistent with these anecdotes, students receiving the nudge indicate a change in the type of farms they plan to look after: 35% less rice fields, 47% less intercropping of roots or vegetables with rice, and 18% less crops in aggregate. In addition, although students receiving the nudge do not report a greater likelihood of perceiving available help in acquiring inputs or managing farms, they are more likely to seek such assistance from their parents. These results suggest that the nudge encourages students to shift time away from subsistence parcels and to discuss changes in farm management with their parents—whom they view as more supportive.

I then examine students' actual farming activities at midline (one year after the nudge) and endline (two years after the nudge). The nudge does not affect students' overall farming participation: it leaves unchanged their total weekly farming hours and their likelihood of looking after a farm. Parcel assignments also remain stable—the nudge does not alter the number of household parcels that students look after or work on. Nor does it change the types of parcels they are assigned to, whether defined by crop (rice, roots, vegetables) or by proxies for subsistence practices (crop diversity and intercropping of roots or vegetables with rice).

Beneath these null effects in parcel assignment, however, lie meaningful shifts in students' involvement with sales-oriented parcels. These shifts become visible when focusing on a designated `"student parcel"`, for which the survey collects more detailed information. I select the parcel using an algorithm that sequentially applies the following priorities, from highest to lowest: (i)

parcels containing roots or vegetables in the corresponding rainy season of the survey (2022 for midline; 2023 for endline), (ii) looked after by students themselves, and (iii) on which they spend the most time. Because the overall assignment of parcels—by crop type, student management status, and their combinations—does not change, any corresponding differences observed on the `"student parcel"` capture shifts in how households allocate students' labor across parcels.

Specifically, students receiving the revelation nudge are more likely to look after the `"student parcel"` (8 percentage points at midline and 13 percentage points at endline; from 25% and 42% respectively) and choose crops on the `"student parcel"` (8 percentage points at midline and 3 percentage points at endline; from 19% and 30% respectively). Consistent with students' plans in the immediate follow-up, the nudge reduces the total number of crops planted on the `"student parcel"` (by 10% at midline and 21% at endline), lowers the likelihood that the `"student parcel"` includes rice (by 19% at midline and 31% at endline), and decreases intercropping of roots or vegetables with rice (by 25% at midline and 26% at endline). Further, the nudge increases the likelihood that the `"student parcel"` is continuously cultivated across the first two rainy seasons by 25%, contrary to the common subsistence practice of fallowing.

Consistent with null effects on aggregate measures across parcels, the fall in the use of subsistence practices on the `"student parcel"` reflects a reassignment of students across parcels designated for different purposes, rather than changes in practices on given parcels. Students receiving the nudge are not more likely to be involved in choosing whether to fallow or whether to intercrop on the `"student parcel"`. Nor are students receiving the nudge more likely to choose whether to adopt promoted soil management techniques. In line with this, I find no evidence of changes in the adoption of soil management techniques on the `"student parcel"`—the promotional video alone is highly successful in increasing the adoption of soil management techniques across parent- and student-managed household parcels, as discussed in Lee (2025).

Besides a shift of students' time away from subsistence parcels, the nudge also triggers adoption of SBAE-promoted methods that generate more and quicker income from school-tested crops on the `"student parcel"`. Cassava is widely cultivated on both school and household farms. Traditionally, it is intercropped with rice—the major staple—serving as a supplementary food source because of its long cultivation cycle. Under subsistence strategies, the crop is typically stored underground for up to 24 months before harvests, and farmers often use planting methods that favor sprouting security—such as inserting multiple stakes per hole—at the cost of reduced tuber size—a key factor favoring sales. Consistent with prioritization of timely income for expenses such as school fees, students receiving the nudge are more likely to follow SBAE practices by having much shorter average and minimum months to root harvests—including cassava—and inserting one stake per hole.

The nudge substantially improves students' welfare. In 2022—the program's second year—students' average income rises by 63% (from 41 USD), driven by higher farm sales. In contrast, without

the nudge, SBAE yields small effects on income that are statistically indistinguishable from zero. Although food and non-food consumption remain unaffected, the nudge leads to a marked increase in annual savings, amplifying the SBAE effect from an 84% increase relative to control schools to 101%. Meanwhile, the nudge sharply reduces student dropouts in both the second and third school years, doubling total days in attendance and thus SBAE's impact per dollar on school attendance. The nudge is thus a critical factor in SBAE being twice as cost-effective as conditional cash transfers. Further, timely payment of school fees increases in both the second and third school years, increasing the fraction of students who pay their school fees on time by 43%-47% in the third school year. Supporting the importance of timely income from farming, the persuasive video-nudge combination that most increases timely harvests also produces the strongest increases in timely school fee payments and school attendance.

Turning to heterogeneity, these nudge effects on students' livelihoods and education hold for both boys and girls, and across four grades. The nudge delivers these broad-based gains by fitting students' different backgrounds and needs: for example, girls earn more through reassignment to sales-oriented parcels, while boys benefit more from adopting SBAE methods that enable quicker income.

The nudge's sizable effects on students' welfare are striking considering the minimal costs it involves—as a five-minute component within surveys. Beyond student outcomes, the current data show no evidence of changes in farming practices or sales on the `"parent parcel"` (defined analogously to the `"student parcel"`). A follow-up survey in 2026 will track (i) parents' farming practices, yields, and welfare, and (ii) indicators of household harmony. Taken together, the current results suggest that signaling to students their parents' openness to changes in resource control can be a highly cost-effective complement to effective school-based programs—by empowering students with diverse capabilities.

I next examine the mechanisms that help interpret these findings. I begin by presenting evidence that experimental information, delivered through the revelation nudge, alters farm management specifically between students and the parent whose beliefs are revealed (randomly chosen by gender), but not with other family elders (including the other parent). In other words, the nudge's effects on farming activities are confined to the particular parent–student dyad involved in the intervention. This indicates that students' responses are driven by information about that particular parent, highlighting the importance of communication frictions in intergenerational resource allocation—contrary to standard household models that assume students would fully internalize parents' preferences and beliefs regardless of the nudge.

I substantiate this interpretation with three pieces of evidence. First, the nudge does not change students' perceived availability of assistance when looking after farms, but changes whom they seek assistance from—indicating that students do not perceive more overall support for managing farms. Second, the nudge does not alter students' plans to look after farms, but whom they

request farms from—showing that it shifts intergenerational interactions rather than students' farming aspirations. Third, the nudge induces changes in farming practices that are concentrated on parcels otherwise looked after by parents whose information is revealed—confirming that behavioral changes are specific to the targeted parent-student dyad.

I address three alternative interpretations of the nudge's effects. First, the nudge, by highlighting students' progress under SBAE, might signal program benefits to students. While I cannot rule out this effect, dyad-specific interactions suggest that communication barriers play a central role in the nudge's impact. Additionally, the nudge has minimal effects on students' program participation or agricultural knowledge acquisition. Second, the nudge might reflect social desirability biases rather than genuine parental beliefs. For instance, parents watching the videos might overstate specific figures, such as the number of tubers students can harvest, to align with perceived expectations. However, the key is not the accuracy of these figures but the nudge's ability to signal parents' openness to delegating tasks. Supporting this, parents who watch the videos express more optimistic beliefs about students' harvests at midline and endline compared to those who do not, despite facing similar social pressures. Furthermore, conditional on households' treatment status, parents' beliefs strongly predict their visits to schools, ability to name field officers, and attendance of parent-teacher meetings, reinforcing the measures' validity. Parents exposed to the persuasive video, which includes testimonials, also show greater willingness to delegate farm tasks, consistent with reduced uncertainty about students' farming skills. Third, as the nudge reveals only parents beliefs when they anticipate growth in students' farm management abilities, it might provide distorted representation of parents' beliefs that students could not obtain in natural interactions. But Section 4.1 shows that after watching the video, only a tiny fraction of parents hold mixed views about students' growth across measures. I also provide robustness checks that indicate the main results hold even if we drop such parents who hold mixed views.

To rationalize the outsized effects of communication frictions, I build a simple model of household production decisions that serves three purposes. First, the model explains why students fail to learn parents' openness to changes in farm management, even after observing parents' visits to schools and technology adoption. Second, the model provides testable insights for the design of agricultural extension when influential household members—in this case, parents—might overlook opportunities for joint decision making with others. Third, the model provides a framework for comparing the revelation nudge with alternative empowerment approaches that aim to encourage joint decision making within households.

I focus on an information asymmetry about parents' beliefs, abstracting from information asymmetry about preferences—"openness" could stem from both. Parents hold two beliefs: first, about the returns to promoted techniques; and second, about the value of delegating farms to students. Parental engagement shifts both beliefs upward. Yet many parents adopt promoted techniques unilaterally to avoid communication costs associated with joint decision making. Importantly, a subset of parents would prefer joint decision making if students were willing to

incur these costs. The revelation nudge operates precisely at this margin: by signaling parents' updated beliefs to students, it increases students' willingness to bear the communication costs, thereby facilitating joint decision making and leading to changes in farm management.

What, then, explains students' inability to learn parents' willingness to delegate farms in the absence of the nudge? After all, parents move first, allowing students to observe their technology adoption decisions. The key lies in the nature of the promoted techniques. The main promoted technique—bed preparation—was already familiar to most parents. What the school farms highlighted was its yield potential when combined with complementary practices such as composting, lower-density cropping, and more attentive farm management. When parents can implement promoted techniques on their own quite effectively—which is true in the current context—their willingness to involve students becomes largely disconnected from their beliefs about whether the techniques work well. As a result, students learn little information about parents' types from parents' actions—leaving the household stuck in an inefficient equilibrium that the nudge can break by directly revealing parents' beliefs.

Social norms further sustain the pooling of parents' types: parcel assignments often follow tradition, so parents may keep students on subsistence parcels even if they privately support delegation of sales-oriented parcels. As a result, students fail to learn parents' willingness to delegate when parents choose to keep students on subsistence parcels, especially when a significant fraction of parcels are tied to lineage or community control—the very settings where I confirm that the nudge has its strongest effects.

The model generates testable implications for the design of SBAE when parents may overlook opportunities for joint decision-making. Specifically, training parents directly (or early) can reduce their benefits from delegation compared to unilateral technology adoption; however, this perverse effect is avoided when the revelation nudge encourages students to incur the communication costs. To test this, I use the randomized assignment of the annual farmer field day (FFD)—a half-day training for parents—across SBAE schools. Consistent with the hypothesis, the FFD significantly increases households' assignment of students to subsistence parcels. For instance, midline use of subsistence practices on the `"student parcel"` increases by 0.3 standard deviations. This effect is fully offset when the nudge is present—and the nudge's effect triples when schools host the FFD. These results highlight the importance of accounting for power dynamics: attempts to broaden benefits from extension can be counterproductive if communication frictions are ignored. More importantly, it highlights the potential of informational nudges to simplify issues of participant targeting in agricultural extension—as such informational nudges enhance its ability to provide broad benefits to all household members and tackle intertwined goals in rural development.

Lastly, the model can also be extended to clarify the distinct merits of the revelation nudge relative to alternative empowerment strategies that encourage joint decision making. First, because students hold private information about what they have learned in the program—and

thus about which farming responsibilities they can shoulder—the revelation nudge leverages students' self-selection into joint decision making. This contrasts with interventions that also assume communication frictions but directly impose discussions, which could be counterproductive in economic empowerment (Lowe & McKelway, 2022; Gazeaud et al., 2023). Second, by signaling parents' willingness to delegate, the nudge lowers the hurdle for students: they need only initiate conversations, without necessarily bearing the burden of finding appropriate transfers. In this sense, the intervention is less demanding than communication or negotiation training programs that seek to alter the nature of household bargaining (Björkman Nyqvist & Jayachandran, 2017; Ashraf et al., 2020; Kala & McKelway, 2025).

Related Literature. This paper contributes to four strands of the literature. First, this paper provides crucial evidence for an emerging body of work that explores the impact of communication frictions within households. The conventional literature on household decisions (Chiappori & Mazzocco, 2017) assumes that information spreads seamlessly within the household. Motivated by cases of inefficient risk sharing in developing countries (Udry, 1996; Duflo & Udry, 2004; Schaner, 2015), family economists have increasingly recognized the importance of asymmetric information in household decisions. The first attempts study cases where spouses have strategic incentives to hide objective information from each other (Ashraf, 2009; Ashraf et al., 2014; Ambler, 2015; Lowe & McKelway, 2022), such as income and use of contraceptives. More recent papers have identified frictions in information transmission (Apedo-Amah et al., 2020; Ashraf et al., 2022; Fehr et al., 2024) and processing (Conlon et al., 2022; Mustafi, 2025) when spouses receive experimentally injected information, such as maternal risks or balls in an urn. This paper makes two key innovations. First, this paper studies the communication of subjective information—preferences and beliefs—that household members naturally form in response to a development program. Second, the paper challenges the notion that frequent interactions could make communication frictions inconsequential, showing that these frictions play a critical role even in high-stakes, real-world environments with ample opportunities for indirect learning.

Second, this paper contributes a novel approach to empower individuals for household economic decisions in developing countries. Much of this work has targeted material and psychological constraints (see Chang et al., 2020; Jayachandran & Voena, 2025), such as expanding women's control over financial accounts or strengthening self-efficacy. Yet many of the most consequential barriers to agency arise not from personal limitations but from dependence on influential household members. In such contexts, efforts to empower individuals unilaterally may prove ineffective—or even counterproductive—if they disrupt household harmony.[1] One line of research thus explores providing communication and negotiation training to help individuals articulate mutual benefits in productive discussions with household members (Björkman Nyqvist & Jayachandran, 2017; Ashraf et al., 2020; Kala & McKelway, 2025). However, such training often requires expertise, tailoring,

[1] Ashraf et al. (2014) show that delivering contraceptives directly to wives in Zambia—rather than jointly with their husbands—increases contraceptive use but reduces women's reported subjective well-being.

and significant resources leading to difficulties in scaling. Another approach seeks to foster joint decision making by organizing discussion sessions (Ashraf et al., 2014; Lowe & McKelway, 2022; Gazeaud et al., 2023)—for example, involving husbands in discussions of work opportunities for their wives. Yet this approach can also backfire, as influential members could reinforce their control when disagreement arises[2] or when time is required for household members to learn about their new opportunities. This paper demonstrates a novel approach that can avoid the pitfalls of these two lines of research. Specifically, when paired with effective programs, using simple messages to signal influential members' openness to changes in resource control can be a low-cost, scalable intervention. By encouraging less powerful individuals to self-select into joint decision making, this approach mobilizes household resources, accommodate individuals with diverse needs and capabilities, and sidesteps the risks of forcing discussions.

Third, this paper provides insights for the design of agricultural extension programs in developing countries. The dominant theoretical frameworks focus on how individuals or households decide whether to adopt a given farming practice (Foster & Rosenzweig, 1995; Conley & Udry, 2010). Yet for smallholder farmers, adoption decisions are often made jointly, as resources are pooled, making communication frictions highly relevant. It is also increasingly recognized that successful extension requires not only the uptake of promoted practices but also their adaptation and integration into farmers' specific environments—rendering the adoption problem multi-dimensional.[3] This paper shows that even when school-based agricultural extension increases household adoption of soil management techniques, the allocation of students' labor to different parcels, the combination of practices used, and the resulting distribution of welfare are very sensitive to nudges mediating communication. In particular, the results imply dyad-specific communication frictions: children's information about fathers shapes decisions on fathers' decision spheres, but not mothers' decision spheres, and vice versa—even when fathers and mothers discuss new programs. This dyad-specificity is relevant for three related literatures: social network effects in agricultural technology diffusion (e.g., Beaman et al., 2021), especially under SBAE; whom to provide cash grants and loans in agriculture (e.g., Beaman et al., 2023); and how to deliver unconditional cash transfers to stimulate education spending (e.g., Haushofer & Shapiro, 2016; Egger et al., 2022; Crosta et al., 2024; Niehaus & Suri, 2024; Weaver et al., 2024).

Fourth, this paper informs the design of informational interventions in education, especially in developing countries. Prior work shows that providing objective information can increase educational investment: informing parents about returns to schooling raises enrollment (Jensen, 2010; Nguyen, 2008), while tools that help parents interpret report cards or monitor assignments improve spending and student performance (Dizon-Ross, 2019; Bergman, 2021). Yet this focus on

[2] Lowe & McKelway (2022) show that explicitly inviting husbands and wives to discuss a new job opportunity in India reduces enrollment by up to 50%. Similarly, Gazeaud et al. (2023) show that large cash grants and financial training stimulate wives' income-generating activities only when wives are not encouraged to bring their husbands to training.

[3] One consideration, for example, is the substantial heterogeneity in farming environments in Africa (Suri & Udry, 2022).

parents overlooks the interaction between parental support and students' own effort. This paper shows that increasing school attendance under SBAE requires both. Further, the sizable effects of the revelation nudge complement two related literatures: first, it shows that interventions that shift individuals' perceptions of others (see Bursztyn & Yang, 2022) could be effective even within households; second, it demonstrates that even when all participants receive dyad-specific encouragement, messages designed around beliefs about program effects—rather than generic content—can improve behavioral responses and offer a theory-informed route to enhancing the replicability of informational nudges (see Congiu & Moscati, 2022; Damgaard & Nielsen, 2018; Coffman et al., 2024; Banerjee et al., 2025).

Outline. This paper is structured as follows. Section 2, introduces agricultural production and livelihoods in rural Liberia, the school-based agricultural extension program, the importance of parental engagement, and contextual factors leading to communication frictions between parents and students. Section 3 explains the experimental design. Section 4 discusses the main empirical results. Section 5 discusses mechanisms that help interpret the results. Section 6 develops a substantive theory of household production decisions with communication frictions, and tests insights for agricultural extension. Section 7 concludes.

2. Study background

2.1. Agricultural production and livelihoods in Liberia

This study is conducted in Liberia where 72% of households live in absolute poverty (Benfica et al., 2018). Agriculture is the primary source of livelihood for about 80% of the population (African Development Bank, 2008). Smallholder farmers in this tropical region traditionally employ a fallow system under collective ownership of the land.[4] This system operates smoothly with minimal inputs to the soil — no tillage nor use of fertilizers — if the fallow period is sufficiently long. However, as in many other sub-Saharan African countries, farmers are forced to shorten the fallow period under economic and population pressures, leading to commonly observed soil degradation and declining agricultural output.

Agricultural intensification — increasing output per unit of land by increasing inputs to the soil — is the common response to soil degradation. Managing the shift to agricultural intensification, however, requires farmers to acquire soil management techniques that are fundamentally different from traditional techniques. In Liberia, this progress is hindered by low levels of coverage by extension services (3.4%) and low levels of education — the average household head has four years of schooling. As a result, Liberia has very low levels of yields in major staples even by the standard

[4] I use the term collective ownership here to refer to communal land tenure or lineage-based land tenure. It should be noted many households and families retain control rights over their parcels, but their farming practices are subject to external scrutiny—an issue discussed in detail in Section 6.7.

of sub-Saharan Africa — FAO (2023) data suggests that rice and cassava yields are only 54% and 62% of the regional average.

Besides low agricultural productivity, Liberia also has an education system with lower school attendance than those in most sub-Saharan African nations. Liberia has a net primary enrollment rate of only 44% in 2017, compared to the regional average of about 80% in sub-Saharan Africa (UNESCO Institute for Statistics, 2020). Overage enrollment is particularly serious — over 40% of primary school students are above 12 years old; and about 50% of junior high students are 18-20 years old (Liberian Ministry of Education, 2017). Another reason is the high annual rates of student dropouts, which stand at 10-15% in upper primary and junior high grades.

Intertwined with the fallow system and high dropout rates is the prevalent practice of child labor in agriculture. As shown in Panel B of Table A1, 98.7% of upper primary and junior high students in our sampled schools farm outside school. Panel A of Table A2 shows that sampled students have a mean age of 16 and 7.9 years of farming experience—meaning that the average student starts farming around age 8. The high land-to-labor ratio in the West African cultivation environment, combined with significant labor demands during agricultural seasons—clearing bush, weeding, and fending off pests and thieves—makes child labor particularly valuable.[5] Section 2.4 further describes how this dependence on child labor for subsistence farming creates tension between students' current situation and their aspirations.

2.2. School-based agricultural extension

School-based agricultural extension (SBAE) has a century-old history in rural America, dating back to the Smith-Hughes Act of 1917. At its core is an integrative model that blends the promotion of modern farming practices with student-centered learning experiences. According to Croom (2008), this model has three interrelated components. Through formal instruction, students learn the scientific basis of modern farming practices through lectures and demonstration. But more important are the other two components that ensure personalized learning experiences for rural students. Through participation in extra-curricular agricultural clubs, students develop their sense of belonging in a national network, meet mentors and role models, and gain leadership experiences and recognition in public services and competitions. Through supervised agricultural experiences, students can develop home projects tailored to their own pace and needs. Rufus Stimson, an early advocate of the method, argues in particular that home projects allow students to profit from farm work and complement school projects that are subject to overcrowding amongst students (Stimson, 1919).

In the sub-Saharan African context, SBAE is an important new model in the delivery of agricultural

[5] Very few farm households can satisfy these high labor demands with family labor alone. According to Benfica et al. (2018), over 80% of farm households in Liberia use hired labor or labor from cooperative groups.

extension and rural education.[6] In particular, SBAE should be distinguished from typical school gardens due to its emphasis on student-centered learning experiences. Typical school gardens are directed by teachers without integration with scientific instruction of modern farming practices, home projects, and a national network of agricultural clubs. By contrast, students participate in SBAE voluntarily, make major decisions on school farms (e.g. choice of crops and use of proceeds), and are empowered to make profits from home projects and gain social recognition from clubs. These distinguishing features of SBAE make it particularly suited to tackle intertwined challenges in agricultural extension and school attendance.

This study examines an SBAE program in Liberia conducted by the local NGO 4-H Liberia in collaboration with the ministries of agriculture and education from January 2021 to December 2023. The program implements the three-component SBAE model through bi-annual teacher training workshops (20 hours each, covering the integrative model and yield-improving technologies) and monthly extension visits for monitoring and instruction. First, formal instruction combines extension officer lessons with experiential teaching by trained teachers, who use structured lesson plans to illustrate scientific concepts through farming practices—such as using seed germination tests to demonstrate experimental methods or composting to explain chemical compounds. Second, extra-curricular 4-H Clubs engage 25-30 students per school in weekly farm work, bi-weekly meetings to discuss crop selection and farm management, and opportunities to compete in national agricultural fairs. Third, the program amplifies technology diffusion by positioning school farms as open demonstration plots for community visitors and encouraging students to start home projects, supported by lessons on agribusiness, farm management, and record keeping, with teachers conducting home and farm visits for supervision.

2.3. The importance of parental engagement

Even with successful training of teachers and students, weak linkages between parents and schools threaten SBAE's ability to achieve its goals. With poor roads and limited infrastructure, only 36% of parents in program schools attend parent-teacher meetings and 44% visit school farms over three years. Parents also lack accurate information about what students learn—focus groups reveal that many parents mistakenly view SBAE as typical school gardens, unaware of its training in agribusiness, yield-improving techniques, and national network of mentors and peers. Without this knowledge, parents may fail to recognize the yield potential of improved techniques, overlook opportunities to delegate farms to students, and underinvest in their children's continued education.

Lee (2025) provides crucial proof of concept that these mechanisms can be activated through light-touch parental engagement treatments: a promotional video session and an annual farmer field

[6] As in the case of agricultural education in early US history, separate provisions of agricultural training and agricultural youth clubs can be found in some sub-Saharan African countries — for example, agricultural youth clubs affiliated with the international 4-H network can be found in 15 African countries. But the integrative SBAE model has only been introduced by the US-based NGO AgriCorps to Liberia and Ghana since the 2010s.

day, described in Section 3.3. These interventions together account for only about 10% of average implementation costs per school (or 30% if the video is extended to all cohorts). By the third year, SBAE alone successfully trains teachers and students but fails to improve technology adoption on any household parcels. Conversely, SBAE augmented with both types of engagement increases adopted techniques by 26%-53% across parent- and student-managed parcels. Parental engagement activates intra-household mechanisms otherwise completely muted: parents visit school farms, learn techniques, observe harvests, and attend parent-teacher meetings, while students gain opportunities to manage household farms, use promoted techniques, and demonstrate improved yields to parents. The augmented program becomes more cost-effective than most rigorously evaluated extension programs in Africa.

Building on this success, this paper examines a sub-treatment designed to overcome intergenerational communication frictions. I show that this intervention accounts for the majority of SBAE's effects on students' economic empowerment, livelihoods, and school attendance.

2.4. Adolescents' aspirations and social norms about farm allocation

Policy reports, alongside a broad anthropology literature on West Africa, document that adolescents in rural Liberia increasingly aspire to shift from assisting on traditional subsistence parcels toward managing sales-oriented parcels (e.g., Scarborough, 2017; Guyer, 1997; Amanor, 1999). Traditional subsistence agriculture typically intercropping roots or vegetables with rice and relies on fallowing to recover soil fertility. In contrast, sales-oriented parcels prioritize crops with market value and employ practices that generate quicker monetary returns—such as monoculture or simplified intercropping, continuous cultivation rather than fallowing, and earlier harvests. This shift in aspirations is driven by both environmental and economic pressures. Declining soil fertility under fallow systems—exacerbated by population growth and shortened fallow periods—makes traditional cultivation increasingly labor-intensive. Meanwhile, rising costs of school fees and other necessities create greater demand for timely cash income that subsistence farming cannot reliably provide.

However, realizing these aspirations depends critically on whether parents accept such changes in farm management. Parents typically support adolescents managing sales-oriented parcels only if they believe adolescents have demonstrated sufficient farming skills—both to ensure productive outcomes and to justify the reallocation of household labor away from subsistence production. SBAE's emphasis on "learning by earning" through agribusiness training further heightens this tension: the program explicitly cultivates students' entrepreneurial aspirations and practical skills for commercial farming, yet whether students can act on this training depends on overcoming communication frictions with parents. The revelation nudge, by signaling parents' optimism about students' growth in farm management skills, offers a potential resolution—increasing students' confidence in proposing such changes to their parents.

2.5. Promoted agricultural techniques

The SBAE program promotes a range of agricultural techniques grouped under soil management, crop protection, improved input use, planting practices, and record keeping. This paper focuses on the adoption of practices that generate sales-oriented farming—practices that maximize yield and sales potential for targeted crops while departing from traditional subsistence strategies. The cornerstone technique is bed preparation (mounds or ridges for root crops, raised beds for vegetables), which loosens soil and enhances drainage to facilitate root penetration. Agronomic trials by the International Institute of Tropical Agriculture suggest that beds can increase cassava yields by 31-46% and sweet potato yields by 70% (Odjugo, 2008; Agbede & Adekiya, 2009). These yield gains become easily visible in stem height and leaf size within three months, and in the number and size of tubers per plant at harvest. Other promoted techniques—composting, row planting, and regular spacing—reinforce these effects when adopted together.

However, maximizing the sales potential of these techniques requires additional departures from traditional subsistence practices. Traditional cultivation in this context is heavily oriented toward food security: farmers typically intercrop multiple crops at close distances, store root crops underground for extended periods (often 12-24 months for cassava), and use planting methods that favor sprouting security over marketability. For instance, when planting cassava, farmers commonly insert multiple stakes per hole, which produces more numerous but smaller tubers that are less suitable for market sales. Similarly, root crops on student-managed parcels typically take 8-9 months to harvest on average, with extended harvest times reflecting the use of crops as stored insurance against food shortages rather than sources of timely cash income.

In contrast, sales-oriented cultivation prioritizes practices that generate quicker monetary returns and larger, more marketable produce. This includes lower-density cropping—such as planting one cassava stake per hole, which enhances tuber size suitable for markets—along with more focused management through fewer crops per parcel, less intercropping of roots or vegetables with rice, and timely harvests (often within 6 months for root crops) to generate cash income when needed, particularly for school expenses. School farms demonstrate the yield and sales potential of this integrated approach, while student home projects allow families to learn about returns under their specific growing conditions with lower subsistence risks.

3. Experimental Design

3.1. Randomized evaluation of SBAE (JMP1; Lee (2025))

Figure 1 presents the experimental design, which is structured in two layers. The outer layer is a randomized evaluation of the SBAE program (Lee, 2025), drawing from a sampling frame of 304 eligible junior high schools across five major food-producing counties in Liberia: Bong,

Lofa, Nimba, Gbarpolu, and Margibi. School eligibility was determined by two criteria: available land and water sources for establishing school farms. These 304 schools were organized into 206 geographical clusters, with schools located within two kilometers of each other grouped together. This clustering strategy ensured a minimum distance of two kilometers between schools in different clusters, limiting spillovers between SBAE and control schools.

The randomization proceeded as follows. Geographical clusters were assigned to SBAE and control conditions within pairs matched on extension officer regions and average school enrollment in 2015/16. All schools within a given cluster received the same treatment assignment. Resource constraints limited recruitment to 100 SBAE schools. The randomization yielded 100 SBAE schools (78 clusters) matched to 97 control schools (77 clusters). All schools assigned to SBAE treatment accepted the program. Figure A1 shows the geographical distribution of sampled schools relative to other junior high schools in Liberia and distinguishes between SBAE and control schools.

Following the school-level randomization, SBAE schools were further randomized into parental engagement conditions through a two-step process. First, 50 SBAE schools were randomized to receive program promotional sessions, stratified by administrative district. Second, independently from the promotional sessions, 44 SBAE schools were randomly selected to host an annual farmer field day treatment, stratified by extension officer region. Both of these randomization procedures included eligibility criteria that we will include as controls in regressions. The criteria are described in Lee (2025).

Panel A of Table A1 shows how the sampled schools compare to others in the targeted counties. The 197 sampled schools were selected through a three-stage process: first, 447 out of 1794 schools were identified as having a functional parent-teacher association (PTA) and water source according to a Water, Sanitation, and Hygiene (WASH) dataset; then, field inspections narrowed this to 304 schools with available land and water; lastly, 197 schools were randomly selected from the 304 schools. As a result, eligible schools tend to be better than county averages in several dimensions: more students, more teachers, and smaller student-teacher ratios.

3.2. Sampling of households and randomization of treatments

Sample of students

The SBAE program targeted upper primary and junior high grades in its recruitment of students. To increase the likelihood of successful follow-up after three school years, I focused on grades 4, 5, 7, and 8, excluding 6th and 9th grade students who would soon transition to different schools.

My sample was drawn from the pool of intended beneficiaries — students who farmed outside school and had parents who farmed. To identify this pool, enumerators conducted screening surveys in these targeted grades during the 2020/21 school year. To maximize the likelihood of capturing actual program participants, enumerators employed a hierarchical sampling strategy within each

school to select approximately 20 students.[7] Students were prioritized in the following order: (1) students who both farmed before 2021 and joined cooperative farming groups (kuu) before 2021; (2) students who joined kuu before 2021; (3) students who farmed before 2021; and (4) other eligible students. Enumerators selected students sequentially down this priority list until reaching the target sample size.[8] Despite this purposive approach, sampled students closely resemble the broader population of eligible students in targeted grades. As shown in Panel B of Table A1, sampled students have only slightly higher rates of prior farming experience (99.4% vs. 95.0%) and prior kuu participation (82.1% vs. 77.1%) compared to all eligible students, suggesting that the sampling strategy successfully captured a representative sample of the target population while increasing the likelihood of observing program participation.

A total of 3,757 students were sampled across 197 schools, averaging approximately 19 students per school and representing 30% of students in targeted grades. The sample was stratified by school, section (upper primary or junior high), and gender. As only one student per household was sampled, the total number of sampled households was also 3,757.

Sample of parents

For every sampled student, enumerators invited one parent or guardian (henceforth, parent) who was planting root crops, vegetables, or both in 2021.[9] In households with both male and female parents meeting this requirement, I randomized which parent was invited to maintain gender balance in the sample. In households with only one eligible parent, that parent was invited. A total of 3,757 parents were invited, corresponding to the 3,757 sampled students. In 96% of cases, the invited parent participated in the survey; in the remaining 4%, a substitute household member participated instead.

Randomization of treatments

1002 households are associated with the 50 SBAE schools that are randomly assigned to host promotional video sessions. As illustrated in Figure 1, I randomly assigned these 1,002 households to one of four treatment conditions in a 2×2 factorial design (described in detail in Section 3.3):

1. Persuasive video with revelation (256 households)
2. Persuasive video without revelation (242 households)
3. Basic video with revelation (250 households)
4. Basic video without revelation (254 households)

[7] Enumerators were instructed to oversample in larger schools and undersample in smaller schools to ensure adequate representation while maintaining feasibility.

[8] Based on a pilot survey, students with prior farming experience and prior kuu participation were substantially more likely to participate in the SBAE program.

[9] This requirement ensured parents were engaged in agricultural activities relevant to the SBAE curriculum. During student sampling, households were pre-screened to ensure at least one parent met this criterion.

The randomization was stratified by school section (i.e., elementary or junior high) and gender of students within each school, enabling us to estimate unbiased heterogeneous treatment effects by these subgroups.

3.3. Details of treatments

3.3.1. The basic and persuasive promotional videos

As discussed in Lee (2025), the parental engagement treatments are intentionally designed to be light-touch, aiming to strengthen weak parent-school linkages by providing parents with program promotional information. Specifically, they help parents (i) understand the interconnections between agriculture and education under SBAE, (ii) reconceive students' roles as active contributors in household production, and (iii) strengthen their aspirations for students' continued education.

The program promotional session treatment was conducted during the first rainy season, approximately 6 months into the program, coinciding with the first student survey (see Section 3.4.1). In 50 SBAE schools, one parent per sampled student was invited to an information session hosted by enumerators outside the school. During 20- to 30-minute sessions, parents watched a promotional video in groups of four to five and completed survey questions before and after viewing.

To understand the barriers hindering parental engagement with the program, we randomized which video parents watched. Half of the parents, randomly selected, watched the *"basic video"* (3 minutes), which described SBAE as an innovative education model that integrates experiential science instruction, extracurricular activities on school farms, and student home projects to advance technology diffusion, educational outcomes, and youth empowerment. The other half watched the *"persuasive video"* (14 minutes), which included all content from the basic video and added testimonials from past participants and locally relevant stakeholders—town chiefs, extension officers, teachers, students, and parents. These persuasive elements highlighted SBAE's effectiveness in empowering students to manage farms, grow more food, and use proceeds productively (e.g., "I paid my own school fees"), demonstrating how the program transforms lives through improved agricultural techniques (e.g., "the cassava I got was big, big").

Both videos were produced by a Liberian company, utilizing footage from SBAE activities in pilot schools, such as students answering judges' questions at the national agricultural fair. To prevent students from directly learning parents' responses—which would interfere with subsequent intergenerational interactions—parents watched the videos outside schools in groups of five under palava huts near school grounds, using laptops. Because many parents could not read or write English, enumerators provided real-time translations. Before and after watching the videos, parents

complete a set of survey questions, most importantly their beliefs about students' growth in farm management abilities under SBAE, which are described in Section 3.3.2.

Crucial for our study of intergenerational interactions, the promotional video does not directly mention any specific promoted techniques, so any treatment effects on household farming practices must operate through parents' interactions with students and the SBAE program.

3.3.2. Measurement of beliefs and consent for sharing

The critical barrier for parents to delegate farm management to students is parents' uncertainty about students' abilities, especially relative to typical adults. To measure the immediate impact of the promotional videos and to prepare for the revelation nudge treatment, I elicited parents' beliefs about students' farm management abilities before and after watching the videos. I also elicited the same beliefs from students to familiarize them with the belief measures. I elicited a total of seven belief measures. The two most important measures were about students' harvests, which best captured the effects of promoted soil management techniques in this context.[10] Both agronomic trials and anecdotes from students suggested that the promoted techniques could increase the number (along with the size) of cassava tubers per plant. I therefore elicited beliefs about the typical number of tubers from 4 cassava plants harvested in 9-10 months, comparing outcomes if planted by a typical adult in the community versus the student.[11] Beliefs about the typical adult are always elicited first, then about the student. When eliciting beliefs about the student, subjects are reminded of their answer to the typical adult question, allowing them to revise it if needed. This procedure produces two key belief measures: (i) students' harvests; and (ii) the difference between students' harvests and typical adults' harvests.

While the measures about harvests are the most important, I also elicited other belief measures to capture a broader set of farming skills that parents might value when considering whether to delegate farm management to students—thereby increasing the likelihood of eliciting positive beliefs that could be revealed to students. The third and fourth belief measures concern agricultural knowledge, captured through two 10-question quizzes that both parents and students complete. The first quiz tests general farm management knowledge, such as optimal timing of crops, while the second focuses on soil management techniques promoted by SBAE, such as how to correctly apply compost. This procedure produces two additional belief measures: (iii) students' score on farm management; and (iv) students' score on agricultural innovations.

Finally, I elicited three additional qualitative belief measures to capture parents' attitudes about the likelihoods that the student (v) would leave a farm untended at some point during the agricultural

[10] Most parents in focus group discussions stressed that it is important for SBAE to train students on the practical aspects of agriculture, rather than just theoretical knowledge. Here, the measure of harvests captures students' practical abilities.

[11] I asked subjects about the *typical* number of tubers from 4 instead of 1 cassava plant because this helps capture the average across plants, rather than simply focusing on the maximum number.

season; (vi) would be able to pay all school fees using money earned from farming; and (vii) would become a farmer after finishing school. These measures are on a 7-point Likert scale and were asked of parents only.

The measurement occurred in three steps—all within the first survey round (June to August 2021)—which are outlined in Figure 1. First, enumerators conducted the baseline survey with students, eliciting their beliefs about their current farm management abilities. Second, during the promotional video sessions, we elicited parents' beliefs about students' current farm management abilities *before* video viewing. After video viewing, we elicited parents' *forward-looking* beliefs about students' farm management abilities in one year. Third, during the follow-up survey with students—conducted within one day after the video sessions—we elicited students' *forward-looking* beliefs about their own farm management abilities in one year, using the same seven measures.

This belief elicitation procedure had two key features. First, it was incentivized. I provided a prize for the 10% of subjects who provided the closest guesses for each of the third and fourth belief measures, which were about performance on the knowledge quizzes. Second, beliefs were carefully elicited to avoid errors. After subjects provided a belief about abilities one year ahead, they were always given a comparison with their corresponding belief about current abilities and a chance to revise.

Parents and students in all four treatment conditions—regardless of the revelation nudge condition—were given the same belief elicitation exercises. Further, at the end of the program promotional session, parents were asked if they agreed to let us share any of their beliefs about students' growth with students. All parents who had messages for us to reveal—those who expected students' growth in at least one belief measure—agreed.

Section 4.1 provides empirical results that further validate this belief elicitation procedure. First, while the revelation nudge selects only parents' beliefs about growth—thus a truncated representation of parents' beliefs—most parents had overwhelmingly positive beliefs about changes in students' farm management abilities. Second, against the concern that parents might provide socially desirable beliefs, I provide evidence that parents' anticipated improvements are large and meaningful, and strongly predict their responses to the program one year later.

3.3.3. The revelation nudge

The revelation nudge—the key treatment of interest—was a randomized add-on to an encouragement message delivered to students in a follow-up survey within one day after parents viewed the video. The survey first activated students' thinking about their potential growth under SBAE by eliciting their beliefs about their abilities one year ahead (Step 3 of the belief elicitation procedure). Upon completing this exercise, a random half of students received the nudge (henceforth, the *revelation nudge*). The nudge contained "revelation messages" for each

belief measure where parents anticipated growth, with each message consisting of two elements: (i) a qualitative statement, such as "your father thinks that if you start growing cassava next year, you can harvest more tubers than if you start now"; and (ii) a number indicating how much parents expected students to improve in one year, such as the increase in the typical number of tubers students could harvest from a cassava plant. The nudge therefore contained one to seven "revelation messages," depending on the number of belief measures for which parents anticipated improvement. Regardless of whether students were assigned the nudge, all students received an encouragement message reminding them about the SBAE program and urging them to share farming practices with their parents. Specifically:

> "You are a student of *[name of school]*.
> The 4-H agricultural education program is currently happening in *[name of school]*.
> We encourage you to talk to your household members about any new knowledge that you have learnt through the program.
> A very good first step would be to talk to *[sampled elder]* (e.g. your uncle)."

A major concern with the revelation nudge treatment is that students could potentially attach negative connotations to the revelation messages. For example, when seeing messages suggesting that elders expected students to be able to grow more cassava next year, students might infer that elders believed they currently had poor farming skills. I alleviated such negative connotations in two ways. First, the introduction to revelation messages emphasized the concept of "learning in farming":

> "The questions we just asked you, show us how much you think you will learn in farming from now until next year. We asked the same questions to *[Sampled parent] (e.g. your father)*. [He/She] will also win the same amount of prize money if [he/she] answered the most questions correctly. Here are [his/her] answers."

Second, every revelation message was read in a positive tone that emphasized students would improve over time. Below is an example of a revelation message (for the measure "cassava harvest relative to a typical adult in the community"):

> "*[Sampled parent] (e.g. your father)* thinks that if you start growing cassava next year, you can harvest more tubers than an adult farmer.
> Specifically, *[sampled parent] (e.g. your father)* thinks that you will harvest ... more tubers from 4 cassava plants than an adult farmer!"

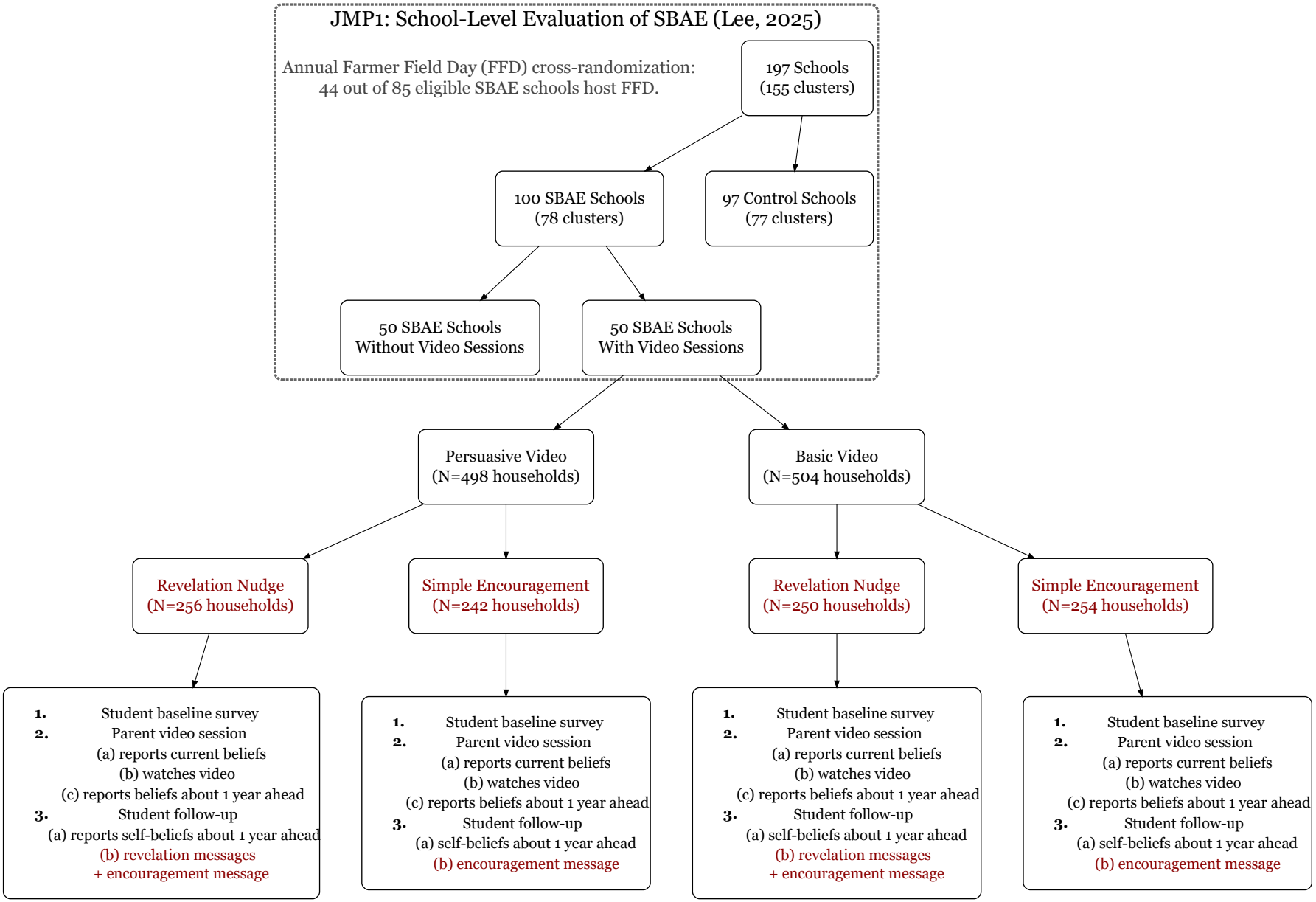


Figure 1: Randomization Tree

3.4. Design of data collection

3.4.1. Timeline of activities

Figure 2 shows a timeline of SBAE activities and randomized treatments.

The SBAE program began in January 2021, which was the beginning of the school year 2020/21,[12] and ended in December 2023. The program covered three school years and three rainy seasons. We conducted a major survey with subjects in the midst of each rainy season. This timing was chosen for two reasons. First, it coincided roughly with the conclusion of a school year, which provided a natural point to track program impact. Second, it coincided with the periods where most Liberian farmers conduct land preparation for their cultivation, which allowed us to ask questions about adoption of promoted soil management techniques.

Mirroring the layered randomization design, the program evaluation was separated into two stages. In the first year, I tracked the program's initial impact in the first survey with students and implemented the treatments described in Section 3.3. Then, in the second and third years, I conducted the midline and endline surveys with teachers, students, and parents to assess the primary outcomes. A future survey is planned to take place in 2026. This future survey tracks parents' livelihoods, households' production indicators (e.g. yields and sales), and indicators of household harmony.

[12] The commencement of the school year was delayed by 3 months due to school closures during the COVID-19 pandemic.

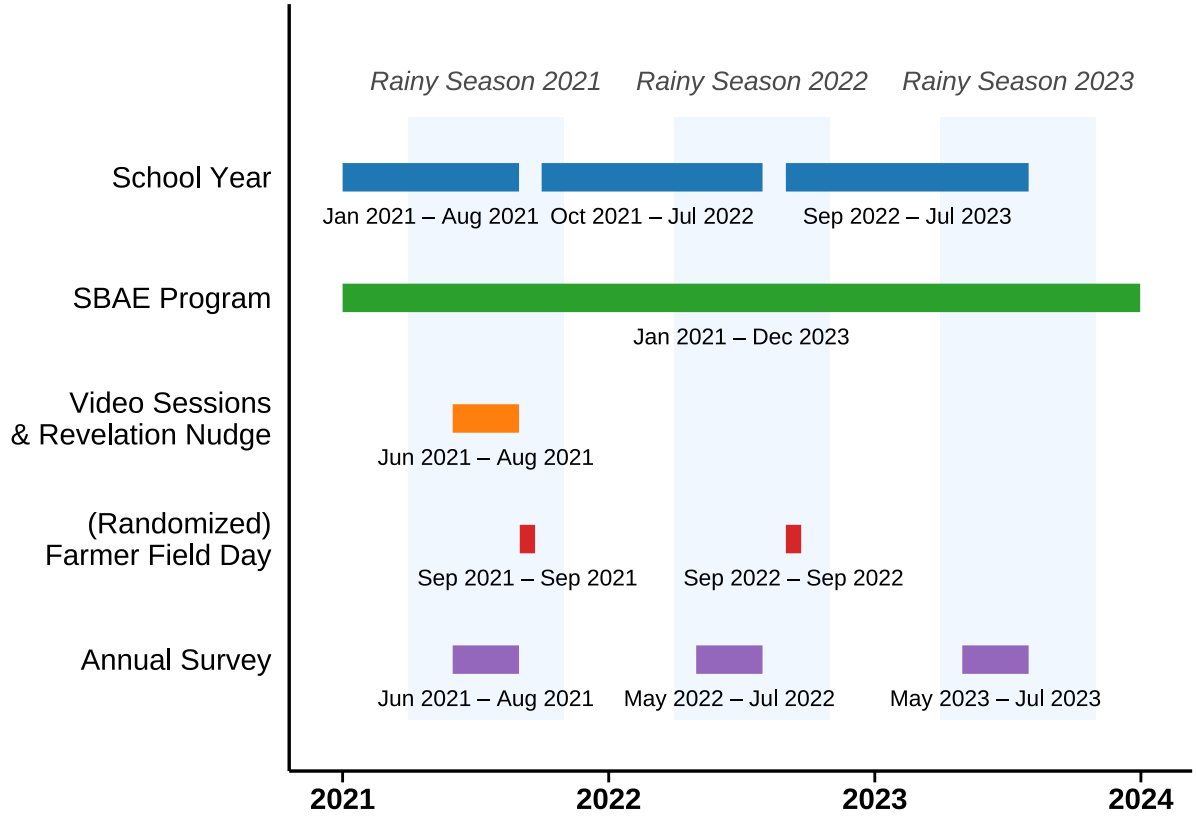


Figure 2: Timeline of SBAE program and data collection activities

3.4.2. Surveys and farm survey modules

Due to budget constraints and funding arriving in stages, survey coverage varied across the three annual rounds. All 1002 households associated with the 50 SBAE schools with promotional video sessions were tracked throughout. For other households, a stronger focus was placed on tracking students. In the first survey (2021), I tracked and interviewed only 2520 students — all sampled students in the 50 SBAE schools assigned to the promotional video treatment, and a random half in the other 147 schools. In the midline survey (2022), all students were tracked but not parents. In the endline survey (2023), all students were tracked, together with 45% of parents — including all sampled parents in the 50 SBAE schools with promotional video sessions.

Besides questions about general farming activities, students at midline and students and parents at endline were given additional, detailed questions about a designated parcel (see Section 3.4.3 for the selection criteria). All 1002 households from the 50 SBAE schools with promotional video sessions received these detailed farm survey modules. Coverage varied among other households: 1388 of 2425 students at midline and 1363 of 1543 parents at endline were selected for the farm survey

module. The details of this random coverage of other households are discussed in Lee (2025).

3.4.3. Designated "student parcel" and "parent parcel"

The hypothesis in this paper is that signaling parents' openness to delegation can cause a shift in students' time away from their traditional roles as assistants in subsistence cultivation toward managing more sales-oriented parcels within the household. This is a complex process involving at least three features: (i) which household parcels students work on; (ii) students' time spent on different household parcels; and (iii) farming practices (i.e., subsistence or sales-oriented) used on parcels on which students work.

To capture these dynamics, I select a designated `"student parcel"` for detailed analysis using an algorithm that sequentially applies the following priorities, from highest to lowest: (i) parcels containing roots or vegetables in the corresponding rainy season of the survey (2022 for midline; 2023 for endline), (ii) parcels looked after by the student, and (iii) parcels on which the student spends the most time. Figure 3 illustrates this selection process. The algorithm first identifies all parcels meeting the highest-priority criterion (containing roots or vegetables), which eliminates P3, P5, and P6 from consideration. Among the remaining parcels (P1, P2, and P4), the algorithm then prioritizes those the student looks after, which eliminates P2. Finally, between P1 and P4—both of which contain roots or vegetables and are looked after by the student—P4 is selected because the student spends more time on it. Parcels that fail to meet earlier criteria are eliminated regardless of whether they satisfy lower-priority conditions.

A `"parent parcel"` is designated analogously for the endline parent survey. The selection algorithm applies the same sequential priorities: (i) parcels containing roots or vegetables in the 2023 rainy season, (ii) parcels looked after by the parent, and (iii) parcels on which the parent spends the most time. This parallel approach allows for comparable analysis of farming practices and time allocation across both student and parent parcels within the same household.

3.5. Measurement of outcomes

A key measure of interest, across various outcomes, is the degree of subsistence orientation. Throughout this paper, I proxy this by (i) the use of rice; (ii) the presence of intercropping of roots or vegetables with rice; and (iii) crop diversity.

3.5.1. Immediate responses following informational treatments

I measure three types of outcomes following the promotional video and revelation nudge treatments. First, I measure parents' beliefs about students' farm management abilities, captured through seven belief measures elicited before video viewing (about current abilities) and after video viewing (about

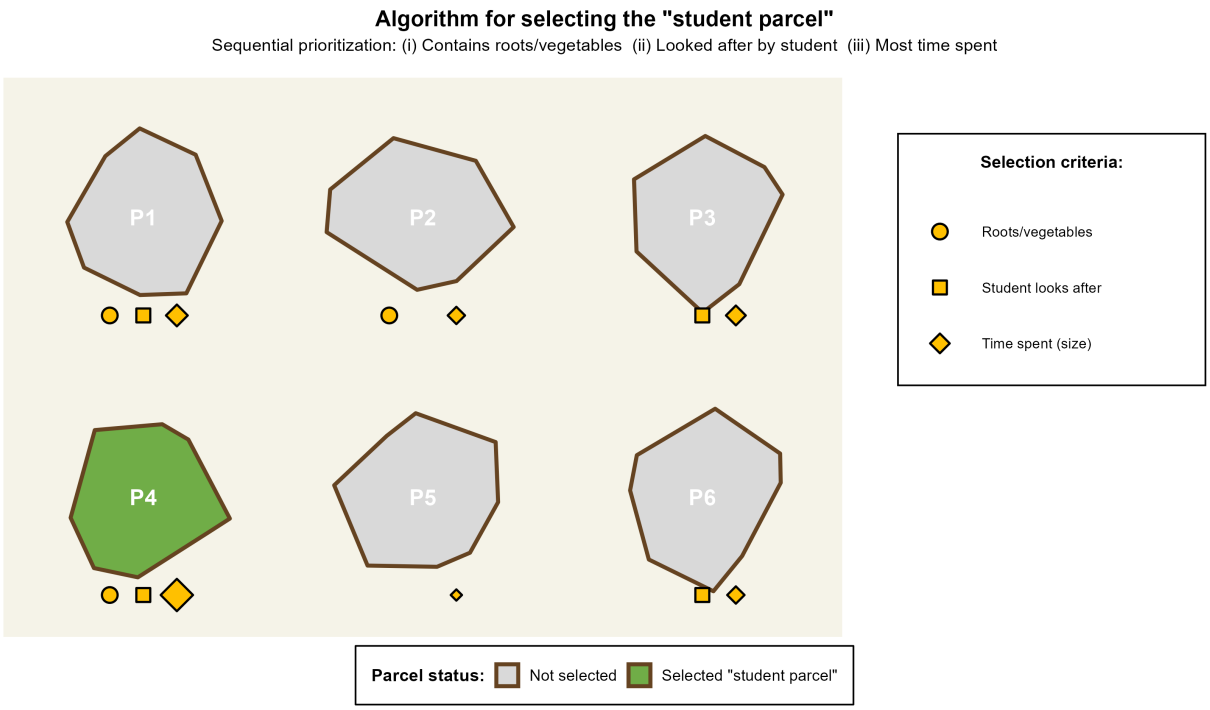


Figure 3: The algorithm for selecting the designated `student parcel`

abilities one year ahead). Second, following the revelation nudge treatment, I ask students about their plans to look after farms in that rainy season and what types of farms they look after, including location, crop types, and degree of subsistence orientation. Third, I ask students whether help is available from family and community elders when looking after farms—such as finding farmland, securing labor, obtaining money for seeds, getting assistance with farm management, or building relationships with town elders. To isolate effects specific to the parent-student dyad targeted by the revelation nudge, I also ask students whom they would seek help from, allowing them to identify multiple people.

3.5.2. Primary outcomes

I measure six types of primary outcomes in the midline and endline student surveys. These outcomes, reported by students, capture the process through which students shift time away from subsistence parcels toward managing sales-oriented parcels, adopt SBAE-promoted commercial farming practices, and benefit more from the SBAE program as a result.

First, I measure students' overall farming activities, including whether they farm outside schools, their weekly hours spent on farming, and the number of household parcels they work on. Second, I

measure how households assign students to parcels, including whether they look after any household parcels, the number of parcels they look after, and the types of household parcels assigned to them—categorized by crops and by degree of subsistence orientation.

Third, I measure whether students gain management authority on the designated `"student parcel"`, including whether students look after it and whether they make independent or joint decisions with elders regarding actions such as choosing crops, selecting soil management techniques, and deciding whether to intercrop or fallow.

Fourth, I measure the degree of subsistence orientation on the designated `"student parcel"`.

Fifth, I measure whether sales-oriented practices are used on the designated `"student parcel"`. This includes three sub-categories: (a) the average months to harvest root crops (across crops and harvests); (b) the use of SBAE-promoted, sales-oriented methods for cassava planting, captured directly by the use of the one-stake-per-hole method and indirectly by parents' retrospective endline reports of observing increased tubers per plant from the designated `"student parcel"`; and (c) the minimum months to harvest root crops.

Sixth, I measure students' welfare outcomes parallel to those in Lee (2025): (a) students' annual income and savings in 2022 (the midline year), retrospectively reported at endline; and (b) students' school attendance, captured by whether students remain enrolled at midline and endline, as well as the imputed total days of school attendance in 2021/22 to 2022/23.[13] I additionally measure timely payment of school fees, reported in the second semester of 2021/22 and the two semesters of 2022/23.

3.5.3. Secondary outcomes

I collect two types of process information to understand how students shift away from subsistence parcels. First, I examine whom students interact with when making this shift. At midline, students report who looks after and makes decisions on the designated `"student parcel"`, allowing me to examine whether shifts concentrate around the targeted parent-student dyad. I also measure the degree of subsistence orientation on parcels looked after by different elders and on parcels owned by treated parents, other household members, and lineage and community members outside the household. At endline, parents retrospectively report whether students have requested to look after farms and make particular decisions on household parcels over the past two years.

Second, I examine whether parents exposed to the SBAE program and promotional video are genuinely open to delegating farm management to students. I assess whether parents' improved beliefs about students' cassava harvests (analogous to those in Section 3.5.1) persist at midline and

[13] This imputed measure is based on students' enrollment status, rate of absence from school, and a total of 200 school days per year. A student who drops out is counted as having zero school days. A student who switches schools is counted as having the same number of school days as the control school average.

endline, and whether these beliefs—elicited after the video sessions—predict parents' subsequent responses to the SBAE program, including their visits to school farms and attendance at parent-teacher meetings. To guard against (differential) social desirability biases, I also ask parents to name the field officers at midline.

3.5.4. Auxiliary outcomes

I collect two types of auxiliary outcomes. The first type assesses broader changes in household welfare beyond the primary outcomes. I examine whether promoted soil management techniques are used on the designated `"student parcel"` and the `"parent parcel"`, and whether there are offsetting changes in the degree of subsistence orientation on the `"parent parcel"`. I also measure students' weekly hours spent on non-farming activities, including studying and household chores, their absenteeism (which is less likely than dropouts to be affected by timely payment of school fees), their annual spending in 2022 and 2023, and food security, captured by the number of breakfasts and lunches consumed over the past week.

The second type tests whether the revelation nudge operates through channels beyond overcoming communication frictions within the targeted parent-student dyad. I examine whether students are more likely to join the SBAE program or spend more weekly hours on school farms. I also assess whether students and parents perceive a shift in community-level opinion on adolescent farm management autonomy, using vignettes that ask subjects to estimate the number of adults (out of 10) who agree with a set of statements.

3.6. Balance of baseline characteristics

Lee (2025) provides strong evidence that baseline characteristics are balanced across SBAE and control schools, as well as SBAE schools assigned to different sub-treatments. Focusing on the 1002 households in schools with promotional video sessions in the current experiment, Table A2 shows that baseline characteristics of sampled students and parents are balanced across treatment arms at baseline, that is, at the beginning of the school year 2020/21. Mirroring our main regressions in Section 4.3), I regress baseline characteristics on binary indicators of the persuasive video (PVideo_h) and the revelation nudge (Rev_h), and their interaction ($\text{PVideo}_h \times \text{Rev}_h$), with strata fixed effects. None of the F-tests of treatment coefficients across baseline characteristics are significant at the 10% significance level.

Panel A presents baseline characteristics of sampled students. Students average 15.6 years of age, are in grade 5.6, have 7.9 years of farming experience, and 51% are male. Treatment arms are well-balanced on these characteristics, with the exception of a minor gender imbalance between the persuasive and basic video conditions. All other pairwise comparisons show no significant differences.

Panel B presents baseline characteristics of sampled parents. Parents average 42 years of age, 46% are male, 63% are biological parents (rather than guardians), and 64% are household heads. All engage in farming, as required by the sampling design. Consistent with typical rural Liberian patterns, 68% are literate in English, 63% completed elementary school, and 45% completed junior high school. Ninety-six percent of survey participants were the originally invited parents. Treatment groups are well-balanced on all characteristics except for a minor imbalance in farming intensity between revelation conditions. All other pairwise comparisons show no significant differences.

3.7. Attrition

3.7.1. Levels and correlates of attrition

Table A3 shows the attrition levels of students in the midline and endline surveys, as well the attrition levels of parents in the endline survey.

Attrition among students is a serious concern in this study. Panel A shows that student attrition rate is 35% by the midline survey and 45% by the endline survey. Panel B shows that this high rate of attrition is mainly due to student dropouts from schools (16% by midline; 22% by endline) or switching to other schools (15% by midline; 21% by endline). The high rate of switching to other schools reflects the fact that many students are progressing to another school section (junior high or senior high); and in addition, students who seek better opportunities in schools closer to urban areas.

While the level of student attrition is high, Panel A of Table A3 shows that it does not vary significantly across treatment conditions. Panel A shows that none of the pairwise comparisons between treatment conditions are statistically significant at the 10% level. Most point estimates are limited to within a pairwise attrition difference of two percentage points. The exception is the interaction between the persuasive video and the revelation nudge (relative to having the persuasive video only)—it reduces midline ane endline attrition by 4.9 percentage points (p=0.44) and 4.2 percentage points (p=0.54) respectively. This is strongly related to the treatment effects, as Panel C and Section 4.4.2) indicate, the reduced dropout rates at midline and endline happen only in the arm with both the persuasive video and the revelation nudge.

Relative to students, the attrition among parents is much lower. Panel A shows that parent attrition rate is only 14% by the endline survey. Panel B shows that the main reasons for parent attrition are due to travelling (5%) or relocation (6%). As in the case of students, endline parent attrition does not vary significantly across treatment arms, with point estimates limited to within a pairwise attrition difference of two percentage points.

In view of the high student attrition rates, Table A4 examines (i) whether student and parent attrition is selective in terms of baseline characteristics; and (ii) whether selective attrition is differential across treatment arms. Panel A shows that students are selected into the endline

sample mainly based on grade, but not based on age, gender or years of farming experiences. An increase of students' grade by one increases attrition probability by 10.4%. But this relationship does not translate into quantiatitvely important differences between the attritors and non-attitors. The attritors are, on average, in 0.05 standard deviations more senior grades than non-attritors. This is a natural reflection of the student sample — upper-grade students are closer to the point where they need to progress to another school section (junior high or senior high). Meanwhile, Wald tests for the interactions of treatments with baseline student characteristics indicate that none of the selective attrition is differential across treatments.

Panel B shows that none of the baseline characteristics predicts parents' selection into the endline sample, except for a slight significance of gender. Wald tests again indicate that selective attrition is not differential across treatment arms.

3.7.2. Summary and countering strategies

In Section 4.3), I will present the main intent-to-treat (ITT) effects from the midline and endline samples. The current section shows that while the student attrition rates at midline and endline is high, posing a threat about internal validity. However, since students and parents are selected into the endline samples in the same manner across treatment conditions, so addressing differential attrition rates is sufficient to preserve the comparability between groups. In the Online Appendix, I will present Lee lower and upper bounds (Lee, 2009) to account for the worse-case and best-case effects of differential attrition.

4. Experimental results

4.1. Parents' beliefs about students' improvements under SBAE

The focus of this paper is the belief revelation nudge, which conveys parents' beliefs about students' improvements under SBAE. Because the nudge is based on parents' beliefs elicited in the video session during the first rainy season, I begin by examining these beliefs. Specifically, I analyze parents' beliefs about their children's current and future farm management abilities, measured before and after video viewing, respectively. The nudge reveals parents' expected improvements to students in seven measures. To understand the nudge treatment, the first question is thus: do parents anticipate improvements?

4.1.1. Do parents anticipate improvements in general?

For each of the seven belief measures, Figure 4 classifies parents' beliefs about one-year changes as improvements, no changes, or declines. For both parents watching the basic or the persuasive

video, 97% express at least one improvement across the seven measures. This forms a strong basis for the revelation nudge, as this implies that there are potential revelation messages for virtually all households in the sample.

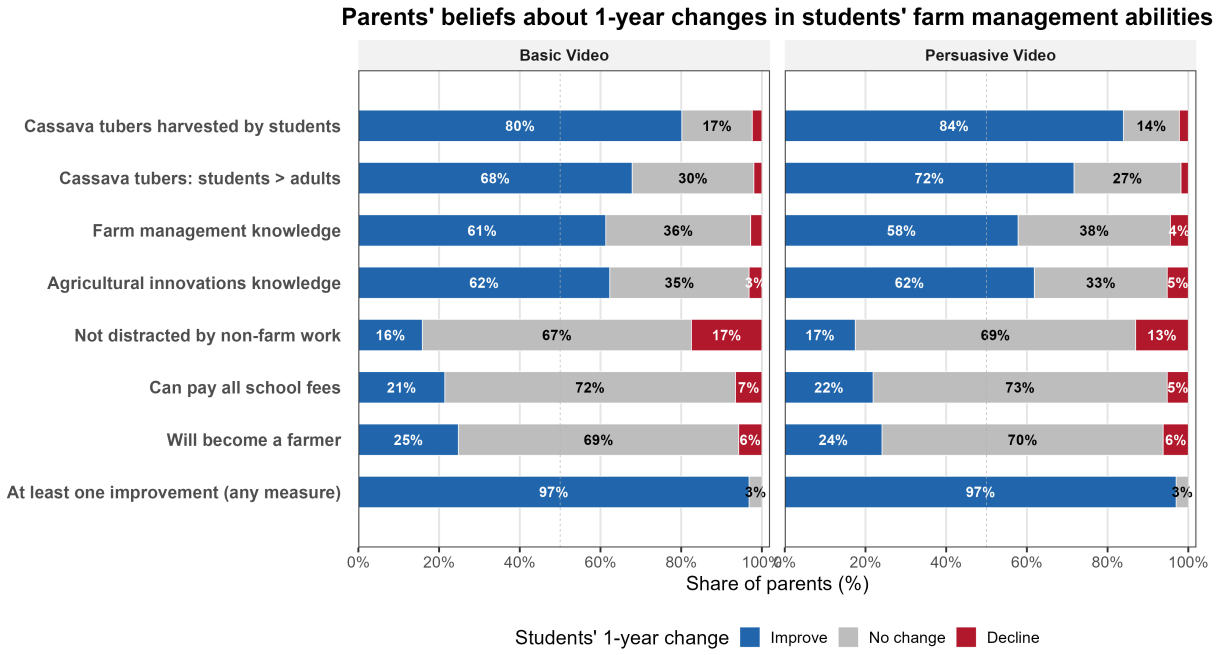
Further inspection suggests that the share of parents who anticipate improvements is highest for measures of students' harvests—a key indicator of their ability to manage farms. Aligned with video messages that highlight the yield-improving potential of techniques from SBAE, 82% of parents anticipate improvements in the number of cassava tubers per plant that students can harvest in 9-10 months. An even more critical factor determining whether parents are open to delegating farm tasks to their adolescent children is students' farm management abilities relative to typical adults in the community. 70% of parents also express optimism on this measure: they anticipate that the number of tubers per plant that students can harvest will exceed that of typical adults.

Many parents also express optimism about students' growth in farming knowledge, which is captured through quizzes on farm management knowledge (0-10) and agricultural innovations knowledge (0-10). 60% anticipate improvements in the first measure and 62% anticipate improvements in the second. The shares of parents who anticipate improvements in these knowledge measures are lower than those for harvests, partly because many parents already express high confidence in students' current knowledge (7.7/10 and 7.6/10 on average, see Table 1). Importantly, fewer than 5% of parents anticipate declines in each of the four measures of knowledge and harvests.

There are far fewer anticipated changes in other aspects of farm management, which are elicited using a 7-point Likert scale. Only 17% anticipate an increase in the likelihood that students will be distracted by non-farm work, 21% anticipate an increase in the likelihood that students can pay all school fees using farm sales, and 26% anticipate an increase in the likelihood that students will become farmers after finishing school. Most parents anticipate no changes in these perceived likelihoods, with roughly equal shares anticipating increases and decreases.

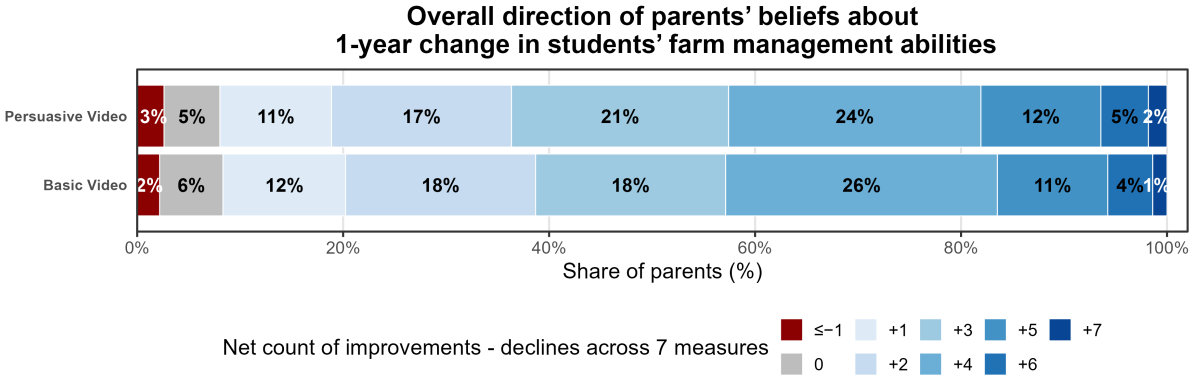
After examining the share of parents who anticipate improvements in each individual measure, I next analyze their aggregate responses in Figure 5. Specifically, I calculate the net count of measures with anticipated improvements minus those with anticipated declines. Across both video arms, 92% of parents have positive net counts, 5-6% have zero, and 2-3% have negative. Furthermore, multiple improvements are common: over 60% have net counts above two, over 40% above three, and about 20% above four.

Overall, the evidence from Figures 4 and 5 indicates that nearly all parents anticipate improvements for students, and in most cases, multiple revelation messages are available.



Notes: This figure shows the share of parents who anticipate “improvements”, “no change”, or “decline” for each of the seven belief measures about students’ farm management abilities: “improvement” if beliefs about students’ abilities one year later, measured after video viewing, are more optimistic than beliefs about students’ current abilities, measured before video viewing. “Cassava tubers harvested by students” refer to the number of cassava tubers from 4 plants, harvested in 9-10 months. “Cassava tubers: students > adults” refer to whether parents believe that the students’ harvested tubers would exceed that of a typical adult in the community: “improvement” if parents believes students’ harvests are superior (regardless of baseline beliefs); “no change” if students’ harvests remain equal or inferior; and “decline” if students’ harvests decline relative to typical adults’. “Farm management knowledge” and “agricultural innovations knowledge” refer to 10-question quizzes that both parents and students complete. The fifth, sixth and seventh measures refer to 7-point Likert scale measures of likelihood: whether students might leave farms untended when they can earn money from non-farm work (reversed scale); whether students can pay off all school fees in the semester following the rainy season; and whether students will become a farmer.

Figure 4: Parents’ anticipated changes in students’ farm management abilities: separate measures



Notes: This figure shows that most parents who watch the promotional video anticipate students to improve their farm management abilities in one year. Parents are placed into a total of nine categories based on the number of measures for which they anticipate improvements, net of the number of measures for which they anticipate declines: A negative net count indicates that pessimistic revisions dominate; a count of zero indicates no overall change; and higher positive counts indicate that optimistic revisions dominate. Net negatives are shown in deep red, zero in gray, and progressively darker blues represent larger net positives.

Figure 5: Parents' anticipated changes in students' farm management abilities: aggregate measures

4.1.2. Are parents' anticipated improvements large and meaningful?

We now know that parents anticipate positive changes for students one year after watching the promotional video in the first rainy season. However, these anticipated changes may not reflect SBAE's training efficacy, as students naturally improve their farming skills as they age. Table 1 shows that the magnitude of anticipated changes is large and meaningful. Specifically, changes attributable to SBAE training should be most evident in students' harvests, where substantial improvements would indicate improved yields under SBAE-promoted techniques. Column (1) of Panel A shows that parents expect students' harvest from 4 plants to increase by 8.4 tubers, equivalent to more than 2 tubers per plant and representing a 36.2% increase relative to the current average. The expectation of 31.4 tubers, or 7.9 tubers per plant, represents a substantial improvement over the typical range of 4-6 tubers per plant under traditional cultivation methods. This magnitude of anticipated change is unlikely to result from measurement noise or natural growth in students' farming skills alone, and more likely to reflect the use of yield-improving techniques under SBAE. Equally importantly, column (2) shows that parents' beliefs shift markedly: while the average parent currently views typical adults as producing higher harvests than students, this same parent expects students to surpass typical adults in future harvests. Panel B shows even larger magnitudes when restricting attention to parents who anticipate improvements: 10.4 tubers from 4 plants, representing a 48.2% increase from the current mean. By contrast, Panel C shows much smaller magnitudes when restricting attention to parents who anticipate declines: 6.4 tubers from 4 plants, or a 23.9% decrease from a relatively higher current mean of 26.7 tubers. This asymmetry is revealing: anticipated improvements more likely reflect expected gains from SBAE techniques,

whereas anticipated declines more likely reflect measurement error or downward corrections of initially overestimated beliefs.

The rest of Table 1 shows the magnitude of anticipated improvements across other belief measures. The average parent expects a 19.7% increase in farm management knowledge scores, a 20.4% increase in agricultural innovation knowledge scores, a 5.3% increase in the likelihood of paying school fees using farm sales, and a 5% increase in the likelihood of becoming a farmer. Column (8) aggregates all belief measures by taking their first principal component and converting beliefs into an index scaled from 0 to 100. This aggregate measure shows an anticipated improvement of 34.6%, further confirming that parents expect substantial improvements overall.

Table 1: Parents' beliefs about students' improvements: magnitudes

	Panel A: Full sample							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mean change	8.352***	8.352***	1.521***	1.558***	-0.025	0.295***	0.292***	16.421***
	(0.253)	(0.253)	(0.058)	(0.060)	(0.048)	(0.033)	(0.031)	(0.399)
Median change	6	6	1	1	0	0	0	16
Min change	-20	-20	-5	-6	-6	-6	-5	-29
Max change	43	43	8	9	6	5	6	72
Current Mean	23.087	-0.646	7.739	7.649	5.071	5.562	5.814	47.503
Current SD	10.939	8.710	2.133	2.120	1.924	1.650	1.263	15.936
Current Min	2	-40	1	1	1	1	1	0
Current Max	60	40	10	10	7	7	7	100
Possible range	[0,60]	[-60,60]	[0,10]	[0,10]	[1,7]	[1,7]	[1,7]	[0,100]
Traditional range	[16,24]	[-8,8]						
Observations	1002	1002	1002	1002	1002	1002	1002	1002

	Panel B: Subsamples with only parents who anticipate improvements							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mean change	10.359***	10.359***	2.683***	2.313***	1.982***	1.774***	1.559***	18.592***
	(0.258)	(0.258)	(0.057)	(0.071)	(0.110)	(0.080)	(0.064)	(0.375)
Median change	8	8	2	2	1	1	1	17
Min change	1	1	1	-6	1	1	1	0
Max change	43	43	8	7	6	5	6	72
Current Mean	21.499	-1.676	6.506	6.759	4.120	4.433	4.820	46.076
Current SD	9.772	8.262	1.672	1.904	1.907	1.699	1.361	15.323
Current Min	2	-40	1	1	1	1	1	0
Current Max	56	36	9	10	6	6	6	92
Possible range	[0,60]	[-60,60]	[0,10]	[0,10]	[1,7]	[1,7]	[1,7]	[0,100]
Traditional range	[16,24]	[-8,8]						
Observations	822	822	597	597	167	217	245	902

	Panel C: Subsamples with only parents who anticipate declines							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mean change	-6.348***	-6.348***	-2.167***	-0.861**	-2.327***	-1.508***	-1.483***	-5.758***
	(0.953)	(0.953)	(0.224)	(0.363)	(0.141)	(0.142)	(0.118)	(0.748)
Median change	-5	-5	-2	-1	-1	-1	-1	-4
Min change	-20	-20	-5	-6	-6	-6	-5	-29
Max change	-1	-1	-1	4	-1	-1	-1	-0
Current Mean	26.652	1.174	8.833	8.167	4.837	6.119	6.250	55.002
Current SD	11.105	9.614	1.648	1.875	1.900	1.327	1.323	14.897
Current Min	10	-20	5	4	2	2	2	23
Current Max	60	18	10	10	7	7	7	80
Possible range	[0,60]	[-60,60]	[0,10]	[0,10]	[1,7]	[1,7]	[1,7]	[0,100]
Traditional range	[16,24]	[-8,8]						
Observations	23	23	36	36	153	59	60	55

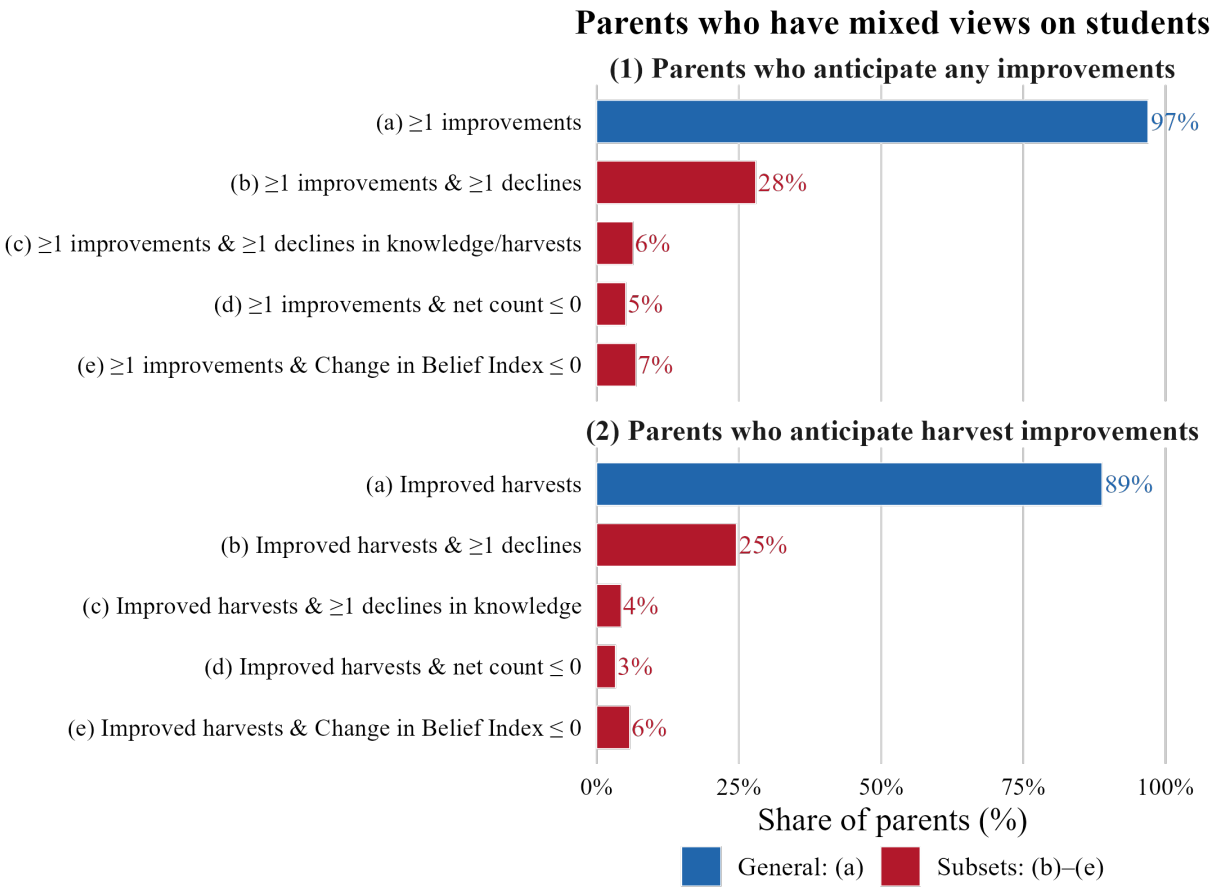
Notes: This table shows parents' beliefs about students' one-yaer change in farming abilities, tested against zero using a one-sample t-test; stars denote significance. Variable 5 (Leave farms untended) is reverse-coded (7 minus value + 1) before differencing so higher indicates more optimistic beliefs. Panel A uses the full sample. Panels B and C restrict the sample *column-wise* to parents who anticipated improvements (1-year-ahead measure - current measure > 0) and parents who anticipate declines (1-year-ahead measure - current measure < 0), so the number of observations differs by column in Panels B and C. In computing the Beliefs Index, I begin by extracting the first principal component of the first seven (current) belief measures, then transform the variation into an 0–100 index.

4.1.3. Are there parents who hold more mixed views about students' improvements?

The revelation nudge selectively reveals only parents' anticipated improvements, not their anticipated declines. This raises the question: does this truncated approach accurately represent parents' overall beliefs about changes in farm management? Figure 6 examines whether there are subsets of parents who hold negative views alongside their anticipated improvements. Panel 1(a) shows that 97% of parents anticipate at least one improvement. Panel 1(b) reveals that 28% of all parents anticipate at least one decline in addition to improvements. However, when we exclude the noisier Likert-scale measures and focus only on beliefs about knowledge and harvests in panel 1(c), this percentage drops substantially to just 6%. Panels 1(d) and 1(e) further confirm that very few parents who anticipate improvements hold overall non-positive views. Only 5% have non-positive net counts, and only 7% show non-positive changes in the aggregate belief index. These findings suggest that the truncated revelation approach captures parents' overall optimism effectively, as the vast majority of parents who anticipate improvements do not simultaneously hold predominantly negative views. Panel 2 presents similar patterns, restricting attention to the 89% of parents who anticipate improvements in harvests. Panel 2(b) shows that 25% of all parents anticipate at least one decline alongside harvest improvements. However, panel 2(c) demonstrates that only 4% of parents remain when we rule out noisier Likert-scale measures and restrict declines to knowledge measures specifically. Panels 2(d) and 2(e) reinforce this pattern, showing that only 3% and 6% of parents hold non-positive net counts and non-positive changes in the aggregate belief index, respectively. These results confirm that even among parents optimistic about harvest improvements—the most critical dimension for farm delegation decisions—the overwhelming majority maintain consistently positive views across multiple belief measures. Overall, except for a small minority, parents anticipate clear positive changes for students across multiple dimensions. The revelation nudge, despite being truncated to show only improvements, thus provides an accurate representation of parents' views rather than a distorted signal of their overall beliefs. In the Online Appendix, I further show that the main results in Section 4.3 remain substantively unchanged when excluding parents who hold mixed views under various definitions.

4.1.4. Are the belief measures reliable?

The above discussion establishes that parents in this experiment—those exposed to the SBAE program and the promotional video—overwhelmingly anticipate improvements for students. This conclusion hinges on the belief measures accurately representing parents' views. In Section 5.2, I provide evidence of strong predictive validity: conditional on treatment status, parents' beliefs about students' harvests, measured in the first rainy season, strongly predict subsequent program responses, including midline school visits, ability to correctly name field officers, and attendance at parent-teacher meetings.



Notes: This figure shows that only a small fraction of parents who hold mixed views on students' one-year changes in students' farm management abilities. 1(a) and 2(a) refer to parents who anticipate any improvements and improvements in harvests respectively. (b), (c), (d), and (e) refer to subgroups of (a) who hold mixed views about students. (b) refers to parents who, alongside improvements, anticipate declines in any measure. (c) refers to parents who, alongside improvements, anticipate declines in knowledge scores or harvests. (d) refers to parents who, alongside improvements, has a non-positive net count of anticipated improvements. (e) refers to parents who, alongside improvements, has a non-positive change in the belief index constructed using the first principal component of the seven measures.

Figure 6: Parents who hold mixed views about students' changes in farm management abilities

4.2. Immediate effects of revelation on students' farming plans

Now we analyze the effects of the revelation treatment on students' responses in the immediate follow-up survey. I run the following regressions:

$$Y_{hs} = \alpha + \beta_1 \text{PVideo}_h + \beta_2 \text{Rev}_h + \gamma_s + \epsilon_{hs} \tag{2}$$

where Y_{hs} is the outcome of interest for household h and school-section-gender strata s, where section refers to whether student is in junior high or elementary school; PVideo_h indicates whether household h is randomly assigned to the persuasive video arm; Rev_h indicates whether household

h is randomly assigned to the revelation nudge; γ_s denotes school-section-gender fixed effects; ϵ_{hs} is an error term. β_2 is the coefficient of interest, capturing the intent-to-treat effects of revelation. The reference group is households who are assigned to the basic video treatment and no revelation.

As emphasized in Section 2.4, adolescents in rural Liberia increasingly aspire to shift from assisting on traditional subsistence parcels—which typically intercrop roots or vegetables with rice and rely on fallowing to recover soil fertility—toward managing sales-oriented parcels that provide quicker monetary returns. This shift is driven by both environmental and economic pressures: declining soil fertility under fallow systems makes traditional cultivation more labor-intensive, while rising costs of school fees and other necessities create greater demand for timely cash income (e.g., Scarborough, 2017; Guyer, 1997; Amanor, 1999). However, parents typically accept such shifts only if adolescents demonstrate sufficient farming skills. Therefore, we examine two types of decisions: (i) the type of farms that students plan to look after (Section 4.2.1); and (ii) whom they would seek help from when looking after farms (Section 4.2.2).

4.2.1. Types of farms to look after

Table 2 examines students' plans to look after farms in the first rainy season. Panel A examines extensive-margin choices in students' decisions to look after farms. The revelation nudge does not significantly affect students' overall plans to look after household farms, decisions to sell crops, the number of crops sold, or the choice of farm locations (non-backyard versus upland).

Panel B examines the types of crops that students plan to cultivate on farms they look after. The revelation nudge significantly reduces the number of crops students plan to cultivate by 0.5 crops, representing an 18% decrease relative to the reference group mean of 2.9 crops. This reduction is driven primarily by a significant decline in rice cultivation. The revelation nudge decreases the likelihood of planning to cultivate rice by 3.4 percentage points, a 35% reduction from 9.8%. The effects on other crops are much smaller (6% for roots and vegetables and 9% for tree crops) and imprecise (p=0.33 and 0.16). The significant drop in crop diversity and the concentrated effect on rice cultivation aligns with students' aspiration to shift away from traditional subsistence cultivation.

Panel C further confirms this by analyzing whether students plan to engage in traditional intercropping of roots or vegetables with rice. The revelation nudge significantly reduces the likelihood of intercropping by 4.4 percentage points, or a 47% decrease from 9.4%. This effect remains robust across multiple specifications that control for whether roots and vegetables are cultivated, whether rice is cultivated, the number of roots and vegetables cultivated, and the total number of crops cultivated. Overall, the revelation effect decreases all three major proxy measures for traditional subsistence cultivation: (i) crop diversity; (ii) rice cultivation; and (iii) intercropping.

Table 2: Immediate responses after encouragement (I):
Intent-to-treat (ITT) estimates - Students' plans to look after household farms

	Look after farms (0/1)	Sell crops (0/1)	# crops sold	Non-backyard (0/1)	Upland (0/1)
	(1)	(2)	(3)	(4)	(5)
Panel A: Extensive margins					
$PVideo_h$	0.008	0.014	0.165	0.008	-0.013
	(0.029)	(0.031)	(0.166)	(0.033)	(0.033)
Rev_h	-0.012	-0.026	-0.207	-0.027	0.008
	(0.029)	(0.031)	(0.166)	(0.033)	(0.033)
Reference group mean	0.724	0.669	1.776	0.583	0.445
Reference group SD	0.448	0.471	2.475	0.494	0.498
Observations	1001	1001	1001	1001	1001
Panel B: What crops to look after					
	# crops	Rice (0/1)	Roots/vegetables (0/1)	Tree crops (0/1)	Other crops (0/1)
	(6)	(7)	(8)	(9)	(10)
$PVideo_h$	0.147	-0.017	0.037	0.016	0.015
	(0.245)	(0.017)	(0.034)	(0.017)	(0.020)
Rev_h	-0.518**	-0.034*	-0.033	-0.024	-0.027
	(0.244)	(0.017)	(0.034)	(0.017)	(0.020)
Reference group mean	2.858	0.098	0.528	0.075	0.110
Reference group SD	4.038	0.298	0.500	0.264	0.314
Observations	1001	1001	1001	1001	1001
Panel C: Proxy for traditional mixedcropping					
	Dependent variable: Planting both roots/vegetables and rice (0/1)				
	(11)	(12)	(13)	(14)	(15)
$PVideo_h$	-0.014	-0.018	0.002	-0.016	-0.017
	(0.017)	(0.016)	(0.005)	(0.016)	(0.015)
Rev_h	-0.044***	-0.040**	-0.013***	-0.039**	-0.030*
	(0.017)	(0.016)	(0.005)	(0.016)	(0.015)
Controls	None	Roots/vegetables (0/1)	Rice (0/1)	# roots/vege	# crops
Reference group mean	0.094	0.094	0.094	0.094	0.094
Reference group SD	0.293	0.293	0.293	0.293	0.293
Observations	1001	1001	1001	1001	1001

Notes: This table reports intent-to-treat (ITT) effects on students' plans to look after farms immediately after the encouragement message, where the randomized revelation nudge is included. These plans refer to farming choices in the rainy season of 2021 - the first rainy season under the SBAE program. Panel A examines extensive-margin choices in students' decisions to look after farms. Panel B examines the type of crops that students would cultivate on farms they look after. Panel C examines a proxy for traditional mixedcropping - whether students plan to cultivate both roots or vegetables AND rice on farms they look after. The reference group is students whose households are assigned to the `basic video` and `no revelation nudge`. All regressions include school-grade section-gender fixed effects (1 observation is dropped due to fixed effects). Robust standard errors are presented in parentheses. Significance: *** p<0.01, ** p<0.05, * p<0.10.

4.2.2. Whom to seek help when looking after farms

Table 3 examines students' perceptions of available help when looking after farms and from whom they would seek assistance among the set of family and community elders. Panel A analyzes students' general perceptions of various types of assistance available from family and community elders. The revelation nudge has close to zero and imprecise effects on any of these perceptions, including the perceived availability of help in finding farmland, finding people to work, obtaining money to buy seeds, asking someone to help look after farms, or building relationships with town elders.

Panel B examines whether students specifically plan to seek help from treated parents, that is, parents from whom they receive information about. The revelation nudge significantly increases students' intentions to seek help from their parents in acquiring inputs and managing farms. Students are 6.1 percentage points more likely to plan seeking money from treated parents to buy seeds, representing a 25% increase from 24.8%. Additionally, students are 5 percentage points more likely to plan asking treated parents to help look after farms when they are busy, an 18% increase from 28.3%. Consistent with students seeking a shift in farm management on existing household farms rather than a search for alternative farms, the revelation nudge does not significantly affect students' plans to seek help from parents in finding farmland, finding people to work, or building relationships with elders.

Panel C demonstrates that students seek assistance for farm management shifts exclusively from treated parents rather than from broader family and community networks. This exclusive focus on treated parents is striking, given that students in the reference group are substantially more likely to seek help from other family and community elders than from their treated parents, with ratios ranging from 1.36 to 1.84 times across the five dimensions. If anything, the revelation nudge reduces students' likelihood of seeking assistance from other family and community elders across the five dimensions. While these estimates are imprecise (p=0.07-0.36), they provide strong evidence that the nudge's positive effects on students' assistance-seeking attempts are concentrated on treated parents.

Overall, the evidence from Section 4.2.1 and Section 4.2.2 suggests that the nudge encourages students to shift time away from subsistence parcels and to discuss changes in farm management with treated parents—whom they view as more supportive.

Table 3: Immediate responses after encouragement (II):
Intent-to-treat (ITT) estimates - Perceived availability of help and intended sources of help

	Panel A: Perception of available help from family and community elders				
	Finding farmland (0/1)	Finding people to work (0/1)	Money to buy seeds (0/1)	Help look after farms (0/1)	Building relationship with elders (0/1)
	(1)	(2)	(3)	(4)	(5)
$PVideo_h$	-0.040	0.013	-0.004	0.002	0.011
	(0.027)	(0.027)	(0.030)	(0.027)	(0.026)
Rev_h	-0.027	-0.017	0.011	0.008	-0.005
	(0.027)	(0.027)	(0.030)	(0.027)	(0.026)
Reference group mean	0.827	0.791	0.705	0.756	0.815
Reference group SD	0.379	0.407	0.457	0.430	0.389
Observations	1001	1001	1001	1001	1001
	Panel B: Whether to seek help from treated parents				
	Finding farmland (0/1)	Finding people to work (0/1)	Money to buy seeds (0/1)	Help look after farms (0/1)	Building relationship with elders (0/1)
	(6)	(7)	(8)	(9)	(10)
$PVideo_h$	-0.039	-0.005	0.000	-0.007	-0.005
	(0.032)	(0.032)	(0.030)	(0.031)	(0.032)
Rev_h	0.032	0.014	0.061**	0.059*	0.030
	(0.032)	(0.032)	(0.030)	(0.031)	(0.032)
Reference group mean	0.350	0.323	0.248	0.283	0.346
Reference group SD	0.478	0.468	0.433	0.452	0.477
Observations	1001	1001	1001	1001	1001
	Panel C: Whether to seek help from other family and community elders				
	Finding farmland (0/1)	Finding people to work (0/1)	Money to buy seeds (0/1)	Help look after farms (0/1)	Building relationship with elders (0/1)
	(11)	(12)	(13)	(14)	(15)
$PVideo_h$	-0.002	0.018	-0.004	0.010	0.016
	(0.033)	(0.033)	(0.033)	(0.033)	(0.033)
Rev_h	-0.060*	-0.030	-0.050	-0.050	-0.035
	(0.033)	(0.033)	(0.033)	(0.033)	(0.033)
Reference group mean	0.476	0.469	0.457	0.472	0.469
Reference group SD	0.500	0.500	0.499	0.500	0.500
Observations	1001	1001	1001	1001	1001

Notes: This table reports intent-to-treat (ITT) effects on students' perceptions immediately after the encouragement message, where the randomized revelation nudge is included. These perceptions are about the assistance they can get from family and community elders when they look after a farm outside school. Panel A examines whether students perceive various types of assistance are available (regardless of who provides assistance). Panel B examines whether students indicate that they would seek assistance from their parents (as one of the selected options). Panel C examines whether students indicate that they would seek assistance from family and community elders other than parents. The reference group is students whose households are assigned to the `basic video` and `no revelation nudge`. All regressions include school-grade section-gender fixed effects (1 observation is dropped due to fixed effects). Robust standard errors are presented in parentheses. Significance: *** p<0.01, ** p<0.05, * p<0.10.

4.3. Revelation effects on students' actual farming activities

Now we analyze the effects of the revelation nudge on students' actual farming activities at midline (one year after the nudge) and endline (two years after the nudge). The `main regression specifications` are as below:

$$Y_{hs} = \alpha + \beta_1 \text{PVideo}_h + \beta_2 \text{Rev}_h + \gamma_s + \epsilon_{hs} \tag{3a}$$

$$Y_{hs} = a + b_1 \text{PVideo}_h + b_2 \text{Rev}_h + b_3 \text{PVideo}_h \times \text{Rev}_h + g_s + e_{hs} \tag{3b}$$

where Y_{hs} is the outcome of interest for household h and school-section-gender strata s, where section refers to whether student is in junior high or elementary school; PVideo_h indicates whether household h is randomly assigned to the persuasive video arm; Rev_h indicates whether household h is randomly assigned to the revelation nudge; γ_s and g_s denote school-section-gender fixed effects; ϵ_{hs} and e_{hs} are error terms. The coefficients of interest are β_2 (the pooled ITT revelation effect, across households assigned to the basic and persuasive video arms), b_2 (the ITT revelation effect in the basic video arm), and $b_2 + b_3$ (the ITT revelation effect in the persuasive video arm). The reference group is households who are assigned to the basic video treatment and no revelation.

I allow for differential revelation effects by video arm type because the persuasive video additionally highlights the transformative potential of SBAE on students' farm management skills and induces more optimistic parental beliefs about students' harvests (Section 4.1). As I show in this section, in the absence of the revelation nudge, parents watching the persuasive version also demonstrate a stronger tendency to delegate farm responsibilities than parents watching the basic version. This difference in video effects could in turn shape the strength of the revelation effect if parents' and students' efforts in initiating shifts in farm management are either substitutes or complements.

An important feature of this experiment is that it is embedded in a larger evaluation of SBAE. To understand the potential importance of communication frictions, we want to know whether the revelation effects are both quantitatively important and pivotal for SBAE effects. Accordingly, I also run the following `reference regression specifications` that align with those in Lee

(2025):

$$Y_{hscp} = c_1 + \rho_1 \text{SBAE}_p + \rho_2 \text{SBAE}_p \times \text{FFD}_s + \rho_3 \text{SBAE}_p \times \text{Session}_s + \lambda \text{Faith-based}_s + g_p + \epsilon_{hscp} \quad (4a)$$

$$\begin{aligned} Y_{hscp} = c_2 &+ r_1 \text{SBAE}_p + r_2 \text{SBAE}_p \times \text{FFD}_s + r_3 \text{SBAE}_p \times \text{Session}_s \\ &+ R_1 \text{SBAE}_p \times \text{Session}_s \times \text{PVideo}_h \\ &+ R_2 \text{SBAE}_p \times \text{Session}_s \times \text{Rev}_h \\ &+ R_3 \text{SBAE}_p \times \text{Session}_s \times \text{PVideo}_h \times \text{Rev}_h \\ &+ \lambda \text{Faith-based}_s + g_p + \epsilon_{hscp} \end{aligned} \quad (4b)$$

where Y_{hscp} is the outcome of interest for household h, school s, geographical cluster c, and pair-of-clusters p; SBAE_p indicates whether pair p is randomly assigned to the SBAE arm; Session_s indicates whether school s hosts program promotional sessions; FFD_s indicates whether school s is randomly assigned to host an annual farmer field day; PVideo_h indicates whether household h is randomly assigned to the persuasive video arm; Rev_h indicates whether household h is randomly assigned to the revelation nudge; Faith-based_s indicates whether school s is funded by faith-based institutions, which are excluded from the randomization of video sessions; g_p denotes randomization pair fixed effects; ϵ_{hscp} is an error term. Standard errors are clustered at the geographical cluster level, which is the highest level at which random assignment took place.[14]

The terms highlighted in red in equations (3b) and (4b) identify the same causal effects of the persuasive video, revelation nudge, and their interaction. In large samples, these two specifications would yield identical estimates. However, in finite samples, the main regression specification (3b) provides more precise estimates because it includes finer school-section-gender strata fixed effects (g_s), which account for more granular sources of variation, compared to the coarser pair-level fixed effects (g_p) in the reference specification (4b). Accordingly, I focus primarily on estimates from specification (3b) when interpreting the revelation effects, while also presenting results from specification (4b) to situate these effects within the broader SBAE evaluation. Specifically, I provide two linear combinations of coefficients to highlight the contributions of the nudge: (i) SBAE effects without revelation ($r_1 + r_2 + \frac{1}{2}(r_3) + \frac{1}{2}(r_3 + R_1)$); and (ii) SBAE effects with revelation ($r_1 + r_2 + \frac{1}{2}(r_3 + R_2) + \frac{1}{2}(r_3 + R_1 + R_2 + R_3)$). The effects of the promotional session and the farmer field day are always included because they are policy-relevant: Lee (2025) has demonstrated them to be

[14] To maximize statistical power, Lee (2025) uses what Muralidharan et al. (2025) refer to as the "short model" in factorial design. r_2 and r_3, for example, give the marginal effects of the promotional session and the farmer field day in a sample in which (roughly) half the households are offered the other treatment. We are underpowered to detect interaction effects between the promotional session and the farmer field day. Lee (2025) shows that the point estimates are similar when using the "long model" that controls for an interaction term between the two parental engagement treatments.

pivotal and they will be included in the scaling of SBAE programs in Liberia.

4.3.1. Overview of results

As hinted in Section 4.2, the revelation nudge shifts students' time away from traditional roles as assistants in subsistence cultivation toward managing more sales-oriented parcels within the household. I demonstrate this shift through five analytical steps illustrated in Figure 7: overall farming activities, parcel assignment, parcel management, use of subsistence practices, and use of sales-oriented practices.

The nudge leaves students' aggregate farming participation unchanged—it does not affect their total weekly farming hours, likelihood of looking after a farm, number of household parcels managed, or types of parcels assigned (Section 4.3.2 and (Section 4.3.3)). However, beneath these null effects lie meaningful shifts in how students allocate time across parcels. To detect these shifts, I focus on the designated `"student parcel"`, which is selected using an algorithm that prioritizes parcels with roots or vegetables that students look after and spend the most time on (see Section 3.4.3 and Figure 3). Since overall parcel assignments remain unchanged, differences on the `"student parcel"` capture time reallocation across household parcels.

Students receiving the revelation nudge demonstrate three key changes on the `"student parcel"`. First, they shift away from subsistence practices: the nudge reduces crop diversity by 10% at midline and 21% at endline, lowers rice cultivation by 19% and 31% respectively, decreases intercropping of roots or vegetables with rice by 25% and 26% respectively, and increases continuous cultivation by 25%. Second, they assume greater management responsibility, becoming more likely to look after the parcel (8 percentage points at midline and 13 at endline) and choose which crops to plant (8 and 3 percentage points respectively). Third, they adopt SBAE-promoted methods for quicker income: for cassava, they reduce time to harvest and follow the one-stake-per-hole recommendation that produces larger, more marketable tubers over the traditional multiple-stake practice.

Critically, these changes reflect students' time reallocation across parcels rather than practice changes on given parcels—students are no more likely to make decisions about fallowing, intercropping, or soil management on the `"student parcel"` itself. As documented in Lee (2025), the promotional video alone increases soil management adoption across all household parcels.

4.3.2. Overall farming activities

Table 4 provides evidence that the revelation nudge does not alter students' overall farming activities at either midline or endline. All revelation effects (whether pooled or under either video arm) are small and imprecise across all key measures: whether students farm outside schools, weekly farming hours, weekly hours farming outside school, and the number of household parcels involved. This stands in contrast to a general tendency for the SBAE program, the promotional video, and the

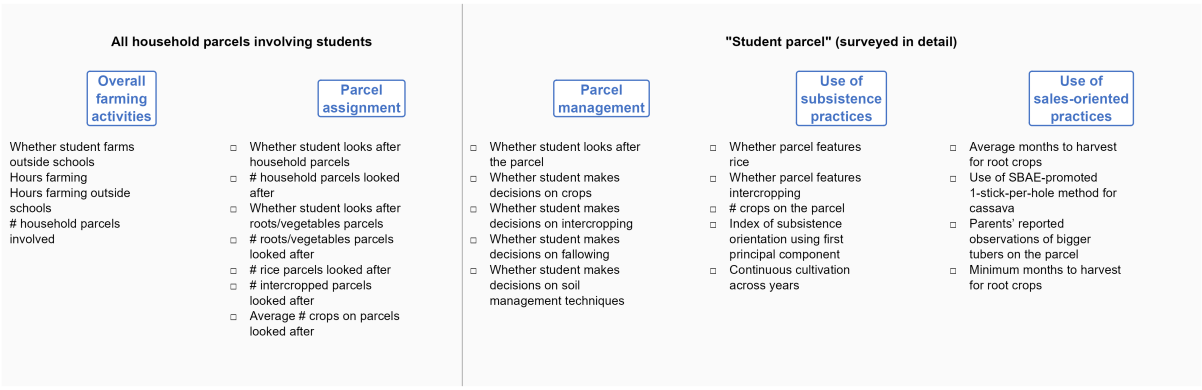


Figure 7: Capturing shifts in students' farming activities in five steps

FFD combined to increase the number of household parcels that students are involved with: by 0.30 parcels, or 18% from 1.68 parcels (Panel A col. (11)), at midline and by 0.22 parcels, or 13% from 1.62 parcels (Panel B col. (11)), at endline.

[Insert Table 4 here]

4.3.3. Parcel assignment

Utilizing more detailed midline data from students, Panel A of Table 5 provides evidence that the revelation nudge does not alter students' role on household parcels. All revelation effects (whether pooled or under either video arm) are small and imprecise across key measures: whether students look after parcels and the number of parcels looked after. This remains true even when restricting attention to parcels featuring roots and vegetables—the focus of SBAE. This stands in contrast to the SBAE program, the promotional video, and the FFD combined, which increase the number of parcels students look after by 0.30 parcels, or 43% from 0.70 parcels (col. (5)), and more specifically, increase the number of roots/vegetables parcels students look after by 0.22 parcels, or 37% from the 0.60 parcels (col. (11)).

Next, following the hint from students' immediate responses Section 4.2.1, Panel B examines whether the revelation nudge makes students look after fewer subsistence parcels, conditional on the number of parcels students look after. I focus on three measures: (i) the number of rice parcels; (ii) the number of intercropped parcels; and (iii) the average number of crops per parcel. When the revelation effect and its interaction with the persuasive video are estimated separately, I find that none of the point estimates are significant at conventional levels (p=0.27-0.4). When pooled together, while all three point estimates (columns (3), (6), and (9)) are negative as expected, they are mostly imprecise: for example, while the number of crops per parcel declines by 10%, it is associated with a p-value of 0.27. While there is indication that students sometimes look

after fewer subsistence parcels, parcel assignment does not appear to be the primary margin of adjustment. In other words, even if students are shifting their work across parcels, it is uncommon that students completely withdraw their involvement with particular parcels. This aligns with how Liberian households typically deploy child labor: as assitants, children are assigned specific tasks across multiple parcels rather than being dedicated to particular parcels.

[Insert Table 5 here]

4.3.4. Management of the designated "student parcel"

Table 6 examines the revelation nudge's effects on students' management of the designated `student parcel`. The reference group means reveal strikingly low levels of student management autonomy at both midline and endline. At midline, only 27% of students look after the `student parcel` and only 24% are involved in choosing crops (Panel A). At endline, these rates improve modestly to 33% and 29% respectively (Panel B).

The revelation nudge significantly increases the fraction of students who shoulder management responsibilities over the `student parcel`. At midline, while the pooled revelation effects are positive but imprecise on the broader measures of whether students look after the parcel (col. 3) and are involved in choosing crops (col. 12), the effects are much stronger when we examine measures that capture more drastic shifts. Specifically, the nudge increases the fraction of students who look after the parcel for the first time by 7.9 percentage points, or a 33% increase from 24%—indicating that the nudge enables students to look after new parcels, consistent with the switch towards sales-oriented parcels that we will discuss in Section 4.3.5 and Section 4.3.6. The nudge also increases the fraction of students who choose crops on their own by 8.2 percentage points, or a 43% increase from 19%. Importantly, this effect is pivotal for SBAE, increasing its effect (col. 8) from 3.9 percentage points (p=0.36) to 8.5 percentage points (p=0.08). This may partly explain why more timely harvests of root crops only increases when the nudge is included (see Panel C of Table 8).

While the nudge's influence on independent crop choice weakens at endline—possibly due to the attrition of upper-grade students who respond more strongly—it continues to significantly increase students' likelihood of looking after the parcel. It increases this likelihood by 12.9 percentage points, or a 31% increase from 41%, and doubles the SBAE effect from 10.1 to 20.5 percentage points. Because the number of parcels that students look after does not change (Table 5), and because the selection of the designated `student parcel` prioritizes parcels students spend most time on, these persistent effects indicate that students reallocate their time toward parcels they look after and away from parcels they work purely as assistants.

Examining the revelation effects by video arm reveals that one arm dominates in each survey round. At midline, the revelation effect under the basic video is substantially larger: 14.2 percentage points

for looking after for the first time (versus 1.6 under persuasive video) and 13 percentage points for choosing crops independently (versus 3.3 under persuasive video). At endline, the revelation effect under the persuasive video dominates: 16.4 percentage points for looking after the parcel (versus 9.6 under basic video). These patterns persist for subsistence Section 4.3.5 and sales-oriented practices Section 4.3.6. Given that the persuasive video induces stronger belief updates about students' harvests and more delegation when the nudge is not included, these results are consistent with parents' and students' communication efforts acting as substitutes in shifting farm management. In Section 5.2 I will present concrete evidence that shifts in farm management are associated with students' requests and concentrated on treated parents' parcels.

[Insert Table 6 here]

4.3.5. Subsistence orientation of the designated "student parcel"

Table 7 examines the revelation nudge's effects on the use of subsistence practices on the designated `student parcel`. As before, I focus on three measures: (i) whether the parcel features rice; (ii) whether the parcel features intercropping; and (iii) the number of crops. I also examine a standardized index using the first principal component of these three measures.

At midline (Panel A), the pooled revelation effect reduces rice cultivation by 5.3 percentage points (p=0.11), or an 18% decrease from 29% (col. 3); intercropping by 6.3 percentage points, or a 24% decrease from 26% (col. 6); the number of crops by 0.53, or a 10% decrease from 5.15 (col. 9); and the overall index by 0.17 standard deviations (col. 12). As in the case of farm management, revelation effects under the basic video dominate: a 30% decrease in rice cultivation, a 39% decrease in intercropping, a 19% decrease in crop diversity; and a 0.29 standard-deviation decrease in the overall index.

At endline (Panel B), the pooled revelation effect reduces rice cultivation by 9.5 percentage points, or a 31% decrease from 31% (col. 3); intercropping by 8.3 percentage points, or a 24% decrease from 34% (col. 6); the number of crops by 0.60, or a 21% decrease from 2.92 (col. 9); and the overall index by 0.26 standard deviations (col. 12). Similar to shifts in farm management, revelation effects under the persuasive video dominate: a 46% decrease in rice cultivation, a 35% decrease in intercropping, a 37% decrease in crop diversity; and a 0.43 standard-deviation decrease in the overall index.

Panel C examines whether the `student parcel` is continuously cultivated across years, as opposed to the traditional subsistence practice of fallowing. Pooling across videos, revelation increases continuous cultivation between the first two years by 6.6 percentage points, or a 9% increase from 73%. Again, this increase is more marked under the basic video arm.

These results at midline and endline complete a circle: they confirm what the immediate follow-up survey suggests—the nudge encourages students to reduce involvement with subsistence

cultivation.[15] Further, given the null effects on overall use of subsistence practices across parcels that students work with, and the selection rules of the `student parcel`, these results indicate that the nudge reallocates students' time away from subsistence parcels. Do they use sales-oriented practices more? We next investigate this issue.

[Insert Table 7 here]

4.3.6. Use of sales-oriented practices on the designated "student parcel"

Table 8 examines the revelation nudge's effects on the use of sales-oriented practices on the designated `student parcel`, utilizing detailed student data at midline. Panel A focuses on months to harvest for root crops, averaging across all root crop harvests and controlling for whether root crops are cultivated. This measure captures market orientation along two distinct margins. First, farmers can adjust their crop portfolio by choosing between short-cycle crops (such as sweet potatoes, typically harvested in 3-6 months) and long-cycle crops (such as cassava, typically harvested in 6-24 months). Second, even within the same crop such as cassava, farmers can exploit the unique storage properties of root crops—which can remain underground for extended periods—to pursue either subsistence strategies, leaving crops in the ground as insurance against food shortages, or sales strategies, harvesting earlier to generate cash income quickly.

The reference group means reveal that student parcels feature relatively long growing cycles oriented toward subsistence along both margins: root crops take an average of 8.28 months to harvest, with only 29% of parcels harvesting root crops within 6 months. For cassava specifically—the primary root crop targeted by SBAE's new techniques on school farms—the average time to harvest is 9.21 months, with only 59% of parcels harvesting cassava within 9 months.

The nudge significantly reduces the time to harvest for root crops, shifting student parcels toward more sales-oriented production. The pooled revelation effect reduces average months to harvest for all root crops by 0.52 months, or 6% from 8.28 months (col. 3), and increases the fraction of parcels harvesting all root crops within 6 months by 8 percentage points, or a 28% increase from 29% (col. 6). As we will see in Panel C, these effects are driven by adjustments along both margins: crop portfolio choices and harvest timing for specific crops. Notably, the effects are equally strong under both video arms. Focusing on cassava specifically—the crop for which SBAE demonstrates improved techniques on school farms—the revelation effect under the basic video further indicates a shift away from subsistence strategies. It reduces average months to harvest cassava by 0.46 months, or 5% from 9.27 months (p=0.06), and increases the fraction of parcels harvesting cassava within 9 months by 10.2 percentage points, or a 19% increase from 53%. This latter increase is particularly meaningful because it represents students moving away from the extended underground storage (12-24 months) that characterizes traditional subsistence farming.

[15] In ongoing data analysis, I show that these effects also hold up in retrospective reports about the first rainy season.

Corroborating these findings, Panel B shows that the nudge also significantly increases the adoption of SBAE-promoted methods that enhance cassava marketability. As a key example, only 1% of reference group student parcels in the promo session without revelation plant one stake per hole (rather than multiple sticks) when cultivating cassava, based on students' midline reports. Both student reports and agronomic trials suggest that this practice produces bigger and more marketable tubers, whereas the traditional multiple-stick method yields smaller, less marketable, but more numerous tubers. Under the basic video, the revelation nudge increases adoption of the one-stick method by 5.6 percentage points (col. 3), representing more than a 5-fold increase. This is one of several lower-density cropping practices that can improve market sales.

To assess whether these practice changes translate into observable improvements, I examine parents' endline reports of harvest outcomes on the `student parcel` over the RCT period. While the SBAE program significantly increases both the likelihood of parents observing bigger tubers and more tubers, the revelation nudge under the basic video increases only the former: by 10.1 percentage points, or 23% from 44% (col. 6), but not the latter (0.9 percentage points, p=0.85, col. 9). This pattern is consistent with the nudge promoting sales-oriented practices that prioritize tuber size and marketability over quantity.

The timing of harvests also matters for students' ability to generate timely income for school fees. Panel C examines the minimum months to harvest across all root crops on the `student parcel`, capturing whether at least one crop can provide quick cash income when needed. The pooled revelation effect reduces minimum months to harvest by 0.38 months, or 5% from 7.91 months (col. 3), and increases the fraction of parcels with at least one root crop harvested within 6 months by 5.9 percentage points, or a 16% increase from 37% (col. 6). Notably, these effects are substantially larger under the persuasive video: the persuasive video reduces minimum months by 0.49 months compared to 0.27 months under the basic video, and increases within-6-month harvesting by 8.8 percentage points compared to 3.1 percentage points under the basic video. Columns 7-12 reveal distinct mechanisms behind these improvements across video arms by examining average months to harvest while controlling for minimum months. Under the persuasive video, once minimum months are controlled for, the revelation effect on average months to harvest becomes small and insignificant at -0.03 months (col. 9). In contrast, the effects remain large under the basic video. This pattern indicates that revelation under the persuasive video primarily works by inducing students to harvest crops more timely, while revelation under the basic video shifts the overall sales orientation of root crop cultivation. This pattern likely explains the more marked increase in students' timely payment of school fees and school attendance under the persuasive video (Section 4.4).

[Insert Table 8 here]

4.4. Revelation effects on welfare

4.4.1. Students' income and savings

Section 4.3 shows that the revelation nudge (i) shifts students' time away from subsistence parcels toward managing sales-oriented parcels; and (ii) increases the adoption of farming practices that enable better sales and more timely income. This section examines whether these behavioral changes translate into improvements in students' welfare. Table 9 examines students' income and savings in 2022—the program's second year—retrospectively reported at endline. The revelation nudge substantially increases students' total income. Column (9) shows an increase of 63%, from 6,070 to 9,893 Liberian dollars (approximately $38 to $62 USD). This gain comes entirely from increased farm sales (col. 3), with non-farm income remaining stable. By contrast, the SBAE program has almost exactly zero and imprecise effects on students' income when the nudge is excluded, as discussed in Lee (2025). Consistent with the patterns observed for farm management, subsistence practices, and sales-oriented practices in 2022, these effects are concentrated in the basic video arm, which shows a 121% increase in income.

As discussed in Lee (2025), savings provide a better measure of students' welfare than income because students both manage their own farms and provide assistance to household members. This is especially true given that total spending remains roughly constant across treatment groups. Column (12) of Table 9 shows that the pooled revelation effect increases annual savings by 62% (p=0.13). As in the case of income, almost all increases happen under the basic video (113%, p=0.07). Placing this in the broader context, column (11) shows that revelation amplifies the SBAE effect from an 84% increase relative to control schools to 101%.

[Insert Table 9 here]

4.4.2. Students' school attendance

As discussed in Section 4.3.6, students' shift toward managing sales-oriented parcels both increases farm sales and generates income at times that align with key spending needs such as school fees. Table 10 shows that revelation substantially increases students' school attendance, alongside dramatic increases in timely payment of school fees.

Panel A examines three attendance measures over the 2021/22 and 2022/23 school years: students' dropout rates in each year, and total days of school attendance across both years, imputed using both dropout and absenteeism data. Consistent with the harvest timing results (Section 4.3.6), revelation increases school attendance most notably under the persuasive video. It reduces dropout rates by 6.5 percentage points in 2021/22 and 9.7 percentage points in 2022/23 (cols. 3 and 6; from baseline rates of 18% and 29%), leading to an increase in total attendance days of 29.7 days, or 12% from 240.58 days.

Panel B examines timely payment of school fees over three periods: the second semester of 2021/22, the first semester of 2022/23, and the second semester of 2022/23. As with attendance, revelation increases timely payment most notably under the persuasive video, raising on-time payment rates by 12.1 percentage points (a 35% increase from 35%), 12.2 percentage points (a 61% increase from 20%), and 11.3 percentage points (a 43% increase from 26%) across the three periods respectively. Alternatively, it increases the number of 2022/23 semesters with on-time payment by 64%. These effects are sufficiently strong that even the pooled revelation effects are sizable: 6.2 percentage points (18% from 35%), 9.3 percentage points (47% from 20%), 11.3 percentage points (43% from 26%), and 0.206 semesters (44% from 0.47) across the same outcomes.

Policy impacts aside, these results reveal important mechanisms. First, revelation effects are consistently concentrated in the persuasive video group for minimum harvest times, timely fee payment, and attendance—but not for average harvest times once minimum times are controlled. This consistent pattern provides strong support for timely farm income as the mechanism for increased school attendance. Second, considering the four possible treatment combinations—(i) basic video without revelation, (ii) persuasive video without revelation, (iii) basic video with revelation, and (iv) persuasive video with revelation—only (iv) significantly increases school attendance and timely fee payment at conventional levels. Given that the persuasive video strengthens parental beliefs about students' farming abilities, this pattern suggests that both parental support for schooling and students' efforts are necessary for improving school attendance—neither alone is sufficient.

[Insert Table 10 here]

4.4.3. SBAE's cost-effectiveness

The results in Section 4.4.1 and Section 4.4.2 dramatically reshape SBAE's cost-effectiveness profile. Lee (2025) demonstrates that SBAE—when augmented with promotional sessions and the annual farmer field day—outperforms agricultural training outside schools in boosting youth savings and proves more than twice as cost-effective as conditional cash transfers in Malawi (Baird et al., 2011) in increasing school attendance.[16] The revelation nudge amplifies these gains substantially, doubling the impact per dollar on attendance.

Figure 8 quantifies this improvement. Without the nudge, SBAE generates $72.87 in additional savings and 29.43 extra attendance days per $100 spent. With the nudge, these figures surge to $87.23 (20% increase) and 59.51 days (102% increase). The marginal effect of the revelation nudge alone—30.09 additional attendance days—exceeds the impact per $100 of conditional cash transfers in Malawi (17.06 to 24.98 days). Alternatively put, the revelation nudge is the key reason SBAE is twice as cost-effective as conditional cash transfers.

[16] The estimates in Lee (2025) are averages over treatment effects across both the revelation and no-revelation conditions in the promotional session treatment arm.

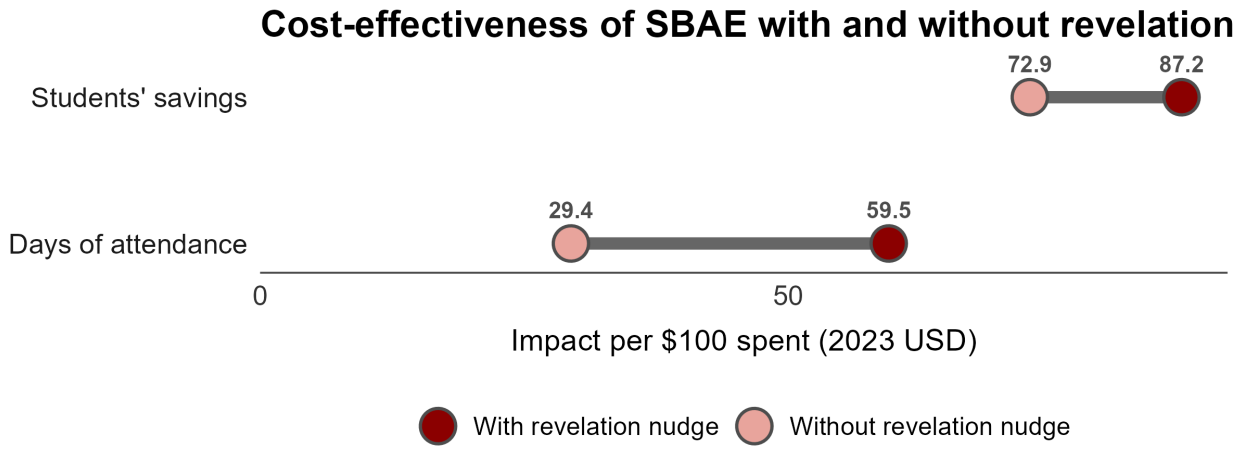


Figure 8: Cost-effectiveness of the SBAE program with and without the revelation nudge

4.4.4. Other considerations about students' and households' welfare

While the revelation nudge substantially increases students' income, savings, school attendance, and timely payment of school fees, other dimensions of student welfare remain largely unchanged. In the Online Appendix, I show that consumption of breakfasts and lunches is unaffected by the nudge at both midline and endline, nor is the rate of absenteeism (defined as days absent over the previous five school days) at either survey round. As shown in Table 4, the revelation nudge does not alter students' weekly hours spent farming at midline and endline. The Online Appendix further confirms that the nudge does not change students' weekly hours spent studying. The only exception is that the nudge increases weekly hours on household chores by 1.3 hours at midline (p=0.07; a 15% increase) under the basic video but not the persuasive video—perhaps reflecting students' compensation for parents' loss of subsistence labor. This effect disappears by endline.

What about changes to household-level welfare? The shifts that the revelation nudge induces involve reassignment of students across household parcels rather than changes in the use of promoted soil management techniques on specific parcels, so it does not affect technology adoption. In the Online Appendix, I confirm this by showing that the nudge has small and imprecise effects on the use of promoted techniques on both the `student parcel` and the `parent parcel`. Nor does it alter the subsistence orientation on the `parent parcel`. Future research should examine whether students' increased control of sales-oriented parcels, combined with the introduction of SBAE-promoted sales-oriented methods, can trigger adoption of these methods by parents over the longer term. Longer-term follow-ups should also track whether parents increase spending on hired labor to substitute for lost subsistence labor, and whether the revelation nudge has implications for household harmony.

4.5. Heterogeneous treatment effects

Section 4.4 shows that the belief revelation nudge substantially improves students' welfare on average. However, not all students necessarily benefit, particularly since the nudge's success depends on students being competent at managing farms. This competence is not guaranteed, as students who self-select into responding to the nudge may be overconfident about their farming abilities. An additional concern is that boys and girls, as well as students of different ages, may be differentially suited to farm management.

Two contextual features mitigate these concerns. First, as I show in Section 5.2, students receiving the nudge typically respond by making requests to parents rather than acting alone. The nudge thus leverages both students' initiative and parents' experienced judgment. Second, the nudge is paired with an effective training program. Lee (2025) demonstrates that SBAE successfully trains teachers and students in farm management skills and the use of agricultural innovations. Importantly, the SBAE-promoted practices that the nudge increases adoption of are straightforward: planting one cassava stake per hole rather than multiple stakes, and harvesting crops more timely.

I now provide empirical evidence by examining treatment effect heterogeneity by students' gender and grade. Stratified sampling on these dimensions enables unbiased estimation of heterogeneous (intent-to-treat) effects. Specifically, I run the following regressions:

$$\begin{aligned} Y_{hs} = a &+ b_1 \text{PVideo}_h + c_1 \text{PVideo}_h \times \text{Female}_h \\ &+ b_2 \text{Rev}_h + c_2 \text{Rev}_h \times \text{Female}_h \\ &+ b_3 \text{PVideo}_h \times \text{Rev}_h + c_3 \text{PVideo}_h \times \text{Rev}_h \times \text{Female}_h \\ &+ g_s + e_{hs} \end{aligned} \tag{5a}$$

$$\begin{aligned} Y_{hs} = a &+ b_1 \text{PVideo}_h + c_1 \text{PVideo}_h \times \text{JHS}_h \\ &+ b_2 \text{Rev}_h + c_2 \text{Rev}_h \times \text{JHS}_h \\ &+ b_3 \text{PVideo}_h \times \text{Rev}_h + c_3 \text{PVideo}_h \times \text{Rev}_h \times \text{JHS}_h \\ &+ g_s + e_{hs} \end{aligned} \tag{5b}$$

where Y_{hs} is the outcome of interest for household h and school-section-gender strata s, where section refers to whether student is in junior high or elementary school; PVideo_h indicates whether household h is randomly assigned to the persuasive video arm; Rev_h indicates whether household h is randomly assigned to the revelation nudge; Female_h and JHS_h are binary indicators for female students and junior high students respectively; g_s denotes school-section-gender fixed effects; e_{hs} is the error term.

For equation (5a), the coefficients of interest are: (i) b_2 (the ITT revelation effect under the basic video for male students); (ii) $b_2 + c_2$ (the ITT revelation effect under the basic video for female

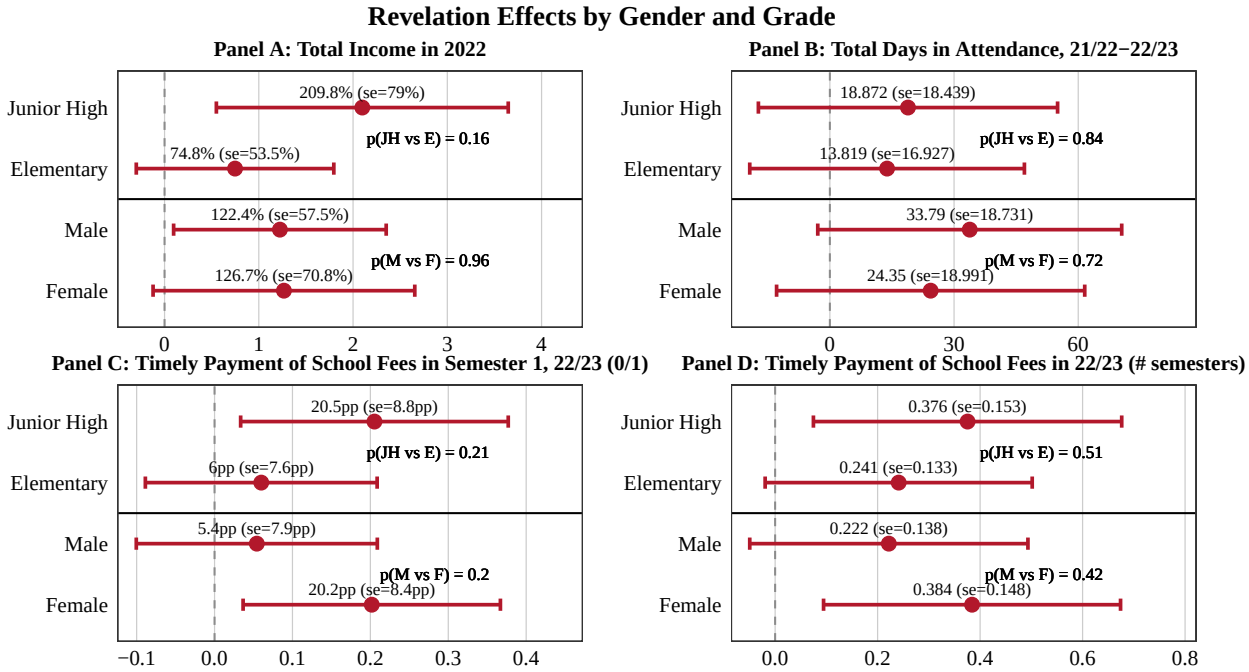
students); (iii) $b_2 + b_3$ (the ITT revelation effect under the persuasive video for male students); and (iv) $b_2 + b_3 + c_2 + c_3$ (the ITT revelation effect under the persuasive video for female students). For equation (5b), analogous coefficients capture effects for elementary versus junior high school students. The reference group is male students in equation (5a) assigned to the basic video without revelation; and elementary school students in equation (5b) assigned to the basic video without revelation.

Figure 9 reports the results, focusing on where the revelation effects are concentrated along the causal chain: under the basic video for farm management and income; and under the persuasive video for the timely harvests of root crops, school attendance, and timely payment of school fees. In the Online Appendix, I show that revelation effects are small and imprecise in the video arm where effects are weaker, and that pooled revelation effects (combining both video arms) preserve the same pattern of differences between subgroups.

Figure 9 yields three key observations. First, all point estimates are positive, and all subgroup comparisons are statistically indistinguishable at conventional levels (the smallest p-value is 0.16). This indicates that all subgroups benefit from the revelation nudge.

Second, both males and females benefit substantially, with only modest differences in effect sizes. Income effects are nearly identical (122% vs. 127%; p=0.96), as are attendance gains (34 vs. 24 days; p=0.72). Females show larger increases in timely fee payment—20 percentage points in the first semester and 0.38 additional semesters with on-time payment, compared to 5 percentage points and 0.22 semesters for males—but these differences are imprecise (p=0.20 and p=0.42).

Third, there is suggestive evidence that junior high students benefit more than elementary students. This pattern is most pronounced for income (210% vs. 75%; p-diff=0.16) and timely payment in the first semester of 2022/23 (21 vs. 6 percentage points; p-diff=0.21).



Note: This figure presents estimates of heterogeneous intent−to−treat effects by subgroups. Effects are estimated based on regressions of outcomes on treatment dummies fully interacted with each other and with subgroups, in the sample of 1002 households that are assigned to watch a promotional video. This figure focuses on revelation effects under the treatment arm where the strongest revelation effects are found. Therefore, for total income, revelation effects under the basic video are presented; for the other outcomes, revelation effects under the persuasive video are presented. Error bars represent 95% confidence intervals.

Figure 9: Heterogeneous treatment effects by students' gender and grade

5. Discussion of mechanisms

Section 4 shows that the revelation nudge shifts students' time from working as assistants on subsistence parcels to managing sales-oriented parcels, and adopt practices enabling timely income from school-tested crops. Sections 4.4 and 4.5 show that the nudge improves substantial welfare—income, savings, school attendance, and timely payment of school fees—across students of different needs and capabilities. The nudge's sizable effects on students' welfare are striking considering the minimal costs it involves—as a five-minute component of *additional* encouragement messages within surveys. The results thus suggest that signaling to students their parents' openness to changes in resource control can be a highly cost-effective complement to effective school-based programs. This section examines various explanations for the outsized effects of the belief revelation nudge.

5.1. Preferred explanation

As discussed in the Introduction, the design of the belief revelation nudge is motivated by concerns about communication frictions—even with frequent interactions and ample opportunities for

indirect learning, students fail to learn parents' openness to changes in farm management. I now present evidence that experimental information, delivered through the revelation nudge, alters farm management specifically between students and the parent whose beliefs are revealed (randomly chosen by gender, see Section 3.2), but not with other family elders (including the other parent). In other words, the nudge's effects on farming activities are confined to the particular parent–student dyad involved in the intervention. This indicates that students' responses are driven by information about that particular parent, highlighting the importance of communication frictions in intergenerational resource allocation—contrary to standard household models that assume students would fully internalize parents' preferences and beliefs regardless of the nudge.

The first set of evidence examines students' plans about which household elders to seek farming assistance from. Table 3 in Section 4.2.2 shows that the nudge does not change whether students perceive assistance as available (e.g., for obtaining farm inputs), but rather changes whom they intend to approach—shifting requests specifically toward the targeted parent.

The second set of evidence examines students' sources of parcels for which they acquire management authority. The belief revelation nudge does not change whether students plan to look after farms (Table 2 in Section 4.2.1), but rather whom they request parcels from. Using detailed midline data, Panel A of Table 11 shows that increases in students' decision authority on the designated `student parcel` are associated with decreases in elders' decision authority, and these decreases are specific to the treated parents. Under the basic video (where midline effects are concentrated), revelation reduces the likelihood of the `student parcel` being looked after by treated parents by 12.8 percentage points, or a 32% decrease from 40%, and reduces their involvement in choosing crops by 18.4 percentage points, or a 30% decrease from 62%. By contrast, these effects are close to zero and imprecise for other elders. Consistent with null results on students' decision authority, changes in parents' involvement in choose soil management techniques are also absent. To strengthen the evidence that students make actual requests to treated parents, Panel B examines endline retrospective reports from parents on requests of changes in farm management from students over the past two years. Revelation increases students' requests to look after farms by 8.3 percentage points (from 60%). Consistent with shifts in farm management at endline, the effects are stronger under the persuasive video arm, though this difference is imprecise (p=0.66).

[Insert Table 11 here]

The third piece of evidence examines where changes in farming practices occur by identifying which household members originally managed the parcels. Specifically, I examine changes in the use of subsistence practices on the designated `student parcel`, categorized by who looks after it: (i) students; (ii) treated parents; and (iii) other elders. A decline in subsistence orientation on parcels of type (i) would indicate that changes in practices happen *within* student-managed parcels without necessarily involving interactions with elders. A decline on types (ii) or (iii), however,

would indicate that changes in practices are associated with shifts in management from elders to students.

Panel A of Table 12 shows that all midline changes fall on parcels originally looked after by treated parents. Under the basic video, revelation reduces rice cultivation by 6.3 percentage points, or 45% from 14%; intercropping by 7.4 percentage points, or 57% from 13%; and crop diversity by 0.91 crops, or 41% from 2.22 crops. Overall, revelation under the basic video reduces the subsistence index on treated-parent-managed parcels by 0.26 standard deviations. By contrast, changes on student-managed and other-elder-managed parcels are consistently close to zero and imprecise. For instance, the reduction in the subsistence index on treated-parent-managed parcels is over four times larger than that on other-elder-managed parcels.

Panel B shows similar patterns at endline: under the persuasive video, revelation reduces crop diversity on treated-parent-managed parcels by 0.62 crops, or 84% from 0.74 crops, and this reduction is about 12 times larger than that on other-elder-managed parcels.

[Insert Table 12 here]

Overall, these three pieces of evidence establish a strong causal chain pinpointing intergenerational interactions unique to the targeted parent-student dyad: whom students seek assistance from for farm management, whom they request parcels from, and whose subsistence parcels they shift away from. Since the gender of the treated parent is randomized, this pattern provides key evidence that changes in students' farming activities are driven by information about the specific treated parent. This sensitivity to experimental information and the subsequent impact on welfare reveals substantial communication frictions, contradicting household models assuming that students can learn the information regardless of the nudge.

5.2. Alternative explanations

Besides communication frictions, there could be alternative explanation for the belief revelation nudge's effects. First, the nudge, by highlighting students' progress under SBAE, might signal program benefits to students and stimulate their participation in sales-oriented parcels. While I cannot rule out this effect, the dyad-specific interactions in Section 5.1 suggest that communication barriers play a central role in the nudge's impact. Additionally, In the Online Appendix, I show that the nudge has minimal effects on students' program participation or agricultural knowledge acquisition.

Second, the nudge might reflect social desirability biases rather than genuine parental beliefs. For instance, parents watching the videos might overstate specific figures, such as the number of tubers students can harvest, to align with perceived expectations. However, the key is not the accuracy of these figures but the nudge's ability to signal parents' openness to delegating tasks.

I therefore present evidence that sampled parents, who are exposed to the SBAE program and the promotional video, become genuinely more open to delegating farm tasks. Panel A of Table 13 provides the first piece of evidence, showing that sampled parents are not only optimistic about students' cassava harvests immediately after watching the video, but remain persistently so throughout the experiment. Relative to SBAE school parents who do not watch the video—who may also face social desirability bias given their exposure to the SBAE program—parents who watch the video believe that students can harvest 1.15 more cassava tubers from 4 plants at midline (a 5% increase from 21.54 tubers) and 0.851 more tubers at endline ($p = 0.17$; a 4% increase from 20.48 tubers).

Next, drawing upon the broader comparison of SBAE and control schools at endline, Panel A further indicates that exposure to the SBAE program overall makes parents more optimistic about students' harvests. They believe that students can harvest 1.45 more tubers from 4 plants (a 7% increase from 20.48 tubers). More importantly, the fraction of parents who believe that students can outperform typical adults in the community increases by 7.3 percentage points (a 21% increase from 35%), and the corresponding fraction who believe that better practices enable students to outperform adults increases by 8.6 percentage points (a 28% increase from 31%). Since the video-treated parents are a subset of the SBAE parents, this evidence further supports that they become more open to delegating farm tasks.

Next, Panel B demonstrates that measured parental beliefs are not merely noise. Specifically, I examine whether variation in first-season beliefs (measured after video viewing) among the 1,002 video-treated households predicts actual responses to the SBAE program, conditional on whether they watch the persuasive video and whether the household is assigned to the revelation nudge. The results indicate that parents' beliefs about the difference between students' and adults' harvests strongly predict their midline responses. A 1-standard-deviation improvement in beliefs is associated with increases in visits to school farms by 14%, with 22% additional visits in the first year; observation of school farm harvest by 22%; work on school farms by 31%; and meetings with the responsible teacher by 12%. To guard against social desirability bias, I also asked parents to name the SBAE field officers in their regions—a 1-standard-deviation improvement in beliefs is associated with a 2.7 percentage point increase in this measure (from 11%). Lastly, more optimistic parents not only pay more attention to the school farms, but also attend more parent-teacher meetings (by 9%). Overall, Panel B suggests that parents' measured beliefs have predictive validity over a wide range of responses to the SBAE program, which could in turn strengthen their openness to delegation.

In addition, consistent with increased openness to farm management changes, parents who attend the video session are more likely to let students look after the designated `student parcel`, even without the revelation nudge (Lee, 2025). Notably, these effects are consistently stronger under the persuasive video, which adds parent testimonials to the basic video content. This pattern aligns with the persuasive video's stronger effect on parental beliefs about student harvests: by reducing

uncertainty about students' farming abilities through testimonials, the persuasive video increases parents' willingness to delegate farm tasks.

Lastly, survey responses from parents and students immediately following the video and revelation nudge treatments indicate no significant shifts in perceived social norms regarding adolescent farm management autonomy. Specifically, as shown in the Online Appendix, there are small and imprecise changes in subjects' estimates of the number of adults (out of 10) in their community who would support adolescents making independent farming decisions. This provides further evidence that the revelation nudge is limited to the targeted parent-student dyad.

[Insert Table 13 here]

6. A substantive theory of communication frictions

6.1. Overview

Lee (2025) establishes that parent engagement with program information is essential for program effectiveness. SBAE alone fails to increase household adoption of soil management techniques, but the program augmented with a promotional video and an annual farmer field day proves highly effective.

Results in Section 4, together with evidence on mechanisms in Section 5, establish a strong case that parental engagement is insufficient for SBAE to tackle goals intertwined with agricultural extension: the economic empowerment of students and school attendance. The revelation nudge dramatically improves these outcomes. This outsized effect of a 5-minute survey component is perhaps puzzling, since parental engagement has already made parents respond to the program by substantially increasing visits to schools and adoption of promoted soil management techniques.[17] The revelation nudge does not alter these measures.] So even without the nudge, students could have learnt its content through direct communication and indirect inference from interactions with parents over the course of two years.

To rationalize these communication frictions, this section develops a model of household production where (i) there is an information asymmetry between students and parents regarding parents' openness to changes in farm management and (ii) it is costly to initiate collective decision making under new technologies. While the general goal is to capture information asymmetries in subjective information, I focus specifically on parents' beliefs about the value of delegation, abstracting from preference-based asymmetries.

[17] Lee (2025) shows that without parental engagement, the SBAE alone does not increase parents' visits to schools and adoption of promoted techniques relative to control schools. SBAE augmented with the promotional video increases parents' visits to schools by 64% and adoption of techniques increases by 0.3 and 0.46 standard deviations on the `"student parcel"` and the `"parent parcel"` respectively.

This model serves three purposes. First, the model explains why students fail to learn parents' openness to changes in farm management, even after observing parents' technology adoption. Second, the model provides testable insights for the design of agricultural extension when influential household members—in this case, parents—might overlook opportunities for joint decision making with others. Third, the model provides a framework for comparing the revelation nudge with alternative empowerment approaches that aim to encourage joint decision making within households.

I begin by rationalizing how the revelation nudge activates two causal mechanismse critical for SBAE's impact. First, it enables students to request transitions from subsistence parcels to sales-oriented parcels, increasing income and sales—what I term the *parcel delegation* channel. Second, it increases the timeliness of root crop harvests, generating more timely school fee payments and improved attendance—what I term the *timely harvest* channel. Focusing on the sample of 1002 households that receive both the SBAE program and the promotional video, I match the following empirical patterns:

(i) The persuasive video is more effective than the basic video on the *parcel delegation* channel;
(ii) The revelation nudge is highly effective on the *parcel delegation* channel, with stronger effects under the basic video arm;
(iii) The *timely harvest* channel is activated only with both the persuasive video and the revelation nudge.

After matching these patterns, I then focus on the model's core insights: (i) students might fail to learn from parents' technology adoption if parents' capacity to implement promoted techniques is high; (ii) since household members often overlook opportunities for joint decision-making under new technologies, technology selection, participant targeting, and informational nudges should be cohesively designed to encourage joint decision making. I test (i) by examining how differential control rights over sales-oriented parcels affect the nudge's effects; and (ii) by utilizing school-level randomization in whether an annual farmer field day is hosted to train parents directly.

6.2. Model Setup

Players and Timing

Consider a household with two players: a parent (P) and a student (S). The student is in a school that receives the SBAE program. The interaction unfolds in the following stages:

Stage 0 (Nature and Signals):

- Nature draws true program quality: $(\theta_R, \theta_{\text{SBAE}})$, where θ_R represents households' returns to the promoted soil management techniques and θ_{SBAE} represents the value to households of

students' learning under SBAE, which includes starting home farming projects and sustained school attendance

- Parent has prior beliefs: $(\mu_R^{\text{prior}}, \mu_{\text{SBAE}}^{\text{prior}}) \sim N(\mu_0, \Sigma_0)$
- Parent observes video signal $s_v = (s_R, s_{\text{SBAE}})$ where $s_v \sim N((\theta_R, \theta_{\text{SBAE}}), \Sigma_v)$
- Signal precision Σ_v depends on video type $v \in \{\text{Basic}, \text{Persuasive}\}$
- Parent forms posterior beliefs: $(\mu_R^{\text{post}}, \mu_{\text{SBAE}}^{\text{post}})$ via Bayesian updating
- If assigned revelation nudge, a noisy signal $\tilde{s} = (\tilde{\mu}_R, \tilde{\mu}_{\text{SBAE}})$ about parent's posterior is transmitted to student

Stage 1 (Parent's Communication Decision): Parent chooses $a_P \in \{C, U, N\}$ where:

- C: Initiate collective decision-making ("communicate")
- U: Unilateral adoption without delegation
- N: Do nothing

Stage 2 (Student's Communication Decision): Student observes a_P and signal $\tilde{s}$ (if nudge assigned), chooses $a_S \in \{C, N\}$. This stage is skipped if $a_P = \{C\}$.

Stage 3 (Household Decisions):

- If collective decision reached (either player chose C):
  - **Stage 3a**: Parent and student fully share their information, preferences, and needs
  - **Stage 3b**: Parent decides whether to accept delegation
  - **Stage 3c**: If delegation accepted, student automatically proposes timely harvest
  - **Stage 3d**: Parent decides whether to accept timely harvest
- Otherwise: Unilateral adoption (if $a_P = U$) or status quo

Primitives and Parameters

Prior Beliefs and True Quality The parent's prior beliefs about program quality are:

$$(\mu_R^{\text{prior}}, \mu_{\text{SBAE}}^{\text{prior}}) \sim N(\mu_0, \Sigma_0) \tag{1}$$

where:

- $\mu_R \in [0, \bar{\mu}_R]$: Expected returns to promoted soil management techniques
- $\mu_{\text{SBAE}} \in [0, \bar{\mu}_{\text{SBAE}}]$: Value of students' learning under SBAE
- $\Sigma_0 = \begin{pmatrix} \sigma^2_{0,R} & 0 \\ 0 & \sigma^2_{0,\text{SBAE}} \end{pmatrix}$

For analytical tractability, we assume the two dimensions are independent in the prior. These priors are commonly known.

Video Treatment as Signal Videos provide noisy signals about true program quality:

$$s_v = (s_R, s_{\text{SBAE}}) \sim N((\theta_R, \theta_{\text{SBAE}}), \Sigma_v) \tag{2}$$

Signal precision matrices:

- **Basic video**: $\Sigma_v^{\text{Basic}} = \begin{pmatrix} \sigma^2_R & 0 \\ 0 & \sigma^2_{\text{SBAE,Basic}} \end{pmatrix}$
- **Persuasive video**: $\Sigma_v^{\text{Persuasive}} = \begin{pmatrix} \sigma^2_R & 0 \\ 0 & \sigma^2_{\text{SBAE,Persuasive}} \end{pmatrix}$

Key assumption: $\sigma^2_{\text{SBAE,Persuasive}} \ll \sigma^2_{\text{SBAE,Basic}}$ but σ^2_R is the same for both videos.

This assumption corresponds to the fact that the persuasive video targets θ_{SBAE} through testimonials on SBAE's transformative impact on students' farming skills, livelihoods, and future careers. The persuasive video also shows stronger impact on parents' beliefs about students' harvests and visits to school farms.

Bayesian Updating The parent's posterior beliefs after observing video signal s_v:

$$\begin{pmatrix} \mu_R^{\text{post}} \\ \mu_{\text{SBAE}}^{\text{post}} \end{pmatrix} = (\Sigma_0^{-1} + \Sigma_v^{-1})^{-1}(\Sigma_0^{-1}\mu_0 + \Sigma_v^{-1}s_v) \tag{3}$$

For independent dimensions, this simplifies to precision-weighted averages:

$$\mu_R^{\text{post}} = \frac{\tau_{0,R} \cdot \mu_R^{\text{prior}} + \tau_{v,R} \cdot s_R}{\tau_{0,R} + \tau_{v,R}} \tag{4}$$

$$\mu_{\text{SBAE}}^{\text{post}} = \frac{\tau_{0,\text{SBAE}} \cdot \mu_{\text{SBAE}}^{\text{prior}} + \tau_{v,\text{SBAE}} \cdot s_{\text{SBAE}}}{\tau_{0,\text{SBAE}} + \tau_{v,\text{SBAE}}} \tag{5}$$

where $\tau = 1/\sigma^2$ denotes precision.

Signal Weight: Define $\eta_v = \frac{\tau_{v,\text{SBAE}}}{\tau_{0,\text{SBAE}}+\tau_{v,\text{SBAE}}}$ as the weight on video signal. Then:

$$\mu^{\text{post}}_{\text{SBAE}} = (1-\eta_v)\mu^{\text{prior}}_{\text{SBAE}} + \eta_v \cdot s_{\text{SBAE}} \tag{6}$$

Key property: $\eta_{\text{Persuasive}} > \eta_{\text{Basic}}$ (persuasive video gets more weight).

Revelation Nudge The nudge generates noisy signal about parent's posterior:

$$\tilde{s} = (\tilde{\mu}_R, \tilde{\mu}_{\text{SBAE}}) \sim N((\mu^{\text{post}}_R, \mu^{\text{post}}_{\text{SBAE}}), \Sigma_{\text{nudge}}) \tag{7}$$

where $\Sigma_{\text{nudge}} = \begin{pmatrix} \sigma^2_{\text{nudge},R} & 0 \\ 0 & \sigma^2_{\text{nudge,SBAE}} \end{pmatrix}$ with small variances.

Payoff Parameters Parent's delegation decision is affected by two parameters. First, we use $\kappa \in (0,1)$ to capture parent's capability to implement promoted techniques independently, relative to delegating to students. This reflects parents' knowledge of promoted techniques and the ease of learning such techniques. Second, we use $\lambda > 0$ to capture the value of student's foregone subsistence labor when delegation happens.

Parent's timely harvest decision is affected by two parameters. I assume that only a fraction $\phi \in (0,1)$ of students' learning value under SBAE is realized under standard timing, relative to timely harvest, since the latter can enhance school attendance. Meanwhile, parent has to weight the benefit of timely harvest against a cost of $\delta > 0$, capturing the cognitive or coordination cost in reorganizing farms.

Finally, two additional parameters affect parent's communication decision: the communication cost $c > 0$ and an exogenous Pareto weight $\beta \in (0,1)$ that specifies parent's share of the joint surplus. Note that the communication cost is common to parent and student, and only needs to be incurred by one party.

Equilibrium Concept

We solve for a Perfect Bayesian Equilibrium (PBE) of this sequential game with incomplete information. An equilibrium consists of strategies and beliefs such that: (i) the parent chooses $a_P \in \{C, U, N\}$ to maximize expected utility given posterior beliefs $(\mu^{\text{post}}_R, \mu^{\text{post}}_{\text{SBAE}})$ and anticipating the student's best response; (ii) the student chooses $a_S \in \{C, N\}$ to maximize expected utility given second-order beliefs $\bar{\mu}^S_{\text{SBAE}}(a_P, \tilde{s})$ about the parent's valuation; (iii) subsequent decisions (delegation acceptance and timely harvest acceptance) are sequentially rational; and (iv) the student's beliefs are consistent with the parent's equilibrium strategy, derived via Bayes' rule on the equilibrium path. Off the equilibrium path, we assume the student maintains prior beliefs. While multiple PBE

may exist due to off-equilibrium belief specifications, our main predictions regarding treatment effects are robust across equilibria. We focus on the equilibrium supported by the refinement that students maintain prior beliefs off-path. The key insight of our analysis is that the informativeness of the parent's action a_P about their posterior beliefs depends critically on the slope of the parent's communication threshold with respect to μ_R, which in turn determines how much the student can learn from observing parental choices.

6.3. Deriving Parent's Decision Thresholds

We begin solving the model by deriving the parent's decision thresholds for three key decisions: (i) communication; (ii) delegation acceptance; and (iii) timely harvest acceptance.

The first step is to define the joint surplus from delegation. Note that once parent and student enter collective decision making, they share all information, so both parties share the same $\{\mu_R, \mu_{SBAE}\}$. Here we assume, for the purpose of the current analysis, that both share parent's posteriors.[18] The joint surplus from delegation (relative to unilateral adoption by parent) is:

$$W^{delegate}(\mu_R, \mu_{\text{SBAE}}, \kappa) = \mu_R(1-\kappa) + \phi \cdot \mu_{\text{SBAE}} - \lambda$$

Given this joint surplus from delegation, we can then derive parent's communication threshold. Parent communicates if $\beta \cdot W^{delegate} > c$:

$$\mu_{\text{SBAE}} > \tilde{\mu}_{\text{SBAE}}^{communicate}(\mu_R, \kappa, \beta) \equiv \frac{1}{\phi}\left[\frac{c}{\beta} + \lambda - \mu_R(1-\kappa)\right] \tag{8}$$

Next we define parent's acceptance thresholds once they are in collective decisions. Parent accepts delegation if $W^{delegate} > 0$:

$$\mu_{\text{SBAE}} > \mu_{\text{SBAE}}^{\text{delegate}}(\mu_R, \kappa) \equiv \frac{\lambda - \mu_R(1-\kappa)}{\phi} \tag{9}$$

Note that both parent's communication and delegation acceptance thresholds are decreasing in μ_R and increasing in κ.

Lastly, when delegation is accepted, student automatically proposes timely harvest. Parent accepts if incremental SBAE benefits exceed coordination cost:

$$(1-\phi) \cdot \mu_{\text{SBAE}} > \delta$$

[18] Alternatively, we can assume that starting from the common prior beliefs, student receives a signal about SBAE that is equivalent to parent receiving the signal from the promotional video.

This defines the timely harvest acceptance threshold, which is a constant:

$$\mu_{\text{SBAE}}^{TH} = \frac{\delta}{1-\phi} \tag{10}$$

For the analysis that follows, we require this threshold ordering to match the empirical patterns: $\mu_{\text{SBAE}}^{\text{delegate}}(\mu_R, \kappa) < \mu_{\text{SBAE}}^{TH} < \tilde{\mu}_{\text{SBAE}}^{communicate}(\mu_R, \kappa, \beta)$. While $\mu_{\text{SBAE}}^{\text{delegate}}(\mu_R, \kappa) < \tilde{\mu}_{\text{SBAE}}^{communicate}(\mu_R, \kappa, \beta)$ is always true, we need parameter restrictions to ensure that the timely harvest threshold is between the other two:

Assumption 1 (Threshold Ordering) *We assume:*

$$(1-\phi)\lambda < \phi\delta \tag{11}$$

$$\frac{\phi\delta}{1-\phi} < \frac{c}{\beta} + \lambda - \bar{\mu}_R(1-\kappa) \tag{12}$$

Condition (11) ensures that the foregone subsistence labor value λ is small relative to the timely harvest coordination cost δ (adjusted by the SBAE realization factor ϕ) that delegation is less demanding than timely harvest. Condition (12) ensures that the communication cost c is large relative to the timely harvest coordination cost that initiating communication is more demanding than accepting timely harvest.

6.4. Why do students fail to learn parents' beliefs?

After deriving parent's decision thresholds, we are now ready to answer the first puzzle of this paper: why do students fail to learn parents' beliefs about μ_{SBAE}, even after observing parents choosing $a_P = U$?

The Student's Inference Problem

In the absence of the nudge, the student only observes the parent's action a_P and knows the prior distribution F. The student needs to form a second-order belief about the parent's valuation of student's learning under SBAE:

$$\bar{\mu}_{\text{SBAE}}^{S}(a_P) = \mathbb{E}[\mu_{\text{SBAE}}^{\text{post}} \mid a_P]$$

To understand this inference, consider what observing $a_P = U$ (unilateral adoption) reveals. The student knows that the parent chose not to communicate, which means:

$$\mu_{\text{SBAE}}^{\text{post}} < \tilde{\mu}_{\text{SBAE}}^{communicate}(\mu_R^{\text{post}}, \kappa, \beta)$$

Here lies the key insight: the communication threshold depends on the parent's beliefs about returns to promoted techniques μ_R^{post}. To infer where the parent's SBAE valuation lies, the student must form beliefs about the parent's μ_R^{post}. The informativeness of the parent's action therefore depends critically on two factors:

1. **The student's beliefs about parent's** μ_R: Does observing $a_P = U$ help the student narrow down what the parent believes about returns to promoted techniques?
2. **The sensitivity of the communication threshold to** μ_R: How much does the communication threshold vary as μ_R changes? This is captured by the slope $\partial \tilde{\mu}_{\text{SBAE}}^{communicate} / \partial \mu_R = -(1-\kappa)/\phi$.

When the communication threshold is strongly downward-sloping in μ_R (large $|-(1-\kappa)/\phi|$), observing $a_P = U$ becomes highly informative because different values of μ_R imply very different ranges for μ_{SBAE}. Conversely, when the threshold is approximately flat (small $|-(1-\kappa)/\phi|$), the student's beliefs about μ_R become irrelevant for inferring μ_{SBAE}, and the parent's action reveals little information.

We now examine these two extreme cases to illustrate the mechanism.

Scenario 1: Strongly downward-sloping threshold (Figure 10)

When κ is low, which means that parents are less capable to implement promoted techniques independently, the threshold has a large negative slope:

$$\left| \frac{\partial \tilde{\mu}_{\text{SBAE}}^{communicate}}{\partial \mu_R} \right| = \frac{1-\kappa}{\phi} \text{ is large}$$

Observing $a_P = U$ is highly informative in this scenario. To see why, note that conditional on choosing $a_P = U$, parents must have $\mu_{\text{SBAE}} < \tilde{\mu}_{\text{SBAE}}^{communicate}(\mu_R)$. When this threshold is strongly downward-sloping, high-μ_R types face a lower threshold and thus must have very low μ_{SBAE} to be constrained by it, while low-μ_R types face a higher threshold and thus can have moderately low μ_{SBAE} while still being constrained.

Specifically, consider two cases:

- **If parent has high returns to promoted techniques (high** μ_R**)**: The communication threshold is substantially lowered by the steep negative slope. Yet the parent still chose unilateral adoption rather than communication. This reveals that the parent's valuation of student learning under SBAE must be quite low—far below an already-reduced threshold.

- **If parent has low returns to promoted techniques (low μ_R)**: The communication threshold remains relatively high since low μ_R doesn't pull it down much. The parent's choice of unilateral adoption reveals that μ_{SBAE} is only moderately low—just below this higher threshold.

Together, these patterns generate a strong negative correlation between μ_R and μ_{SBAE} conditional on observing $a_P = U$: $\text{Corr}(\mu_R, \mu_{\text{SBAE}} \mid a_P = U) < 0$ with large magnitude. This negative correlation makes the parent's action highly informative about their beliefs regarding student learning under SBAE. Even if the student is uncertain about the parent's exact μ_R, observing $a_P = U$ allows the student to substantially update their beliefs about μ_{SBAE} by integrating over possible μ_R values weighted by this strong negative correlation.

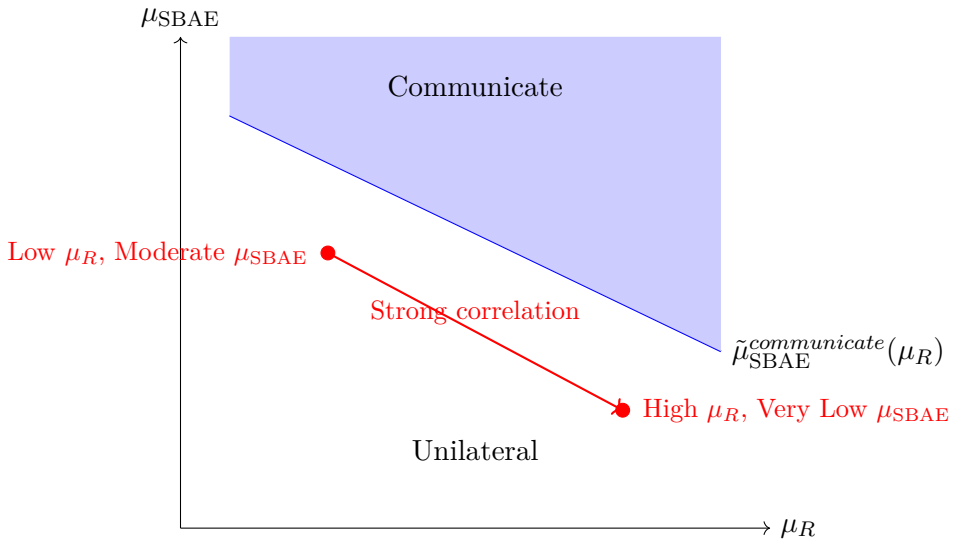


Figure 10: Informative threshold: Strongly downward-sloping $\tilde{\mu}_{\text{SBAE}}^{communicate}(\mu_R)$.

Notes: Observing $a_P = U$ provides substantial information about μ_{SBAE} because types choosing unilateral adoption exhibit strong negative correlation between μ_R and μ_{SBAE}.

Scenario 2: Approximately flat threshold (Figure 11)

When the parent's unilateral capability κ is high, $(1-\kappa)$ is small and the threshold is approximately flat:

$$\frac{\partial \tilde{\mu}_{\text{SBAE}}^{communicate}}{\partial \mu_R} = -\frac{(1-\kappa)}{\phi} \approx 0$$

As $\kappa \to 1$:

$$\tilde{\mu}_{\text{SBAE}}^{communicate} \to \frac{1}{\phi}\left[\frac{c}{\beta} + \lambda\right] \equiv \bar{\mu}_{\text{SBAE}}^{P} \text{ (constant)}$$

Observing $a_P = U$ is uninformative in this scenario. To see why, note that conditional on choosing $a_P = U$, parents must have $\mu_{\text{SBAE}} < \tilde{\mu}_{\text{SBAE}}^{communicate}(\mu_R)$. When the threshold is approximately

constant at $\bar{\mu}^P_{\text{SBAE}}$, this constraint becomes simply $\mu_{\text{SBAE}} < \bar{\mu}^P_{\text{SBAE}}$, which is independent of μ_R. This means that among all parents choosing unilateral adoption, those with high returns to promoted techniques (high μ_R) are just as likely to appear as those with low returns (low μ_R)—the only requirement is that their SBAE valuation falls below the fixed threshold. Since the constraint does not vary with μ_R, the distribution of μ_R among parents choosing U is the same as the unconditional distribution, creating no correlation between the two belief dimensions.

This generates approximately zero correlation between μ_R and μ_{SBAE} conditional on observing $a_P = U$: $\text{Corr}(\mu_R, \mu_{\text{SBAE}} \mid a_P = U) \approx 0$. As a result, the student learns very little from observing the parent's action. Upon observing unilateral adoption, the student can only infer that $\mu_{\text{SBAE}} < \bar{\mu}^P_{\text{SBAE}}$, but gains no information about: (i) where within this range the parent's SBAE valuation lies (it could be just barely below the threshold or far below it); or (ii) what the parent believes about returns to promoted techniques (the parent's μ_R could be high or low with equal likelihood). The parent's action simply reveals that SBAE valuation is insufficient to justify the communication cost, without revealing the underlying reasons why. Critically, the student's own beliefs about the parent's μ_R—whether based on priors or other observations—become irrelevant for inferring μ_{SBAE} because the communication threshold does not depend on μ_R.

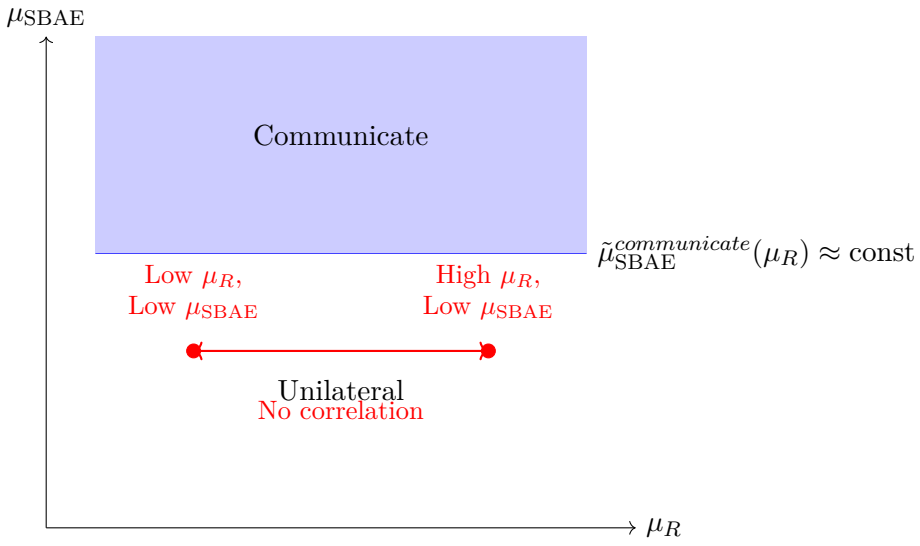


Figure 11: Uninformative threshold: Approximately flat $\tilde{\mu}^{communicate}_{\text{SBAE}}(\mu_R)$.

Notes: When parents have high unilateral capability, observing $a_P = U$ provides little information about μ_{SBAE} because types choosing unilateral adoption are uniformly distributed across μ_R values.

Student's Communication Decision

In what follows, we use the following assumption to rationalize the outsized effects of the revelation nudge in this experiment.

Assumption 1 (Weak Learning): Parents have sufficiently high unilateral adoption capability

(κ) that the communication threshold is relatively flat. Therefore, after observing $a_P = U$, the student's posterior $\bar{\mu}^S_{\text{SBAE}}(U)$ remains close to the prior mean and below the threshold that would make the student willing to initiate communication.

This corresponds to the fact that the main promoted technique—bed preparation—was already familiar to most parents. At endline, it is applied on the `parent parcel` in 75% of control school households. What the school farms highlighted was its yield potential when combined with complementary practices such as composting, lower-density cropping, and more attentive farm management.

We next analyze the student's communication decision. The student initiates collective decision-making if their expected net benefit from communication exceeds the cost c. For communication to be worthwhile, the student must believe there is sufficient probability that: (i) the parent will accept delegation; and (ii) the resulting joint surplus will justify incurring the communication cost. This requires the student's second-order belief about the parent's SBAE valuation to exceed a threshold related to the delegation acceptance threshold.

Under Assumption 1 (weak learning), after observing $a_P = U$, the student's updated belief $\bar{\mu}^S_{\text{SBAE}}(U)$ remains too low to justify initiating communication. The student does not initiate collective decision-making, and the household remains at the status quo or with unilateral adoption by the parent.

6.5. How the Revelation Nudge Helps Overcome Weak Learning

When the student receives the revelation nudge, he observes signal $\tilde{s} = (\tilde{\mu}_R, \tilde{\mu}_{\text{SBAE}})$. Using Bayesian updating:

$$\bar{\mu}^S_{\text{SBAE}}(\tilde{s}, a_P) = \lambda_{\tilde{s}} \cdot \tilde{\mu}_{\text{SBAE}} + (1 - \lambda_{\tilde{s}}) \cdot \mathbb{E}[\mu^{\text{post}}_{\text{SBAE}} \mid a_P] \tag{13}$$

where $\lambda_{\tilde{s}} = \frac{\tau_{\text{nudge}}}{\tau_{\text{post}} + \tau_{\text{nudge}}} \in (0, 1)$ is the precision weight.

When the signal is sufficiently precise ($\sigma^2_{\text{nudge,SBAE}}$ small):

$$\bar{\mu}^S_{\text{SBAE}}(\tilde{s}, a_P) \approx \tilde{\mu}_{\text{SBAE}} \approx \mu^{\text{post}}_{\text{SBAE}} \tag{14}$$

The nudge overcomes the weak learning problem by directly revealing the parent's posterior beliefs.

The student communicates if $\bar{\mu}^S_{\text{SBAE}}(\tilde{s}, U) > \bar{\mu}^{\text{delegate}}_{\text{SBAE}}(U)$, identifying "marginal types" with $\mu_{\text{SBAE}} \in (\bar{\mu}^{\text{delegate}}_{\text{SBAE}}, \tilde{\mu}^{communicate}_{\text{SBAE}})$.

6.6. Treatment effects

This section presents predictions that correspond to the main treatment effects in Section 4.

Video Effect on Delegation

We begin by characterizing how the video treatments affect the equilibrium likelihood of delegation. Without the nudge, delegation only occurs when the parent initiates communication, which happens when their posterior belief about SBAE value exceeds the communication threshold. The probability of delegation is:

$$\Pi_v^{(i)} = \Pr(\mu_{\text{SBAE}}^{\text{post}} > \tilde{\mu}_{\text{SBAE}}^{communicate}(\mu_R^{\text{post}}) \mid \text{Video } v)$$

Prediction 1 (Video Main Effect on Delegation): The persuasive video increases the probability of delegation, and more so than the basic video.

This is via through two channels. First, assuming the video signals contain positive information on average (i.e., $\mathbb{E}[s_{\text{SBAE}}] > \mu_{\text{SBAE}}^{\text{prior}}$), the persuasive video shifts more weight onto the signal, moving the expected posterior belief $\mathbb{E}[\mu_{\text{SBAE}}^{\text{post}}]$ higher and thus increasing the probability of crossing the communication threshold. Formally:

$$\frac{\partial \Pi_v^{delegate}}{\partial \eta_v} = \frac{\partial \Pi_v^{delegate}}{\partial \mathbb{E}[\mu_{\text{SBAE}}^{\text{post}}]} \cdot \frac{\partial \mathbb{E}[\mu_{\text{SBAE}}^{\text{post}}]}{\partial \eta_v} > 0 \tag{15}$$

where $\frac{\partial \mathbb{E}[\mu_{\text{SBAE}}^{\text{post}}]}{\partial \eta_v} = \mathbb{E}[s_{\text{SBAE}}] - \mu_{\text{SBAE}}^{\text{prior}} > 0$. Second, the persuasive video reduces posterior variance, which tightens the distribution of beliefs around the mean. This "precision effect" also increases the probability that beliefs exceed the threshold, as there is less probability mass in the lower tail of the distribution. Figure 12 demonstrates these two channels for the persuasive video, in relation to the basic video.

[Insert Figure 12 here]

Nudge Effect on Delegation

Having characterized how videos affect delegation directly, we now examine how the revelation nudge affects delegation outcomes, and how this effect varies across video treatments. The key insight is that the nudge increases delegation for a specific subset of households: those where the parent would accept delegation but not initiate communication on their own, but the student would be willing to initiate communication after learning about the parent's beliefs. We define these households as "marginal types". Formally, marginal types are those where the parent's posterior belief $\mu_{\text{SBAE}}^{\text{post}}$ falls in the range:

$$MT_v = \{\mu_{\text{SBAE}} \in (\bar{\mu}_{\text{SBAE}}^{\text{delegate}}, \tilde{\mu}_{\text{SBAE}}^{communicate}(\mu_R))\} \tag{16}$$

Households in this interval are precisely those who benefit from the nudge: the parent's valuation of SBAE is high enough that the student would want to communicate if informed, but not quite high enough for the parent to initiate communication unilaterally. The effect of the nudge on delegation probability is simply the probability mass in the marginal type region:

$$\Delta_v^{delegation} = \Pr(\mu_{\text{SBAE}}^{\text{post}} \in MT_v \mid \text{Video } v) \tag{17}$$

Prediction 2 (Videos and Nudges as Substitutes for Delegation): The nudge effect is larger for the basic video than for the persuasive video:

$$\Delta_{\text{Basic}}^{delegate} > \Delta_{\text{Persuasive}}^{delegate} \tag{18}$$

This substitution pattern arises because persuasive videos and revelation nudges work through related mechanisms—both help overcome information frictions that prevent communication. When one intervention is strong (persuasive video), it "crowds out" the potential impact of the other (nudge), as there are fewer marginal households left to benefit. Conversely, when one intervention is weak (basic video), the other intervention has more scope to make a difference. Figure 13 how the stronger rightward shift of parents' distribution under persuasive video leaves less marginal types for the nudge to influence.

[Insert Figure 13 here]

Nudge Effect on Timely Harvest

We now turn to timely harvest adoption, where the interaction between videos and nudges produces a strikingly different pattern. Unlike delegation, where persuasive videos and nudges act as substitutes, for timely harvest they act as strong complements. The key difference is that timely harvest represents a more demanding practice than simple delegation. While delegation merely reassigns an existing parcel to the student, timely harvest requires reorganizing farm operations and coordinating harvest timing differently—a more costly change that requires higher conviction about SBAE's value.

Recall that the threshold ordering in our model is: $\mu_{\text{SBAE}}^{\text{delegate}}(\mu_R, \kappa) < \mu_{\text{SBAE}}^{TH} < \tilde{\mu}_{\text{SBAE}}^{communicate}(\mu_R, \kappa, \beta)$.For timely harvest, the relevant marginal households are those in what we call the "unlockable pool"—parents whose beliefs are high enough to accept timely harvest (and thus high enough for acceptance of delegation), but not high enough to initiate communication. Formally:

$$UP_v = \{\mu_{\text{SBAE}}^{\text{post}} \in (\mu_{\text{SBAE}}^{TH}, \tilde{\mu}_{\text{SBAE}}^{communicate}(\mu_R^{\text{post}}))\} \tag{19}$$

The nudge effect on timely harvest is the probability mass in this unlockable pool:

$$\Delta_v^{TH} = \Pr(\mu_{\text{SBAE}}^{\text{post}} \in UP_v \mid \text{Video } v) \tag{20}$$

Prediction 3 (Videos and Nudges as Complements for Timely Harvest): The nudge effect on timely harvest is much larger for persuasive videos than for basic videos:

$$\Delta_{\text{Persuasive}}^{TH} \gg \Delta_{\text{Basic}}^{TH} \approx 0 \tag{21}$$

The intuition is clear, as can be seen in Figure 14: the persuasive video shifts beliefs upward into the critical zone between the timely harvest threshold and the communication threshold, concentrating probability density where households are willing to adopt the demanding practice but won't initiate conversation themselves. The nudge then unlocks this pool by enabling communication. Neither intervention alone is sufficient—persuasive videos without nudges leave many willing households unable to communicate, while nudges without persuasive videos enable communication in households whose beliefs are too low to justify the costly reorganization of timely harvest. Together, they unlock timely harvest.

[Insert Figure 14 here]

6.7. Testing Heterogeneous Learning with Parcel Control Rights

In matching the treatment effects, we have so far utilized a core insight of the model: the revelation nudge works because it helps students overcome a learning problem about parents' beliefs, and the strength of learning is inversely related to the size of the nudge effects. Now we test this insight by examining an important mechanism that affects learning: who has control rights over the designated `student parcel`.

As in most West African contexts, there is a significant amount of variation in the control rights over parcels used by a household. Focusing on the 1002 households that receive the promotional video, only 29% of the designated `student parcel` are controlled by treated parents. Even if other household members are included, this percentage rises only to 48%. What matters for this section is that 38% are controlled by family members who live outside the household, lineage members, or under community control.[19]

I utilize this contextual feature, which creates variation in how much discretion parents have over students' role on the `student parcel`, to test the learning mechanism. The key idea is that when parents have less discretion due to traditional norms governing externally-controlled

[19] Other reported sources include land bought, borrowed, or rented by the student.

parcels, students learn less from observing parents' actions. This differential learning then generates heterogeneous nudge effects across types of parcel control rights.

Selection Effects and Asymmetric Nudge Impacts

When students observe their parents choosing unilateral adoption ($a_P = U$) rather than communication, what they learn depends critically on the ownership of sales-oriented parcels.

If sales-oriented parcels are controlled by the household, parents have full discretion, so $a_P = U$ (retaining students on subsistence parcels) is informative. Students rationally revise beliefs downward: $\bar{\mu}^{S,H}_{\text{SBAE}}(U) < \mu^{\text{prior}}_{\text{SBAE}}$. Parents choosing $a_P = U$ on household parcels are predominantly those with low SBAE valuations.

If sales-oriented parcels are controlled by people external to the household, traditional norms constrain choices. Even high-valuation parents may choose $a_P = U$ to maintain traditional assignments rather than because they're unwilling to delegate. Students cannot distinguish constrained from unconstrained parents, so beliefs remain at the prior: $\bar{\mu}^{S,E}_{\text{SBAE}}(U) \approx \mu^{\text{prior}}_{\text{SBAE}}$. This sustains a pooling equilibrium where many high-valuation parents are mixed with low-valuation parents.

This differential selection generates asymmetric nudge effects across parcel types. For household-controlled sales-oriented parcels, the nudge provides information revelation about parents whom students have already correctly inferred to have low SBAE valuations—the potential treatment effect is limited. For externally-owned sales-oriented parcels, the nudge breaks the pooling equilibrium by separating high-valuation parents constrained by traditional norms from genuinely low-valuation parents. This separates a larger mass of high-type parents who were previously unidentified due to the binding traditional constraints, generating larger treatment effects on the margin.

Let $\pi_H \in [0, 1]$ denote the fraction of households with household control rights, where parents have full discretion over assignments. Let $\pi_E = 1 - \pi_H$ denote the fraction with lineage- or community-controlled sales-oriented parcels. We have the following prediction, visually depicted in Figure 15.

Prediction 4 (Concentrated Reassignments): Let $\Delta^H_{delegate}$ denote the nudge effect on household-owned parcels and $\Delta^E_{delegate}$ denote the nudge effect on externally-controlled parcels. The overall nudge effect is:

$$\Delta_{delegate} = \pi_H \cdot \Delta^H_{delegate} + \pi_E \cdot \Delta^E_{delegate} \tag{22}$$

We predict $\Delta^E_{delegate} > \Delta^H_{delegate}$, meaning the nudge has a larger marginal effect on

externally-controlled parcels. This implies:

$$\frac{\text{Share of delegations to external parcels}}{\text{Share of delegations to household parcels}} > \frac{\pi_E}{\pi_H} \tag{23}$$

[Insert Figure 15 here]

Table 14 tests Prediction 4 by examining whether the nudge effects on the midline use of subsistence practices on the designated `student parcel` are concentrated on parcels that are looked after by treated parents but controlled by lineage and community members external to the household. This builds upon the results of Table 11 (Panel A), which suggest that the shifts in subsistence orientation are specific to parcels originally looked after by treated parents. Table 14 splits the decline in crop diversity on these treated-parent-managed parcels by parcel control rights: (i) controlled by treated parents; (ii) controlled by the household; and (iii) externally controlled by lineage or community members.

Columns (4) to (6) show that the declines in crop diversity (under the basic video) are heavily concentrated on externally controlled parcels (52%; p=0.04), with substantially weaker and imprecise effects on treated-parent-controlled parcels (29%; p=0.44) and household-controlled parcels (34%; p=0.3). Consistent with the general nudge effects in Table 11, this concentration also only happens on parcels originally looked after by treated parents, not on those looked after by students (col. 1-3) or other household members (col. 7-9). Thus, taken together, we have validated Prediction 4: the nudge effects are stronger when parcels are controlled external to the household.

[Insert Table 14 here]

6.8. Testing Implications for participant targeting with Farmer Field Day (FFD)

Having established the learning mechanism, we now examine its implications for extension program design. A key question is whom to target with direct training. In the model, training influential household members who control key resources can reduce their incentives for collective decision-making, as increased capability for unilateral implementation may dominate the gains from delegation. This can narrow program benefits by foreclosing opportunities to empower other household members through joint decision-making. These findings suggest that training design must account for intrahousehold bargaining to avoid inadvertently concentrating benefits among already-powerful members while missing intertwined objectives in rural development, such as youth empowerment and school attendance.

Specifically, I leverage the school-level randomization of the annual FFD—a half-day training for parents on soil management techniques—to examine how direct training of parents affects the

assignment of students to subsistence-oriented parcels, and, consequently, the size of the nudge effects. I focus on the a perverse effect in which training parents directly can reduce their benefits from delegation compared to unilateral technology adoption.

Note that the FFD provides both training and demonstration of harvests, so it induces two effects for the parent. First, the FFD increases the parent's capacity to unilaterally implement promoted techniques: $\kappa^{\text{FFD}} > \kappa$. Second, the FFD increases the parent' expected returns from promoted techniques: $\mu_R^{\text{post,FFD}} > \mu_R^{\text{post}}$. Since we focus on the sample of parents who have watched the promotional video and thus visited the school farm subsequently, we assume that the FFD does not affect the parent's beliefs about SBAE value: $\mu_{\text{SBAE}}^{\text{post,FFD}} = \mu_{\text{SBAE}}^{\text{post}}$.

Perverse Effect Condition and Nudge Effects

Note that increasing κ unambiguously reduce incentives to delegate, whereas increasing μ_R can either increase or decrease delegation incentives, depending on parameter values. We thus analyze the condition under which there would be a perverse effect.

Note that the change in surplus from delegation with FFD training is:

$$\Delta W^{delegate} = \Delta_R^{\text{FFD}}(1-\kappa) - \mu_R \cdot \Delta_\kappa^{\text{FFD}} - \Delta_R^{\text{FFD}} \cdot \Delta_\kappa^{\text{FFD}}$$

Perverse effect condition: If the terms involving capability improvement dominate:

$$\mu_R \cdot \Delta_\kappa^{\text{FFD}} + \Delta_R^{\text{FFD}} \cdot \Delta_\kappa^{\text{FFD}} > \Delta_R^{\text{FFD}}(1-\kappa)$$

or,

$$\mu_R \cdot \Delta_\kappa^{\text{FFD}} > \Delta_R^{\text{FFD}}(1-\kappa-\Delta_\kappa^{\text{FFD}}) \tag{24}$$

then surplus from delegation actually *falls*, and the communication threshold shifts upward:

$$\tilde{\mu}_{\text{SBAE}}^{communicate,\text{FFD}} > \tilde{\mu}_{\text{SBAE}}^{communicate}$$

The perverse effect condition is likely met in the current context for three reasons. First, κ is high, as most parents are already familiar with the main promoted technique, bed preparation, so that they only need training on how to combine that with other techniques. Second, μ_R tends to be large relative to Δ_R^{FFD}, as parents have already watched the promotional video and many have visited the school farm subsequently. Third, evidence from Lee (2025) indicates that $\Delta_\kappa^{\text{FFD}}$ is likely large, as the FFD improves parents' knowledge of promoted techniques in a quiz.

Can the revelation nudge reverse this perverse effect? Note that the FFD training increases the parent's unilateral capability κ, making the communication threshold flatter (weaker learning). Under Assumption 1 (weak learning), this means:

$$\left|\frac{\partial \tilde{\mu}_{\text{SBAE}}^{communicate,\text{FFD}}}{\partial \mu_R}\right| = \frac{1-\kappa^{\text{FFD}}}{\phi} < \frac{1-\kappa}{\phi} = \left|\frac{\partial \tilde{\mu}_{\text{SBAE}}^{communicate}}{\partial \mu_R}\right| \tag{25}$$

Hence the FFD creates a larger gap between the parent's delegation acceptance threshold $\bar{\mu}_{\text{SBAE}}^{\text{delegate}}$ (unchanged by FFD) and the parent's communication threshold $\tilde{\mu}_{\text{SBAE}}^{delegate,\text{FFD}}$ (shifted upward by FFD).

The set of marginal types that can only be identified through the nudge expands:

$$MT^{\text{FFD}} = \{\mu_{\text{SBAE}} \in (\bar{\mu}_{\text{SBAE}}^{\text{delegate}}, \tilde{\mu}_{\text{SBAE}}^{communicate,\text{FFD}})\} \supset MT^{\text{baseline}} \tag{26}$$

The nudge effect with FFD is therefore larger:

$$\Delta_{delegate}^{\text{FFD}} = \Pr(\text{marginal types identified} \mid \text{FFD}) > \Delta_{delegate}^{\text{baseline}} \tag{27}$$

Hence we the following prediction.

Prediction 5 (FFD-Nudge Interaction for Delegation): Suppose the perverse effect condition (24) holds. Then the revelation effect on delegation is stronger when FFD training is provided.

Table 15 tests this prediction by examining how FFD affects the nudge effects on the midline use of subsistence practices on the designated `student parcel`. Specifically, I run the following regressions:

$$Y_{hs} = a + b_1\text{PVideo}_h + b_2\text{Rev}_h + b_3\text{PVideo}_h \times \text{Rev}_h + b_4\text{FFD}_s + e_{hs} \tag{6a}$$

$$\begin{aligned} Y_{hs} = a &+ b_1\text{PVideo}_h + c_1\text{PVideo}_h \times \text{FFD}_s \\ &+ b_2\text{Rev}_h + c_2\text{Rev}_h \times \text{FFD}_s \\ &+ b_3\text{PVideo}_h \times \text{Rev}_h + c_3\text{PVideo}_h \times \text{Rev}_h \times \text{FFD}_s + e_{hs} \end{aligned} \tag{6b}$$

where Y_{hs} is the outcome of interest for household h and section-gender strata s, where section refers to whether student is in junior high or elementary school; PVideo_h indicates whether household h is randomly assigned to the persuasive video arm; Rev_h indicates whether household h is randomly assigned to the revelation nudge; FFD_s is a binary indicator of whether schools are randomly assigned to host an annual FFD; g_s denotes section-gender fixed effects; e_{hs} is the error term.

I first test if the perverse effect holds by estimating regression (6a), focusing on the FFD effect pooled across treatment conditions. Table 15 shows that the evidence for a perverse effect is overwhelmingly strong, as the FFD strongly increases the subsistence orientation of the designated `student parcel`. It increases the fraction of rice parcels by 9.7 percentage points (% increase from 20%), the fraction of intercropped parcels by 9.9 percentage points (% increase from 18%), crop diversity by 1 (% increase from 4.76%), and the subsistence index by 0.3 standard deviations. Note

that the pooled FFD effect on the subsistence index is almost twice in magnitude compared to the pooled revelation effect in Table 7 (0.172 standard deviations; Panel A).

Given that the perverse effect condition holds, I now test **Prediction 5** by estimating regression (6b). The results are also presented in Table 15. The first thing to notice is that under the basic video, the FFD effects are large and positive, similar to the pooled FFD effects above (FFD_s, cols 2, 5, 8, and 11). Next, we focus on the revelation effect under the basic video, which has been shown to be the driving force at midline. Without FFD, this effect (Rev_h, cols 1, 4, 7, and 10) is negative as expected, but small and imprecise. Once FFD is provided, however, this effect ($Rev_d \times FFD_s$, cols 2, 5, 8, and 11) becomes much more negative as suggested by **Prediction 5**: each of the coefficients are large and significant at the 5% level. The effect on the subsistence index, in particular, is about 2.4 times bigger than the effect when FFD is not hosted. In other words, the revelation effect under the basic video is strongly driven by effects under the FFD. Lastly, notice that this revelation effect fully offsets the FFD effect. The total effect of the basic video and the FFD on the subsistence index is 0.41 standard deviations. Adding the revelation nudge pulls this total effect down to -0.032 standard deviations (p=0.84). Overall speaking, therefore, **Prediction 5** is validated. These results highlight the importance of accounting for power dynamics in participant targeting: attempts such as the FFD, which aim to broaden benefits from agricultural extension, can be counterproductive if communication frictions are ignored; meanwhile, such perverse effects can be eliminated with simple informational nudges.

[Insert Table 15 here]

7. Concluding remarks

Positioning schools as hubs for rural development offers a promising solution to the logistical challenges that constrain development programs in rural areas. By leveraging existing infrastructure, teachers, students' initiative, and social ties, schools can reach dispersed households at scale while strengthening rural education through practical, experiential learning. Lee (2025) achieves half of this promise: by engaging parents with program information, school-based agricultural extension can drive household technology adoption and become more cost-effective than most rigorously evaluated extension programs in sub-Saharan Africa. This paper unlocks the other half—students' economic empowerment and increased school attendance. How we achieve this is perhaps subtle and unexpected: through a simple nudge—a five-minute survey component—that overcomes communication frictions and signals to students their parents' openness to changes in farm management.

This is surprising partly because parents are already made more open to changes in farm management through parental engagement, so one would expect that any additional interventions

tackling communication frictions would only modify the degree of delegation to students. Yet the results are dramatic: the SBAE program only improves students' annual income with the nudge, while the nudge doubles SBAE's impact per dollar on days of school attendance. Even more surprising, Lee (2025) shows that parental engagement substantially improves parents' visits to school farms and adoption of promoted soil management techniques. So it is puzzling why students fail to infer parents' openness to delegation without the nudge.

The answer likely lies in the behavioral nature of household decision making, and this nature is amplified in the agricultural context of West Africa. Many empirical contributions (e.g. Udry (1996); Duflo & Udry (2004)) have elucidated how the control of different productive resources within households is rigidly tied to different persons. This mode of decision making is best elucidated by the Lundberg-Pollak separate spheres model (Lundberg & Pollak, 1993), in which the noncooperative equilibrium reflects how the prevailing social norms assign resources. This paper focuses on the *change* aspect of decision making. It is human nature to exhibit a preference for the status quo and overlook opportunities for joint decision making. This is especially true when there are many margins of change in a cultivation system adapting its way to more diverse practices. Assigning students to look after traditional rice fields is one thing, assigning them to manage sales-oriented parcels is another. Being receptive to SBAE-promoted techniques is one thing, making the more demanding adjustments required for early harvests of root crops is another. The revelation nudge is inspirational in nature: it overcomes the natural inertia in household decision making by signaling to students that their parents are open to changes in resource control, something that they perhaps have not imagined before.

The failure of students to learn from parents' responsiveness to the SBAE program and technology adoption, however, is far from necessary. I offer a substantive theory of communication frictions that has important implications for agricultural extension and rural development. Specifically, parents' responsiveness to the SBAE program reflects a mix of their beliefs about the returns to promoted techniques and their willingness to delegate farms to students. When parents can implement promoted techniques on their own quite effectively—which is true in the current context—their willingness to involve students becomes largely disconnected from their beliefs about whether the techniques work well. As a result, students learn little information about parents' types from parents' actions.

This pooling of parents' types in equilibrium is more severe in two cases. First, when household parcels are controlled by lineage or community, social norms can inhibit parents who are willing to delegate from doing so. Second, when a farmer field day inadvertently strengthens parents' capacity to unilaterally implement promoted techniques, parents become more likely to bypass joint decision making despite their willingness to delegate. I provide strong evidence that the revelation nudge works much more strongly in these cases of learning failure.

The nature of promoted techniques, along with these cases, provides generalizable insights:

technology selection, participant targeting, and informational nudges should be designed cohesively, especially when third-party approval for changes is required—a common feature in smallholder agriculture where resources are pooled. This subtle encouragement of joint decision making could in turn broaden the scope of benefits from agricultural extension, enabling a holistic solution to intertwined development goals.

Having clearly understood the importance of communication frictions, why they persist, and that the revelation nudge works, moving forward we should find logistically convenient ways to overcome communication frictions. As SBAE scales to all rural schools in Liberia by 2035—and expands to Côte d'Ivoire and Kenya—it opens a powerful testing ground for variants that tackle intergenerational communication frictions. These include: (i) eliciting parents' beliefs in front of students; (ii) involving students in parental engagement sessions; (iii) having students draft home project plans that require discussion with parents about their growth in farm management under SBAE; (iv) assigning joint home projects to students and parents; and (v) hosting structured discussions where students and parents reflect together on students' development in farm management skills.

The expansion of SBAE also creates opportunities to leverage its maturing system of high-frequency student data collection, developed over the course of the RCT. This infrastructure could address key limitations in the current study, including high student attrition and resource constraints that restricted the breadth of outcome measurement. The exploratory hypotheses examined in this paper—spanning multiple dimensions of farm management, intergenerational dynamics, and welfare—would benefit from more comprehensive data collection than survey costs permitted. A key limitation is that the current study relies on retrospective measures of income, savings, and timely payment of school fees, without directly linking these outcomes to the application of specific practices such as sales-oriented cultivation and timely harvests of root crops. Further, advancing the theoretical framework in Section 6 requires distinguishing between students' second-order uncertainty about parents' preferences versus beliefs, as this distinction is critical for understanding the mechanisms of social learning.[20]

Lastly, this research agenda aims to embed the model of intergenerational interactions within the broader framework of agricultural technology diffusion. A first avenue is to reinterpret the μ_R parameter in Section 6 as capturing the appropriateness of promoted techniques in a highly heterogeneous environment, thereby linking this framework to the companion model in (Lee & Udry, 2025). In this extension, μ_R evolves through standard social learning dynamics, allowing intergenerational interactions and social learning to jointly shape diffusion. A second avenue is to take dyad-specific communication frictions seriously and trace their implications within social networks—particularly how they mediate diffusion patterns and shape gendered outcomes.

[20] I measured students' second-order beliefs about parents' beliefs regarding their agricultural knowledge, which yielded no important findings. In retrospect, this measurement should have focused on parents' beliefs about students' cassava harvests, which proved crucial throughout the study.

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Tables to be included in the main text

Table 4: Treatment effects on students' overall farming activities

	Panel A: During rainy season 2022 (midline)											
	Farm outside schools (0/1)			Weekly hours farming			Weekly hours farming outside school			# household parcels involved		
Specification	Reference		Main	Reference		Main	Reference		Main	Reference		Mai
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$SBAE_p$	0.000	0.000		0.555	0.540		0.185	0.176		-0.037	-0.039	
	(0.013)	(0.013)		(0.609)	(0.608)		(0.587)	(0.587)		(0.075)	(0.075)	
$SBAE_p \times FFD_s$	0.011	0.011		-0.827	-0.804		-0.724	-0.707		0.099	0.103	
	(0.013)	(0.013)		(0.644)	(0.638)		(0.660)	(0.654)		(0.075)	(0.073)	
$SBAE_p \times Session_s$	0.010	-0.005		0.658	0.356		0.073	-0.173		0.218**	0.154	
	(0.013)	(0.017)		(0.567)	(0.698)		(0.522)	(0.598)		(0.087)	(0.104)	
$SBAE_p \times Session_s \times PVideo_h$		0.020	0.014		1.184*	0.975		0.790	0.622		0.165	0.09
		(0.016)	(0.016)		(0.668)	(0.909)		(0.686)	(0.801)		(0.118)	(0.13
$SBAE_p \times Session_s \times Rev_h$		0.012	0.008		-0.175	-0.002		0.029	0.069		0.044	0.03
		(0.015)	(0.016)		(0.700)	(0.877)		(0.587)	(0.773)		(0.103)	(0.12
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		-0.002	0.011		-0.747	-0.639		-0.617	-0.342		-0.158	0.00
		(0.018)	(0.023)		(1.009)	(1.273)		(0.956)	(1.122)		(0.139)	(0.18
Summary: Intent-to-treat effects of the revelation nudge												
Rev Effect given Basic Video		0.012	0.008		-0.175	-0.002		0.029	0.069		0.044	0.03
		(0.015)	(0.016)		(0.700)	(0.877)		(0.587)	(0.773)		(0.103)	(0.12
Rev Effect given Persuasive Video		0.009	0.019		-0.922	-0.641		-0.588	-0.273		-0.114	0.04
		(0.009)	(0.016)		(0.823)	(0.888)		(0.827)	(0.783)		(0.117)	(0.13
Pooled Rev Effect		0.010	0.014		-0.549	-0.317		-0.280	-0.100		-0.035	0.03
		(0.009)	(0.011)		(0.573)	(0.611)		(0.534)	(0.538)		(0.085)	(0.09
Summary: Intent-to-treat effects of SBAE program (with video and FFD)												
SBAE Effect w/o Rev		0.016			0.684			-0.310			0.301***	
		(0.012)			(0.621)			(0.623)			(0.073)	
SBAE Effect w/ Rev		0.026**			0.135			-0.590			0.266***	
		(0.012)			(0.566)			(0.551)			(0.087)	
Summary Statistics for Reference Groups												
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Prom sessio
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video witho Rev
Reference group mean	0.96	0.96	0.97	9.46	9.46	9.95	8.21	8.21	8.09	1.68	1.68	1.84
Reference group SD	0.20	0.20	0.18	7.26	7.26	7.11	6.24	6.24	6.42	0.93	0.93	1.11
Observations	2425	2425	662	2425	2425	662	2425	2425	662	2425	2425	662

Notes: This table reports treatment effects on students' overall farming activities during the rainy seasons of 2022 (midline; Panel A) and 2023 (endline; Panel B). The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. The third portion reports the intent-to-treat effects of the SBAE program (augmented with program promotional video and the annual farmer field day) (i) when the revelation nudge is absent; and (ii) when the revelation nudge is present. Again, these effects are computed using linear combinations of estimates. The reference regression specifications include pair of clusters fixed effects and standard errors are clustered at the cluster of schools level. The main regression specification includes school grade section gender fixed

Table 4: Treatment effects on students' overall farming activities *(continued)*

	Panel B: During rainy season 2023 (endline)											
	Farm outside schools (0/1)			Weekly hours farming			Weekly hours farming outside school			# household parcels involved		
Specification	Reference		Main	Reference		Main	Reference		Main	Reference		Main
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$SBAE_p$	-0.017	-0.017		0.487	0.506		0.321	0.339		0.107*	0.104*	
	(0.019)	(0.019)		(0.782)	(0.784)		(0.732)	(0.733)		(0.062)	(0.062)	
$SBAE_p \times FFD_s$	0.044*	0.045*		-0.618	-0.620		-0.805	-0.806		0.065	0.068	
	(0.023)	(0.024)		(0.743)	(0.747)		(0.720)	(0.723)		(0.111)	(0.109)	
$SBAE_p \times Session_s$	0.017	0.004		0.825	0.519		0.404	0.100		0.016	-0.010	
	(0.027)	(0.038)		(0.714)	(0.860)		(0.679)	(0.852)		(0.112)	(0.123)	
$SBAE_p \times Session_s \times PVideo_h$		0.033	0.047		-0.309	-0.778		-0.278	-0.815		0.109	0.053
		(0.037)	(0.044)		(0.896)	(1.234)		(0.897)	(1.157)		(0.140)	(0.171
$SBAE_p \times Session_s \times Rev_h$		0.008	-0.005		1.666*	1.721		1.568*	1.163		-0.099	-0.123
		(0.037)	(0.043)		(0.909)	(1.200)		(0.873)	(1.125)		(0.136)	(0.166
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		-0.029	-0.015		-1.633	-1.901		-1.497	-1.392		0.101	0.089
		(0.047)	(0.062)		(1.347)	(1.716)		(1.299)	(1.609)		(0.208)	(0.238
Summary: Intent-to-treat effects of the revelation nudge												
Rev Effect given Basic Video		0.008	-0.005		1.666*	1.721		1.568*	1.163		-0.099	-0.123
		(0.037)	(0.043)		(0.909)	(1.200)		(0.873)	(1.125)		(0.136)	(0.166
Rev Effect given Persuasive Video		-0.021	-0.020		0.033	-0.180		0.071	-0.229		0.002	-0.033
		(0.038)	(0.043)		(0.974)	(1.189)		(0.932)	(1.115)		(0.179)	(0.165
Pooled Rev Effect		-0.006	-0.013		0.850	0.762		0.820	0.461		-0.048	-0.078
		(0.029)	(0.030)		(0.659)	(0.832)		(0.627)	(0.779)		(0.120)	(0.115
Summary: Intent-to-treat effects of SBAE program (with video and FFD)												
SBAE Effect w/o Rev		0.049*			0.251			-0.506			0.216**	
		(0.026)			(0.803)			(0.741)			(0.092)	
SBAE Effect w/ Rev		0.043*			1.101			0.313			0.167*	
		(0.022)			(0.779)			(0.733)			(0.089)	
Summary Statistics for Reference Groups												
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev
Reference group mean	0.91	0.91	0.93	10.17	10.17	11.45	9.20	9.20	9.68	1.62	1.62	1.76
Reference group SD	0.28	0.28	0.26	8.04	8.04	7.91	7.67	7.67	7.47	1.01	1.01	0.95
Observations	1488	1488	396	1487	1487	396	1487	1487	396	1488	1488	396

Notes: This table reports treatment effects on students' overall farming activities during the rainy seasons of 2022 (midline; Panel A) and 2023 (endline; Panel B). The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. The third portion reports the intent-to-treat effects of the SBAE program (augmented with program promotional video and the annual farmer field day) (i) when the revelation nudge is absent; and (ii) when the revelation nudge is present. Again, these effects are computed using linear combinations of estimates. The reference regression specifications include pair of clusters fixed effects and standard errors are clustered at the cluster of schools level. The main regression specification includes school grade section gender fixed

Table 5: Treatment effects on parcel assignments during rainy season 2022 (midline)

	Panel A: Students' assignment to look after household parcels											
	Look after parcels (0/1)			# parcels looked after			Look after roots/vegetables parcels (0/1)			# roots/vegetables parcels looked after		
Specification	Reference		Main	Reference		Main	Reference		Main	Reference		M
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(1
$SBAE_p$	0.033	0.032		0.045	0.043		0.026	0.024		0.045	0.043	
	(0.042)	(0.042)		(0.064)	(0.064)		(0.040)	(0.040)		(0.057)	(0.057)	
$SBAE_p \times FFD_s$	0.030	0.033		0.050	0.054		0.038	0.041		0.052	0.054	
	(0.040)	(0.040)		(0.057)	(0.057)		(0.041)	(0.041)		(0.052)	(0.052)	
$SBAE_p \times Session_s$	0.073*	0.027		0.151**	0.100		0.051	0.004		0.089	0.048	
	(0.043)	(0.053)		(0.070)	(0.085)		(0.042)	(0.051)		(0.060)	(0.070)	
$SBAE_p \times Session_s \times PVideo_h$		0.117**	0.080		0.194*	0.094		0.126**	0.094		0.141	0.
		(0.051)	(0.060)		(0.106)	(0.113)		(0.053)	(0.061)		(0.091)	(0
$SBAE_p \times Session_s \times Rev_h$		0.019	0.019		-0.037	-0.102		-0.018	-0.007		-0.041	-0
		(0.059)	(0.058)		(0.092)	(0.109)		(0.057)	(0.059)		(0.089)	(0
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		-0.088	-0.041		-0.104	0.076		-0.026	0.018		-0.030	0.
		(0.082)	(0.084)		(0.150)	(0.158)		(0.084)	(0.085)		(0.143)	(0
Summary: Intent-to-treat effects of the revelation nudge												
Rev Effect given Basic Video		0.019	0.019		-0.037	-0.102		-0.018	-0.007		-0.041	-0
		(0.059)	(0.058)		(0.092)	(0.109)		(0.057)	(0.059)		(0.089)	(0
Rev Effect given Persuasive Video		-0.068	-0.022		-0.141	-0.026		-0.044	0.011		-0.071	0.
		(0.057)	(0.059)		(0.114)	(0.110)		(0.055)	(0.060)		(0.098)	(0
Pooled Rev Effect		-0.025	-0.001		-0.089	-0.065		-0.031	0.002		-0.056	-0
		(0.041)	(0.040)		(0.071)	(0.076)		(0.037)	(0.041)		(0.061)	(0
Summary: Intent-to-treat effects of SBAE program (with video and FFD)												
SBAE Effect w/o Rev		0.150***			0.295***			0.132***			0.215***	
		(0.041)			(0.072)			(0.041)			(0.059)	
SBAE Effect w/ Rev		0.125***			0.206***			0.101**			0.159**	
		(0.040)			(0.068)			(0.042)			(0.062)	
Summary Statistics for Reference Groups												
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Pr ses
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Ba Vi wit Re
Reference group mean	0.53	0.53	0.60	0.70	0.70	0.87	0.48	0.48	0.53	0.60	0.60	0
Reference group SD	0.50	0.50	0.49	0.79	0.79	0.93	0.50	0.50	0.50	0.71	0.71	0
Observations	2425	2425	662	2425	2425	662	2425	2425	662	2425	2425	6

Notes: This table reports treatment effects on students' overall farming activities during the rainy seasons of 2022 (midline; Panel A) and 2023 (endline; Panel B). The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. The third portion reports the intent-to-treat effects of the SBAE program (augmented with program promotional video and the annual farmer field day) (i) when the revelation nudge is absent; and (ii) when the revelation nudge is present. Again, these effects are computed using linear combinations of estimates. The reference regression specifications include pair of clusters fixed effects and standard errors are clustered at the cluster of schools level. The main regression specification includes school grade section gender fixed

Table 5: Treatment effects on parcel assignments during rainy season 2022 (midline) *(continued)*

	Panel B: Proxies—students' assignment to look after subsistence-oriented parcels								
	# rice parcels			# intercropped parcels			average # crops per parcel		
Specification	Reference		Main	Reference		Main	Reference		Main
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$SBAE_p$	0.011	0.011		0.002	0.003		-0.228	-0.232	
	(0.027)	(0.027)		(0.027)	(0.027)		(0.183)	(0.182)	
$SBAE_p \times FFD_s$	-0.002	-0.002		0.004	0.004		0.365**	0.369**	
	(0.033)	(0.033)		(0.029)	(0.029)		(0.174)	(0.173)	
$SBAE_p \times Session_s$	-0.016	-0.005		-0.012	0.002		0.171	0.145	
	(0.026)	(0.043)		(0.027)	(0.042)		(0.169)	(0.226)	
$SBAE_p \times Session_s \times PVideo_h$		0.011	-0.024		0.004	-0.027		0.318	0.148
		(0.042)	(0.051)		(0.044)	(0.049)		(0.246)	(0.286)
$SBAE_p \times Session_s \times Rev_h$		-0.004	-0.047		-0.003	-0.047		-0.345	-0.345
		(0.057)	(0.049)		(0.052)	(0.047)		(0.268)	(0.276)
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		-0.056	-0.019		-0.054	-0.017		0.171	0.294
		(0.073)	(0.071)		(0.070)	(0.068)		(0.340)	(0.401)
# parcels looked after	0.218***	0.221***	0.171***	0.168***	0.171***	0.136***	1.607***	1.614***	1.270***
	(0.018)	(0.018)	(0.019)	(0.017)	(0.017)	(0.018)	(0.112)	(0.112)	(0.099)
	Summary: Intent-to-treat effects of the revelation nudge								
Rev Effect given Basic Video		-0.004	-0.047		-0.003	-0.047		-0.345	-0.345
		(0.057)	(0.049)		(0.052)	(0.047)		(0.268)	(0.276)
Rev Effect given Persuasive Video		-0.060	-0.065		-0.058	-0.064		-0.174	-0.051
		(0.042)	(0.050)		(0.040)	(0.048)		(0.226)	(0.280)
Pooled Rev Effect		-0.032	-0.056		-0.031	-0.055*		-0.259	-0.199
		(0.035)	(0.034)		(0.030)	(0.033)		(0.180)	(0.192)
	Summary: Intent-to-treat effects of SBAE program (with video and FFD)								
SBAE Effect w/o Rev		0.009			0.011			0.441**	
		(0.038)			(0.034)			(0.202)	
SBAE Effect w/ Rev		-0.022			-0.019			0.181	
		(0.034)			(0.032)			(0.184)	
	Summary Statistics for Reference Groups								
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev
Reference group mean	0.18	0.18	0.18	0.15	0.15	0.16	2.03	2.03	2.15
Reference group SD	0.43	0.43	0.43	0.40	0.40	0.40	2.66	2.66	2.50
Observations	2425	2425	662	2425	2425	662	2425	2425	662

Notes: This table reports treatment effects on students' assignment to look after household parcels (Panel A) and, more specifically, subsistence-oriented parcels (Panel B) during the rainy seasons of 2022 (midline). The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. The third portion reports the intent-to-treat

Table 6: Treatment effects on students' management of the designated `student parcel`

	Panel A: During rainy season 2022 (midline)											
	Look after (0/1)			Look after: first time (0/1)			Choose crops: independently (0/1)			Choose crops: involved (0/1)		
Specification	Reference		Main	Reference		Main	Reference		Main	Reference		Ma
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$SBAE_p$	0.047	0.044		0.018	0.015		0.035	0.032		0.039	0.036	
	(0.053)	(0.052)		(0.051)	(0.050)		(0.050)	(0.048)		(0.058)	(0.057)	
$SBAE_p \times FFD_s$	-0.000	0.007		0.058	0.064*		-0.006	0.000		0.017	0.023	
	(0.041)	(0.040)		(0.039)	(0.038)		(0.059)	(0.058)		(0.064)	(0.063)	
$SBAE_p \times Session_s$	0.051	-0.030		0.057	-0.021		0.031	-0.047		0.015	-0.055	
	(0.053)	(0.065)		(0.052)	(0.062)		(0.040)	(0.048)		(0.060)	(0.073)	
$SBAE_p \times Session_s \times PVideo_h$		0.128**	0.145**		0.092**	0.110*		0.107***	0.077		0.117**	0.13
		(0.052)	(0.060)		(0.045)	(0.057)		(0.041)	(0.056)		(0.053)	(0.0
$SBAE_p \times Session_s \times Rev_h$		0.091	0.097*		0.117**	0.142**		0.106**	0.130**		0.085	0.12
		(0.057)	(0.058)		(0.049)	(0.056)		(0.044)	(0.054)		(0.060)	(0.0
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		-0.115	-0.124		-0.109*	-0.125		-0.121*	-0.097		-0.125	-0.1
		(0.072)	(0.084)		(0.063)	(0.081)		(0.070)	(0.078)		(0.087)	(0.0
Summary: Intent-to-treat effects of the revelation nudge												
Rev Effect given Basic Video		0.091	0.097*		0.117**	0.142**		0.106**	0.130**		0.085	0.12
		(0.057)	(0.058)		(0.049)	(0.056)		(0.044)	(0.054)		(0.060)	(0.0
Rev Effect given Persuasive Video		-0.024	-0.026		0.008	0.016		-0.015	0.033		-0.039	-0.0
		(0.052)	(0.058)		(0.045)	(0.056)		(0.051)	(0.054)		(0.055)	(0.0
Pooled Rev Effect		0.033	0.035		0.063*	0.079**		0.046	0.082**		0.023	0.0
		(0.040)	(0.040)		(0.035)	(0.039)		(0.033)	(0.037)		(0.037)	(0.0
Summary: Intent-to-treat effects of SBAE program (with video and FFD)												
SBAE Effect w/o Rev		0.085			0.103**			0.039			0.063	
		(0.051)			(0.049)			(0.043)			(0.055)	
SBAE Effect w/ Rev		0.118**			0.166***			0.085*			0.086	
		(0.048)			(0.047)			(0.048)			(0.062)	
Summary Statistics for Reference Groups												
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Pro sessi
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basi Vide with Rev
Reference group mean	0.27	0.27	0.29	0.21	0.21	0.24	0.18	0.18	0.19	0.24	0.24	0.2
Reference group SD	0.45	0.45	0.46	0.41	0.41	0.43	0.38	0.38	0.39	0.43	0.43	0.4
Observations	1384	1384	650	1384	1384	650	1384	1384	650	1384	1384	65

Notes: This table reports treatment effects on students' management of the designated `student parcel` during the rainy seasons of 2022 (midline; Panel A) and 2023 (endline; Panel B). The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. The third portion reports the intent-to-treat effects of the SBAE program (augmented with program promotional video and the annual farmer field day) (i) when the revelation nudge is absent; and (ii) when the revelation nudge is present. Again, these effects are computed using linear combinations of estimates. The reference regression specifications include pair of clusters fixed effects and standard errors are clustered at the cluster of schools level. The main regression specification includes school grade section gender

Table 6: Treatment effects on students' management of the designated `student parcel`
(continued)

	Panel B: During rainy season 2023 (endline)								
	Look after (0/1)			Choose crops: independently (0/1)			Choose crops: involved (0/1)		
Specification	Reference		Main	Reference		Main	Reference		Main
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$SBAE_p$	0.113***	0.113***		0.027	0.027		0.011	0.011	
	(0.041)	(0.041)		(0.046)	(0.046)		(0.043)	(0.043)	
$SBAE_p \times FFD_s$	-0.039	-0.039		0.027	0.027		0.031	0.031	
	(0.048)	(0.048)		(0.042)	(0.042)		(0.045)	(0.044)	
$SBAE_p \times Session_s$	0.082*	0.035		0.057	0.065		0.071	0.064	
	(0.049)	(0.065)		(0.047)	(0.065)		(0.046)	(0.066)	
$SBAE_p \times Session_s \times PVideo_h$		-0.015	0.010		-0.023	-0.009		0.006	0.047
		(0.072)	(0.082)		(0.071)	(0.075)		(0.075)	(0.082)
$SBAE_p \times Session_s \times Rev_h$		0.063	0.094		-0.004	0.028		-0.009	0.017
		(0.075)	(0.080)		(0.064)	(0.073)		(0.067)	(0.080)
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		0.084	0.070		0.021	0.002		0.035	0.003
		(0.119)	(0.114)		(0.083)	(0.105)		(0.095)	(0.114)
	Summary: Intent-to-treat effects of the revelation nudge								
Rev Effect given Basic Video		0.063	0.094		-0.004	0.028		-0.009	0.017
		(0.075)	(0.080)		(0.064)	(0.073)		(0.067)	(0.080)
Rev Effect given Persuasive Video		0.146*	0.164**		0.017	0.031		0.026	0.020
		(0.081)	(0.079)		(0.066)	(0.073)		(0.068)	(0.079)
Pooled Rev Effect		0.104**	0.129**		0.007	0.029		0.009	0.018
		(0.050)	(0.055)		(0.050)	(0.051)		(0.048)	(0.055)
	Summary: Intent-to-treat effects of SBAE program (with video and FFD)								
SBAE Effect w/o Rev		0.101**			0.107**			0.108**	
		(0.048)			(0.044)			(0.044)	
SBAE Effect w/ Rev		0.205***			0.114***			0.117***	
		(0.051)			(0.042)			(0.043)	
	Summary Statistics for Reference Groups								
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev
Reference group mean	0.33	0.33	0.41	0.23	0.23	0.28	0.29	0.29	0.34
Reference group SD	0.47	0.47	0.49	0.42	0.42	0.45	0.46	0.46	0.48
Observations	1488	1488	396	1488	1488	396	1488	1488	396

Notes: This table reports treatment effects on students' management of the designated `student parcel` during the rainy seasons of 2022 (midline; Panel A) and 2023 (endline; Panel B). The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. The third portion reports the intent-to-treat effects of the SBAE program (augmented with program promotional video and the annual farmer field day) (i) when the revelation nudge is absent; and (ii) when the revelation nudge is present. Again, these effects are computed using linear combinations of estimates. The reference

Table 7: Treatment effects on the use of subsistence practices on the designated `student parcel`

	Panel A: During rainy season 2022 (midline)										
	Rice (0/1)			Intercropping (0/1)			# crops			First principal component (SD)	
Specification	Reference		Main	Reference		Main	Reference		Main	Reference	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
$SBAE_p$	-0.063	-0.061		-0.020	-0.018		-0.631**	-0.625**		-0.145	-0.140
	(0.049)	(0.049)		(0.049)	(0.049)		(0.244)	(0.245)		(0.110)	(0.109)
$SBAE_p \times FFD_s$	-0.014	-0.018		-0.025	-0.028		0.267	0.246		-0.017	-0.027
	(0.036)	(0.036)		(0.041)	(0.040)		(0.271)	(0.271)		(0.087)	(0.086)
$SBAE_p \times Session_s$	-0.023	0.030		-0.054	-0.004		0.184	0.448		-0.066	0.068
	(0.052)	(0.065)		(0.050)	(0.062)		(0.272)	(0.351)		(0.123)	(0.152)
$SBAE_p \times Session_s \times PVideo_h$		-0.062	-0.066		-0.046	-0.065		-0.046	-0.255		-0.117
		(0.056)	(0.050)		(0.054)	(0.048)		(0.416)	(0.407)		(0.134)
$SBAE_p \times Session_s \times Rev_h$		-0.076	-0.088*		-0.084*	-0.102**		-0.724**	-0.980**		-0.237**
		(0.049)	(0.048)		(0.046)	(0.046)		(0.347)	(0.396)		(0.116)
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		0.064	0.070		0.063	0.077		0.518	0.902		0.183
		(0.071)	(0.070)		(0.067)	(0.067)		(0.561)	(0.571)		(0.170)
Summary: Intent-to-treat effects of the revelation nudge											
Rev Effect given Basic Video		-0.076	-0.088*		-0.084*	-0.102**		-0.724**	-0.980**		-0.237**
		(0.049)	(0.048)		(0.046)	(0.046)		(0.347)	(0.396)		(0.116)
Rev Effect given Persuasive Video		-0.012	-0.017		-0.021	-0.025		-0.206	-0.077		-0.054
		(0.049)	(0.048)		(0.044)	(0.046)		(0.334)	(0.395)		(0.105)
Pooled Rev Effect		-0.044	-0.053		-0.052*	-0.063**		-0.465**	-0.528*		-0.146**
		(0.034)	(0.033)		(0.030)	(0.032)		(0.192)	(0.274)		(0.071)
Summary: Intent-to-treat effects of SBAE program (with video and FFD)											
SBAE Effect w/o Rev		-0.080*			-0.074*			0.046			-0.158
		(0.045)			(0.044)			(0.330)			(0.109)
SBAE Effect w/ Rev		-0.124***			-0.126***			-0.419			-0.304***
		(0.047)			(0.047)			(0.302)			(0.112)
Summary Statistics for Reference Groups											
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools
Reference group mean	0.30	0.30	0.29	0.26	0.26	0.26	5.05	5.05	5.15	0.16	0.16
Reference group SD	0.46	0.46	0.45	0.44	0.44	0.44	3.09	3.09	3.35	1.04	1.04
Observations	1384	1384	650	1384	1384	650	1384	1384	650	1384	1384

Notes: This table reports treatment effects on proxies for the use of subsistence practices on the designated `student parcel` during the rainy seasons of 2022 (midline; Panel A) and 2023 (endline; Panel B). The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. The third portion reports the intent-to-treat effects of the SBAE program (augmented with program promotional video and the annual farmer field day) (i) when the revelation nudge is absent; and (ii) when the revelation nudge is present. Again, these effects are computed using linear combinations of estimates. The reference regression specifications include pair-of-clusters fixed effects and standard errors are clustered at the cluster-of-schools level. The main regression specification includes school-grade section-gender fixed effects and robust standard errors are presented. Significance: *** p<0.01, ** p<0.05, * p<0.10

Table 7: Treatment effects on the use of subsistence practices on the designated `student parcel`
(continued)

	Panel B: During rainy season 2023 (endline)											
	Rice (0/1)			Intercropping (0/1)			# crops			First principal component (SD)		
Specification	Reference		Main	Reference		Main	Reference		Main	Reference		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(
$SBAE_p$	-0.006	-0.004		-0.038	-0.037		-0.188	-0.184		-0.068	-0.064	
	(0.037)	(0.037)		(0.041)	(0.041)		(0.196)	(0.198)		(0.082)	(0.082)	
$SBAE_p \times FFD_s$	0.033	0.031		0.043	0.042		0.252	0.252		0.108	0.105	
	(0.037)	(0.037)		(0.038)	(0.039)		(0.214)	(0.217)		(0.084)	(0.085)	
$SBAE_p \times Session_s$	-0.102***	-0.051		-0.061*	-0.029		-0.092	0.028		-0.166**	-0.071	
	(0.038)	(0.050)		(0.033)	(0.049)		(0.199)	(0.256)		(0.073)	(0.105)	
$SBAE_p \times Session_s \times PVideo_h$		-0.077	-0.056		-0.026	0.003		0.017	0.375		-0.094	
		(0.052)	(0.069)		(0.057)	(0.070)		(0.334)	(0.348)		(0.124)	(
$SBAE_p \times Session_s \times Rev_h$		0.001	-0.046		0.011	-0.019		-0.027	-0.115		0.007	
		(0.057)	(0.068)		(0.059)	(0.068)		(0.259)	(0.338)		(0.116)	(
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		-0.057	-0.097		-0.096	-0.127		-0.448	-0.961**		-0.209	-(
		(0.080)	(0.097)		(0.083)	(0.097)		(0.431)	(0.483)		(0.171)	(
Summary: Intent-to-treat effects of the revelation nudge												
Rev Effect given Basic Video		0.001	-0.046		0.011	-0.019		-0.027	-0.115		0.007	
		(0.057)	(0.068)		(0.059)	(0.068)		(0.259)	(0.338)		(0.116)	(
Rev Effect given Persuasive Video		-0.056	-0.144**		-0.086	-0.146**		-0.474	-1.076***		-0.202	-0.
		(0.048)	(0.067)		(0.061)	(0.067)		(0.332)	(0.335)		(0.129)	(
Pooled Rev Effect		-0.028	-0.095**		-0.037	-0.083*		-0.250	-0.600**		-0.098	-0
		(0.034)	(0.047)		(0.043)	(0.047)		(0.205)	(0.236)		(0.088)	(
Summary: Intent-to-treat effects of SBAE program (with video and FFD)												
SBAE Effect w/o Rev		-0.062*			-0.037			0.104			-0.077	
		(0.034)			(0.040)			(0.237)			(0.087)	
SBAE Effect w/ Rev		-0.090**			-0.074*			-0.146			-0.174*	
		(0.039)			(0.044)			(0.251)			(0.097)	
Summary Statistics for Reference Groups												
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	P se
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Ba Vi wi Re
Reference group mean	0.30	0.30	0.31	0.34	0.34	0.33	2.91	2.91	2.92	0.07	0.07	
Reference group SD	0.46	0.46	0.46	0.47	0.47	0.47	2.46	2.46	2.36	1.01	1.01	
Observations	1488	1488	396	1488	1488	396	1488	1488	396	1488	1488	

Notes: This table reports treatment effects on proxies for the use of subsistence practices on the designated `student parcel` during the rainy seasons of 2022 (midline; Panel A) and 2023 (endline; Panel B). The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. The third portion reports the intent-to-treat effects of the SBAE program (augmented with program promotional video and the annual farmer field day) (i) when the revelation nudge is absent; and (ii) when the revelation nudge is present. Again, these effects are computed using linear combinations of estimates. The

Table 7: Treatment effects on the use of subsistence practices on the designated `student parcel`
(continued)

	Panel C: Continuous cultivation in the RCT period					
	Across 2021 and 2022 (0/1)			Across 2022 and 2023 (0/1)		
Specification	Reference		Main	Reference		Main
	(1)	(2)	(3)	(4)	(5)	(6)
$SBAE_p$	-0.023	-0.024		-0.050	-0.051	
	(0.050)	(0.050)		(0.048)	(0.047)	
$SBAE_p \times FFD_s$	-0.008	-0.005		0.054	0.056	
	(0.044)	(0.045)		(0.042)	(0.042)	
$SBAE_p \times Session_s$	0.037	-0.016		0.032	0.009	
	(0.050)	(0.054)		(0.053)	(0.065)	
$SBAE_p \times Session_s \times PVideo_h$		0.031	0.019		0.047	-0.017
		(0.043)	(0.052)		(0.076)	(0.085)
$SBAE_p \times Session_s \times Rev_h$		0.097***	0.098*		-0.004	-0.079
		(0.034)	(0.051)		(0.072)	(0.083)
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		-0.051	-0.064		0.010	0.142
		(0.059)	(0.073)		(0.123)	(0.119)
	Summary: Intent-to-treat effects of the revelation nudge					
Rev Effect given Basic Video		0.097***	0.098*		-0.004	-0.079
		(0.034)	(0.051)		(0.072)	(0.083)
Rev Effect given Persuasive Video		0.047	0.034		0.006	0.063
		(0.047)	(0.051)		(0.078)	(0.082)
Pooled Rev Effect		0.072**	0.066*		0.001	-0.007
		(0.028)	(0.035)		(0.044)	(0.058)
	Summary: Intent-to-treat effects of SBAE program (with video and FFD)					
SBAE Effect w/o Rev		-0.031			0.037	
		(0.043)			(0.045)	
SBAE Effect w/ Rev		0.041			0.038	
		(0.044)			(0.043)	
	Summary Statistics for Reference Groups					
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev
Reference group mean	0.76	0.76	0.73	0.40	0.40	0.38
Reference group SD	0.43	0.43	0.45	0.49	0.49	0.49
Observations	1384	1384	650	1488	1488	396

Notes: This table reports treatment effects on proxies for the use of subsistence practices on the designated `student parcel` during the rainy seasons of 2022 (midline; Panel A) and 2023 (endline; Panel B). Panel C summarizes continuous cultivation outcomes over the RCT period. The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge; the third portion reports the intent-to-treat effects of the SBAE program (augmented with program promotional video and the annual farmer field day) with and without the revelation nudge. These

Table 8: Treatment effects on the use of sales-oriented practices the designated `student parcel` during rainy season 2022 (midline)

	Panel A: Average months to harvest root crops											
	All root crops: Average months			All root crops: Within 6 months (0/1)			Cassava: Average months			Cassava: Within 9 months (0/1)		
Specification	Reference		Main	Reference		Main	Reference		Main	Reference		Mai
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$SBAE_p$	-0.021	-0.014		-0.015	-0.016		0.068	0.074		-0.015	-0.017	
	(0.320)	(0.322)		(0.036)	(0.036)		(0.266)	(0.268)		(0.036)	(0.036)	
$SBAE_p \times FFD_s$	-0.120	-0.130		0.031	0.032		-0.130	-0.139		0.014	0.017	
	(0.282)	(0.285)		(0.039)	(0.039)		(0.206)	(0.210)		(0.028)	(0.028)	
$SBAE_p \times Session_s$	-0.052	0.236		0.006	-0.030		-0.294	-0.096		0.028	-0.014	
	(0.307)	(0.333)		(0.031)	(0.038)		(0.295)	(0.306)		(0.033)	(0.040)	
$SBAE_p \times Session_s \times PVideo_h$		-0.112	-0.044		-0.006	-0.015		-0.166	-0.265		0.044	0.04
		(0.212)	(0.304)		(0.039)	(0.045)		(0.196)	(0.247)		(0.032)	(0.04
$SBAE_p \times Session_s \times Rev_h$		-0.440*	-0.598**		0.068*	0.082*		-0.259	-0.464*		0.071**	0.102
		(0.259)	(0.296)		(0.039)	(0.044)		(0.203)	(0.241)		(0.033)	(0.04
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		-0.002	0.151		0.011	-0.004		0.080	0.419		-0.064	-0.10
		(0.313)	(0.427)		(0.054)	(0.064)		(0.262)	(0.346)		(0.054)	(0.05
Summary: Intent-to-treat effects of the revelation nudge												
Rev Effect given Basic Video		-0.440*	-0.598**		0.068*	0.082*		-0.259	-0.464*		0.071**	0.102
		(0.259)	(0.296)		(0.039)	(0.044)		(0.203)	(0.241)		(0.033)	(0.04
Rev Effect given Persuasive Video		-0.442*	-0.447		0.079*	0.079*		-0.179	-0.045		0.007	-0.00
		(0.260)	(0.295)		(0.043)	(0.044)		(0.212)	(0.240)		(0.038)	(0.04
Pooled Rev Effect		-0.441**	-0.522**		0.073**	0.080**		-0.219	-0.254		0.039	0.050
		(0.207)	(0.204)		(0.031)	(0.030)		(0.161)	(0.167)		(0.024)	(0.02
Summary: Intent-to-treat effects of SBAE program (with video and FFD)												
SBAE Effect w/o Rev		0.036			-0.016			-0.244			0.008	
		(0.326)			(0.041)			(0.267)			(0.036)	
SBAE Effect w/ Rev		-0.405			0.058			-0.463*			0.047	
		(0.290)			(0.037)			(0.241)			(0.033)	
Summary Statistics for Reference Groups												
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Prom sessio
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video witho Rev
Reference group mean | Cultivated	8.28	8.28	8.71	0.29	0.29	0.23	9.21	9.21	9.27	0.59	0.59	0.53
Reference group SD | Cultivated	3.44	3.44	2.97	0.46	0.46	0.42	3.36	3.36	2.64	0.49	0.49	0.50
Observations	1384	1384	650	1384	1384	650	1384	1384	650	1384	1384	650

Notes: This table reports treatment effects on proxies for the use of sales-oriented practices on the designated `student parcel` during the rainy season of 2022 (midline). Panel A reports the average months it takes for root crops and cassava to be harvested. The top portion reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge; the third portion reports the intent-to-treat effects of the SBAE program (augmented with program promotional video and the annual farmer field day) with and without the revelation nudge. These effects are computed using linear combinations of estimates. All regressions include controls for whether the corresponding crops are cultivated. The reference regression specifications include pair-of-clusters fixed effects and standard

Table 8: Treatment effects on the use of sales-oriented practices the designated `student parcel` during rainy season 2022 (midline) *(continued)*

	Panel B: SBAE-promoted, sales-oriented methods for cassava: adoption and impact								
	Students' midline reports			Parents' endline reports					
	One stick per hole (0/1)			Observed bigger tubers on `student parcel` (0/1)			Observed more tubers on `student parcel` (0/1)		
Specification	Reference		Main	Reference		Main	Reference		Main
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$SBAE_p$	0.008	0.006		-0.003	-0.002		0.030	0.030	
	(0.017)	(0.016)		(0.048)	(0.048)		(0.038)	(0.038)	
$SBAE_p \times FFD_s$	0.004	0.009		0.088**	0.089**		-0.013	-0.013	
	(0.021)	(0.020)		(0.041)	(0.041)		(0.048)	(0.048)	
$SBAE_p \times Session_s$	0.004	-0.036*		0.051	-0.003		0.090*	0.079	
	(0.013)	(0.019)		(0.053)	(0.059)		(0.049)	(0.059)	
$SBAE_p \times Session_s \times PVideo_h$		0.072***	0.047*		0.087*	0.086*		0.030	0.042
		(0.023)	(0.025)		(0.049)	(0.049)		(0.050)	(0.049)
$SBAE_p \times Session_s \times Rev_h$		0.069**	0.056**		0.070	0.101**		-0.011	0.009
		(0.026)	(0.024)		(0.045)	(0.048)		(0.050)	(0.048)
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		-0.122***	-0.084**		-0.097	-0.093		0.008	-0.028
		(0.033)	(0.035)		(0.064)	(0.070)		(0.069)	(0.070)
Additional Controls				Observed on own parcels	Observed on own parcels	Observed on own parcels	Observed on own parcels	Observed on own parcels	Observed on own parcels
Summary: Intent-to-treat effects of the revelation nudge									
Rev Effect given Basic Video		0.069**	0.056**		0.070	0.101**		-0.011	0.009
		(0.026)	(0.024)		(0.045)	(0.048)		(0.050)	(0.048)
Rev Effect given Persuasive Video		-0.053**	-0.027		-0.027*	0.008		-0.003	-0.019
		(0.020)	(0.024)		(0.050)	(0.050)		(0.043)	(0.050)
Pooled Rev Effect		0.008	0.014		0.022	0.057		-0.007	-0.005
		(0.016)	(0.017)		(0.035)	(0.034)		(0.031)	(0.034)
Summary: Intent-to-treat effects of SBAE program (with video and FFD)									
SBAE Effect w/o Rev		0.016			0.127**			0.110**	
		(0.019)			(0.051)			(0.043)	
SBAE Effect w/ Rev		0.024			0.149**			0.103**	
		(0.019)			(0.052)			(0.044)	
Summary Statistics for Reference Groups									
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev
Reference group mean | Cultivated	0.05	0.05	0.01	0.43	0.43	0.44	0.50	0.50	0.61
Reference group SD | Cultivated	0.22	0.22	0.11	0.50	0.50	0.50	0.50	0.50	0.49
Observations	1384	1384	650	1543	1543	786	1543	1543	786

Notes: This table reports treatment effects on proxies for the use of sales-oriented practices on the designated `student parcel` during the rainy season of 2022 (midline). Panel B reports measures that capture whether SBAE-promoted methods for cassava are used and how this affects parents' observations of crop performance. The top

Table 8: Treatment effects on the use of sales-oriented practices the designated `student parcel` during rainy season 2022 (midline) *(continued)*

	Panel C: Cultivation of short-cycle root crops											
	All root crops: Minimum months to harvest			All root crops: Within 6 months (0/1)			All root crops: Average months to harvest			All root crops: Within 6 months (0/1)		
Specification	Reference		Main	Reference		Main	Reference		Main	Reference		M
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(1
$SBAE_p$	0.150	0.154		-0.077*	-0.078*		-0.151	-0.148		0.041*	0.041*	
	(0.335)	(0.336)		(0.039)	(0.039)		(0.099)	(0.101)		(0.021)	(0.022)	
$SBAE_p \times FFD_s$	-0.312	-0.313		0.083**	0.082**		0.150*	0.140*		-0.029	-0.028	
	(0.283)	(0.287)		(0.038)	(0.038)		(0.079)	(0.080)		(0.020)	(0.020)	
$SBAE_p \times Session_s$	-0.176	0.020		0.041	0.016		0.101	0.219		-0.023	-0.041	
	(0.286)	(0.342)		(0.032)	(0.041)		(0.138)	(0.152)		(0.020)	(0.026)	
$SBAE_p \times Session_s \times PVideo_h$		-0.025	0.076		-0.014	-0.021		-0.090	-0.108		0.004	0.
		(0.236)	(0.332)		(0.041)	(0.048)		(0.107)	(0.118)		(0.028)	(0
$SBAE_p \times Session_s \times Rev_h$		-0.216	-0.273		0.035	0.031		-0.253**	-0.368***		0.042*	0.0
		(0.283)	(0.323)		(0.041)	(0.047)		(0.095)	(0.115)		(0.023)	(0
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		-0.262	-0.216		0.050	0.057		0.225	0.334**		-0.025	-0
		(0.389)	(0.467)		(0.063)	(0.067)		(0.142)	(0.166)		(0.033)	(0
Additional Controls							Min. months	Min. months	Min. months	Min. months within 6	Min. months within 6	Mi mo wit
	Summary: Intent-to-treat effects of the revelation nudge											
Rev Effect given Basic Video		-0.216	-0.273		0.035	0.031		-0.253**	-0.368***		0.042*	0.0
		(0.283)	(0.323)		(0.041)	(0.047)		(0.095)	(0.115)		(0.023)	(0
Rev Effect given Persuasive Video		-0.479	-0.489		0.085*	0.088*		-0.028	-0.034		0.017	0.
		(0.311)	(0.323)		(0.048)	(0.046)		(0.101)	(0.115)		(0.024)	(0
Pooled Rev Effect		-0.348	-0.381*		0.060*	0.059*		-0.141**	-0.201**		0.030*	0.0
		(0.225)	(0.223)		(0.032)	(0.032)		(0.068)	(0.080)		(0.017)	(0
	Summary: Intent-to-treat effects of SBAE program (with video and FFD)											
SBAE Effect w/o Rev		-0.151			0.014			0.166			-0.026	
		(0.334)			(0.043)			(0.116)			(0.023)	
SBAE Effect w/ Rev		-0.498			0.074*			0.025			0.004	
		(0.298)			(0.039)			(0.106)			(0.023)	
	Summary Statistics for Reference Groups											
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Pr ses
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Ba Vi wit Re
Reference group mean | Cultivated	7.72	7.72	7.91	0.42	0.42	0.37	8.28	8.28	8.71	0.29	0.29	0
Reference group SD | Cultivated	3.67	3.67	3.44	0.49	0.49	0.48	3.44	3.44	2.97	0.46	0.46	0
Observations	1384	1384	650	1384	1384	650	1384	1384	650	1384	1384	6

Notes: This table reports treatment effects on proxies for the use of sales-oriented practices on the designated `student parcel` during the rainy season of 2022 (midline).

Table 9: Treatment effects on students' income and savings in 2022 (retrospective reports in endline; Liberian dollars)

Specification	Log (1 + Farm sales)			Log (1 + Non-farm income)			Log (1 + Total income)			Log (1 + Total savings)		
	Reference		Main	Reference		Main	Reference		Main	Reference		Main
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$SBAE_p$	0.229	0.222		-0.490	-0.494*		0.135	0.128		-0.171	-0.169	
	(0.346)	(0.338)		(0.299)	(0.295)		(0.310)	(0.298)		(0.357)	(0.357)	
$SBAE_p \times FFD_s$	-0.109	-0.091		0.605	0.618		-0.009	0.014		0.541	0.550	
	(0.488)	(0.469)		(0.415)	(0.405)		(0.482)	(0.458)		(0.432)	(0.434)	
$SBAE_p \times Session_s$	0.565*	0.189		-0.489	-0.585		-0.022	-0.534		0.550	0.348	
	(0.322)	(0.405)		(0.319)	(0.457)		(0.270)	(0.435)		(0.477)	(0.620)	
$SBAE_p \times Session_s \times PVideo_h$		0.587	0.082		0.481	0.780		0.743	0.428		0.231	-0.088
		(0.532)	(0.457)		(0.565)	(0.573)		(0.519)	(0.502)		(0.586)	(0.582)
$SBAE_p \times Session_s \times Rev_h$		0.590*	0.479		0.129	0.495		0.919**	1.208**		0.614	1.130*
		(0.339)	(0.399)		(0.473)	(0.527)		(0.366)	(0.463)		(0.555)	(0.613)
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		-0.836	-0.149		-0.787	-1.164*		-1.268**	-1.148*		-0.897	-1.014
		(0.654)	(0.589)		(0.574)	(0.664)		(0.576)	(0.624)		(0.894)	(0.928)
Summary: Intent-to-treat effects of the revelation nudge												
Rev Effect given Basic Video		0.590*	0.479		0.129	0.495		0.919**	1.208**		0.614	1.130*
		(0.339)	(0.399)		(0.473)	(0.527)		(0.366)	(0.463)		(0.555)	(0.613)
Rev Effect given Persuasive Video		-0.245	0.331		-0.658	-0.669		-0.350	0.060		-0.282	0.116
		(0.466)	(0.403)		(0.470)	(0.538)		(0.352)	(0.349)		(0.580)	(0.628)
Pooled Rev Effect		0.172	0.404		-0.265	-0.092		0.285	0.629**		0.166	0.619
		(0.243)	(0.272)		(0.374)	(0.422)		(0.215)	(0.269)		(0.350)	(0.408)
Summary: Intent-to-treat effects of SBAE program (with video and FFD)												
SBAE Effect w/o Rev		0.614*			-0.221			-0.020			0.844**	
		(0.333)			(0.420)			(0.336)			(0.358)	
SBAE Effect w/ Rev		0.786**			-0.486			0.264			1.011**	
		(0.382)			(0.394)			(0.320)			(0.395)	
Summary Statistics for Reference Groups												
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev
Reference group mean	3,407	3,407	4,524	1,860	1,860	1,499	5,367	5,367	6,070	2,952	2,952	3,429
Reference group SD	4,491	4,491	5,591	3,272	3,272	3,595	6,266	6,266	7,584	4,237	4,237	4,534
Observations	1488	1488	95	1488	1488	396	1488	1488	396	1488	1488	396

Notes: This table reports treatment effects on students' income and savings in the year of 2022, retrospectively reported in the endline survey. The top portion reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge; the third portion reports the intent-to-treat effects of the SBAE program (augmented with program promotional video and the annual farmer field day) with and without the revelation nudge. These effects are computed using linear combinations of estimates. All regressions include controls for whether the corresponding crops are cultivated. The reference regression specifications include pair-of-clusters fixed effects and standard errors are clustered at the cluster-of-schools level. The main regression specification includes school-grade section-gender fixed effects and robust standard errors are presented. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 10: Treatment effects on students' school attendance

	Panel A: School enrollment and days in attendance								
	Midline			**Endline**			**Imputed**		
	Enrolled in school (0/1)			Enrolled in school (0/1)			Total days in attendance		
Specification	Reference		Main	Reference		Main	Reference		Main
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$SBAE_p$	0.008	0.008		-0.001	-0.001		-3.572	-3.714	
	(0.023)	(0.023)		(0.028)	(0.027)		(7.964)	(7.909)	
$SBAE_p \times FFD_s$	-0.007	-0.007		-0.011	-0.010		1.957	2.076	
	(0.023)	(0.023)		(0.031)	(0.030)		(9.505)	(9.371)	
$SBAE_p \times Session_s$	0.037	0.045		0.050*	0.049		19.198**	16.477	
	(0.025)	(0.030)		(0.028)	(0.042)		(9.075)	(11.645)	
$SBAE_p \times Session_s \times PVideo_h$		-0.028	-0.029		-0.032	-0.063		-6.465	-7.517
		(0.034)	(0.038)		(0.047)	(0.049)		(11.611)	(13.238)
$SBAE_p \times Session_s \times Rev_h$		-0.029	-0.008		-0.007	-0.012		-3.017	0.529
		(0.031)	(0.037)		(0.051)	(0.047)		(12.334)	(12.891)
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		0.086*	0.074		0.080	0.109		29.766*	29.215
		(0.045)	(0.053)		(0.062)	(0.069)		(17.854)	(18.799)
	Summary: Intent-to-treat effects of the revelation nudge								
Rev Effect given Basic Video		-0.029	-0.008		-0.007	-0.012		-3.017	0.529
		(0.031)	(0.037)		(0.051)	(0.047)		(12.334)	(12.891)
Rev Effect given Persuasive Video		0.056*	0.065*		0.073*	0.097**		26.749**	29.745**
		(0.031)	(0.038)		(0.041)	(0.049)		(11.616)	(13.308)
Pooled Rev Effect		0.014	0.027		0.033	0.041		11.866	14.685
		(0.022)	(0.026)		(0.034)	(0.033)		(7.989)	(9.130)
	Summary: Intent-to-treat effects of SBAE program (with video and FFD)								
SBAE Effect w/o Rev		0.038*			0.033			11.606	
		(0.022)			(0.028)			(9.145)	
SBAE Effect w/ Rev		0.052**			0.065**			23.472**	
		(0.021)			(0.031)			(9.984)	
	Summary Statistics for Reference Groups								
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev
Reference group mean	0.78	0.78	0.82	0.67	0.67	0.71	229.09	229.09	240.58
Reference group SD	0.42	0.42	0.39	0.47	0.47	0.46	144.02	144.02	137.39
Observations	3728	3728	990	3115	3115	834	3728	3728	990

Notes: This table reports treatment effects on students' school attendance in the midline and endline surveys. Panel A reports whether students remain enrolled during the midline (2021/22) and endline (2022/23) surveys and an imputed number of school days in 2021/22 to 2022/23. This imputed measure is based on students' enrollment status, (measured) rate of absence from school, and a total number of 200 school days per school year. A student who drop outs from school is counted as having zero school days. A student who switches schools is counted as having the same number of school days as control school average. The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. The third portion reports the intent-to-treat effects of the SBAE program

Table 10: Treatment effects on students' school attendance *(continued)*

	Panel B: Timely payment of school fees											
	Midline Semester 2 (0/1)			**Endline** Semester 1 (0/1)			**Endline** Semester 2 (0/1)			**Endline** # semesters		
Specification	Reference		Main	Reference		Main	Reference		Main	Reference		Main
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$SBAE_p$	0.021	0.021		0.055**	0.054**		0.067**	0.067**		0.122***	0.121***	
	(0.032)	(0.032)		(0.026)	(0.025)		(0.032)	(0.032)		(0.044)	(0.044)	
$SBAE_p \times FFD_s$	-0.037	-0.038		-0.046*	-0.045*		-0.084***	-0.084***		-0.130***	-0.129***	
	(0.030)	(0.030)		(0.026)	(0.025)		(0.030)	(0.030)		(0.045)	(0.045)	
$SBAE_p \times Session_s$	-0.029	-0.032		-0.029	-0.055		-0.016	-0.020		-0.045	-0.075	
	(0.038)	(0.050)		(0.031)	(0.045)		(0.036)	(0.051)		(0.054)	(0.077)	
$SBAE_p \times Session_s \times PVideo_h$		-0.034	-0.045		-0.006	-0.017		-0.077	-0.074		-0.083	-0.091
		(0.057)	(0.060)		(0.047)	(0.058)		(0.057)	(0.062)		(0.093)	(0.102)
$SBAE_p \times Session_s \times Rev_h$		-0.028	0.005		0.045	0.066		0.037	0.054		0.082	0.120
		(0.050)	(0.058)		(0.042)	(0.055)		(0.049)	(0.058)		(0.070)	(0.096)
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		0.128*	0.117		0.020	0.057		0.083	0.122		0.103	0.179
		(0.077)	(0.084)		(0.062)	(0.081)		(0.070)	(0.086)		(0.109)	(0.142)
Summary: Intent-to-treat effects of the revelation nudge												
Rev Effect given Basic Video		-0.028	0.005		0.045	0.066		0.037	0.054		0.082	0.120
		(0.050)	(0.058)		(0.042)	(0.055)		(0.049)	(0.058)		(0.070)	(0.096)
Rev Effect given Persuasive Video		0.100*	0.121**		0.065	0.122**		0.119**	0.177***		0.184**	0.299***
		(0.057)	(0.059)		(0.041)	(0.057)		(0.054)	(0.061)		(0.081)	(0.100)
Pooled Rev Effect		0.036	0.062		0.055**	0.093**		0.078**	0.113***		0.133**	0.206***
		(0.037)	(0.040)		(0.027)	(0.039)		(0.038)	(0.041)		(0.052)	(0.068)
Summary: Intent-to-treat effects of SBAE program (with video and FFD)												
SBAE Effect w/o Rev		-0.028			-0.004			0.009			0.005	
		(0.037)			(0.030)			(0.034)			(0.050)	
SBAE Effect w/ Rev		0.008			0.051*			0.087**			0.138**	
		(0.030)			(0.029)			(0.034)			(0.056)	
Summary Statistics for Reference Groups												
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev
Reference group mean	0.35	0.35	0.35	0.20	0.20	0.20	0.24	0.24	0.26	0.44	0.44	0.47
Reference group SD	0.48	0.48	0.48	0.40	0.40	0.40	0.43	0.43	0.44	0.70	0.70	0.70
Observations	2434	2434	666	1970	1970	530	1970	1970	530	1970	1970	530

Notes: This table reports treatment effects on students' school attendance in the midline and endline surveys. Panel B reports whether students' school fees are paid on time at various points of time. The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. The third portion reports the intent-to-treat effects of the SBAE program (augmented with program promotional video and the annual farmer field day) (i) when the revelation nudge is absent; and (ii) when the revelation nudge is present. Again, these effects are computed using linear combinations of estimates. The reference regression specifications include pair-of-clusters fixed effects and standard errors are clustered at the cluster-of-schools level. The main regression specification includes school-grade section-gender fixed effects and robust standard errors are presented. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 11: Process information about changes in the management of household farms

	Look after (0/1)	Look after (0/1)	Choose crops: involved (0/1)	Choose crops: involved (0/1)	Choose techniques: involved (0/1)	Choose techniques: involved (0/1)
	Panel A: Changes in management of the designated `student parcel` (midline)					
	Treated parents	Other household members	Treated parents	Other household members	Treated parents	Other household members
	(1)	(2)	(3)	(4)	(5)	(6)
$PVideo_h$	-0.140**	-0.017	-0.103*	0.034	-0.034	0.023
	(0.059)	(0.057)	(0.062)	(0.048)	(0.061)	(0.046)
Rev_h	-0.128**	0.006	-0.184***	0.000	-0.030	0.016
	(0.057)	(0.056)	(0.060)	(0.046)	(0.058)	(0.045)
$PVideo_h \times Rev_h$	0.190**	-0.008	0.244***	-0.047	0.095	-0.047
	(0.083)	(0.080)	(0.087)	(0.067)	(0.085)	(0.065)
	Summary: Intent-to-treat effects of the revelation nudge					
Rev Effect given Basic Video	-0.128**	0.006	-0.184***	0.000	-0.030	0.016
	(0.057)	(0.056)	(0.060)	(0.046)	(0.058)	(0.045)
Rev Effect given Persuasive Video	0.062	-0.001	0.059	-0.046	0.065	-0.031
	(0.057)	(0.056)	(0.061)	(0.047)	(0.059)	(0.045)
Pooled Rev Effect	-0.033	0.002	-0.064	-0.023	0.017	-0.007
	(0.040)	(0.038)	(0.042)	(0.032)	(0.041)	(0.031)
	Summary Statistics for Reference Groups					
Sample	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session
Reference group	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev
Reference group mean	0.40	0.29	0.62	0.58	0.49	0.47
Reference group SD	0.49	0.46	0.49	0.39	0.50	0.42
Observations	650	650	650	650	650	650

Notes: This table reports process information about changes in the management of household farms. Panel A reports midline changes in the involvement of treated parents and other household members on the designated `student parcel`. The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. All regressions include school-grade section-gender fixed effects and robust standard errors are presented. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 11: Process information about changes in the management of household farms *(continued)*

	Panel B: Treated parents' retrospective reports of requests from students (endline)											
	Look after farms (0/1)			Plant particular crops (0/1)			Change intercropping (0/1)			Change fallowing (0/1)		
Specification	Reference		Main	Reference		Main	Reference		Main	Reference		M
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(1
$SBAE_p$	0.122***	0.122***		0.001	0.001		0.040	0.040		0.000	0.000	
	(0.044)	(0.045)		(0.043)	(0.043)		(0.048)	(0.048)		(0.049)	(0.050)	
$SBAE_p \times FFD_s$	-0.049	-0.049		-0.021	-0.020		0.002	0.002		0.021	0.020	
	(0.048)	(0.048)		(0.039)	(0.039)		(0.042)	(0.042)		(0.049)	(0.049)	
$SBAE_p \times Session_s$	0.022	-0.002		0.020	0.039		0.103*	0.078		0.143**	0.159**	
	(0.041)	(0.050)		(0.047)	(0.062)		(0.055)	(0.063)		(0.067)	(0.073)	
$SBAE_p \times Session_s \times PVideo_h$		-0.006	0.012		-0.006	-0.000		0.024	0.056		-0.042	-0
		(0.042)	(0.053)		(0.051)	(0.056)		(0.052)	(0.054)		(0.046)	(0
$SBAE_p \times Session_s \times Rev_h$		0.039	0.067		-0.015	-0.004		0.037	0.041		-0.035	-0
		(0.051)	(0.052)		(0.059)	(0.054)		(0.054)	(0.053)		(0.046)	(0
$SBAE_p \times Session_s \times PVideo_h \times Rev_h$		0.031	0.034		-0.031	-0.011		-0.020	-0.035		0.087	0.
		(0.067)	(0.076)		(0.085)	(0.080)		(0.069)	(0.077)		(0.075)	(0
Summary: Intent-to-treat effects of the revelation nudge												
Rev Effect given Basic Video		0.039	0.067		-0.015	-0.004		0.037	0.041		-0.035	-0
		(0.051)	(0.052)		(0.059)	(0.054)		(0.054)	(0.053)		(0.046)	(0
Rev Effect given Persuasive Video		0.070	0.101*		-0.046	-0.015		0.016	0.006		0.052	0.
		(0.045)	(0.054)		(0.060)	(0.057)		(0.048)	(0.055)		(0.053)	(0
Pooled Rev Effect		0.054	0.083**		-0.031	-0.010		0.026	0.024		0.008	0.
		(0.034)	(0.037)		(0.041)	(0.039)		(0.038)	(0.038)		(0.033)	(0
Summary: Intent-to-treat effects of SBAE program (with video and FFD)												
SBAE Effect w/o Rev		0.068			0.016			0.132**			0.159***	
		(0.049)			(0.051)			(0.052)			(0.056)	
SBAE Effect w/ Rev		0.122**			-0.015			0.158***			0.167***	
		(0.052)			(0.048)			(0.052)			(0.052)	
Summary Statistics for Reference Groups												
Sample	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Promo session	Full sample	Full sample	Pr ses
Reference group	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Basic Video without Rev	Control Schools	Control Schools	Ba Vi wit Re
Reference group mean	0.50	0.50	0.60	0.54	0.54	0.57	0.40	0.40	0.50	0.28	0.28	0
Reference group SD	0.50	0.50	0.49	0.50	0.50	0.50	0.49	0.49	0.50	0.45	0.45	0
Observations	1543	1543	786	1543	1543	786	1543	1543	786	1543	1543	7

Notes: This table reports process information about changes in the management of household farms. Panel B reports treated parents' endline retrospective reports on whether students make various requests in the past two years. The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. The third portion reports the intent-to-treat effects of the SBAE program (augmented with program promotional video and the annual farmer field day) (i) when the revelation nudge is absent; and (ii) when the revelation nudge is present. Again, these effects are computed using linear combinations of estimates. The reference regression specifications include pair-of-clusters fixed effects and standard errors are clustered at the cluster-of-schools level. The main regression specification includes school-grade section-gender fixed effects and robust standard errors are presented. Significance: *** p<0.01, ** p<0.05, * p<0.10

Table 12: Use of subsistence practices on the designated `student parcel`, by management status

	Panel A: During rainy season 2022 (midline)											
	Rice (0/1)			Intercropping (0/1)			# crops			First principal component (SD)		
	Student-managed	Treated-parent-managed	Other-managed	Student-managed	Treated-parent-managed	Other-managed	Student-managed	Treated-parent-managed	Other-managed	Student-managed	Treated-parent-managed	Other-managed
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$PVideo_h$	0.000	-0.033	-0.033	-0.002	-0.043	-0.020	0.447	-0.706*	0.004	0.045	-0.166	-0.080
	(0.024)	(0.036)	(0.036)	(0.024)	(0.034)	(0.034)	(0.326)	(0.390)	(0.383)	(0.135)	(0.111)	(0.138)
Rev_h	-0.018	-0.063*	-0.007	-0.018	-0.074**	-0.009	0.200	-0.909**	-0.271	-0.070	-0.260**	-0.057
	(0.023)	(0.035)	(0.035)	(0.023)	(0.033)	(0.033)	(0.316)	(0.379)	(0.372)	(0.132)	(0.108)	(0.134)
$PVideo_h \times Rev_h$	0.005	0.061	0.004	0.016	0.064	-0.003	-0.285	1.279**	-0.092	0.024	0.282*	-0.010
	(0.034)	(0.050)	(0.050)	(0.033)	(0.048)	(0.047)	(0.457)	(0.548)	(0.537)	(0.190)	(0.156)	(0.194)
Summary: Intent-to-treat effects of the revelation nudge												
Rev Effect given Basic Video	-0.018	-0.063*	-0.007	-0.018	-0.074**	-0.009	0.200	-0.909**	-0.271	-0.070	-0.260**	-0.057
	(0.023)	(0.035)	(0.035)	(0.023)	(0.033)	(0.033)	(0.316)	(0.379)	(0.372)	(0.132)	(0.108)	(0.134)
Rev Effect given Persuasive Video	-0.013	-0.002	-0.003	-0.002	-0.010	-0.012	-0.085	0.370	-0.363	-0.046	0.023	-0.067
	(0.023)	(0.035)	(0.035)	(0.023)	(0.033)	(0.033)	(0.316)	(0.379)	(0.372)	(0.131)	(0.108)	(0.134)
Pooled Rev Effect	-0.015	-0.032	-0.005	-0.010	-0.042*	-0.011	0.058	-0.269	-0.317	-0.058	-0.118	-0.062
	(0.016)	(0.024)	(0.024)	(0.016)	(0.023)	(0.023)	(0.218)	(0.263)	(0.257)	(0.091)	(0.075)	(0.093)
Summary Statistics for Reference Groups												
Sample	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session
Reference group	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev
Reference group mean	0.05	0.14	0.10	0.05	0.13	0.08	1.38	2.22	1.55	0.07	0.07	0.09
Reference group SD	0.22	0.34	0.30	0.21	0.34	0.28	2.75	3.32	3.17	1.25	1.08	1.13
Observations	650	650	650	650	650	650	650	650	650	650	650	650

Notes: This table reports the use of subsistence practices on the designated `student parcel` during the rainy seasons of 2022 (midline; Panel A) and 2023 (endline; Panel B), with separate measures for `student parcel` looked after by students, treated parents, and other household members. The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. All regressions include school-grade section-gender fixed effects and robust standard errors are presented. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 12: Use of subsistence practices on the designated `student parcel`, by management status
(continued)

	Panel B: During rainy season 2023 (endline)											
	Rice (0/1)			Intercropping (0/1)			# crops			First principal component (SD)		
	Student-managed	Treated-parent-managed	Other-managed	Student-managed	Treated-parent-managed	Other-managed	Student-managed	Treated-parent-managed	Other-managed	Student-managed	Treated-parent-managed	Other-managed
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$PVideo_h$	-0.034	-0.016	0.018	-0.053	0.031	0.021	0.023	0.375	-0.137	-0.152	0.092	0.037
	(0.032)	(0.043)	(0.032)	(0.036)	(0.046)	(0.035)	(0.289)	(0.285)	(0.197)	(0.149)	(0.154)	(0.165)
Rev_h	-0.001	0.012	0.016	-0.032	0.072	0.002	0.078	0.287	-0.175	-0.040	0.164	-0.014
	(0.032)	(0.042)	(0.031)	(0.035)	(0.044)	(0.034)	(0.281)	(0.277)	(0.191)	(0.145)	(0.149)	(0.161)
$PVideo_h \times Rev_h$	-0.019	-0.056	-0.033	0.020	-0.114*	-0.034	0.054	-0.902**	0.121	0.008	-0.397*	-0.094
	(0.045)	(0.060)	(0.044)	(0.050)	(0.063)	(0.048)	(0.402)	(0.396)	(0.273)	(0.208)	(0.214)	(0.229)
Summary: Intent-to-treat effects of the revelation nudge												
Rev Effect given Basic Video	-0.001	0.012	0.016	-0.032	0.072	0.002	0.078	0.287	-0.175	-0.040	0.164	-0.014
	(0.032)	(0.042)	(0.031)	(0.035)	(0.044)	(0.034)	(0.281)	(0.277)	(0.191)	(0.145)	(0.149)	(0.161)
Rev Effect given Persuasive Video	-0.020	-0.045	-0.017	-0.012	-0.042	-0.032	0.132	-0.615**	-0.054	-0.032	-0.233	-0.108
	(0.031)	(0.042)	(0.031)	(0.035)	(0.044)	(0.033)	(0.279)	(0.275)	(0.189)	(0.144)	(0.148)	(0.159)
Pooled Rev Effect	-0.010	-0.017	-0.001	-0.022	0.015	-0.015	0.105	-0.168	-0.114	-0.036	-0.036	-0.061
	(0.022)	(0.029)	(0.021)	(0.024)	(0.031)	(0.023)	(0.195)	(0.193)	(0.132)	(0.100)	(0.104)	(0.111)
Summary Statistics for Reference Groups												
Sample	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session
Reference group	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev
Reference group mean	0.04	0.09	0.04	0.07	0.08	0.05	1.07	0.74	0.41	0.01	-0.02	0.03
Reference group SD	0.20	0.29	0.20	0.26	0.28	0.22	1.66	1.97	1.42	0.90	1.05	1.09
Observations	396	396	396	396	396	396	396	396	396	396	396	396

Notes: This table reports the use of subsistence practices on the designated `student parcel` during the rainy seasons of 2022 (midline; Panel A) and 2023 (endline; Panel B), with separate measures for `student parcel` looked after by students, treated parents, and other household members. The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. All regressions include school-grade section-gender fixed effects and robust standard errors are presented. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 13: Evidence that video-treated parents are genuinely open to delegating farm tasks

	Panel A: Treated parents are persistently optimistic about students' cassava harvests							
	Midline		Endline		Endline		Endline	
	Harvested tubers from 4 cassava plants		Harvested tubers from 4 cassava plants		Students' tubers exceed adults' (0/1)		Students' tubers exceed adults': due to practices (0/1)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$SBAE_s$			1.452***		0.073**		0.086***	
			(0.496)		(0.029)		(0.026)	
$SESSION_s$		1.149**		0.851		-0.001		-0.007
		(0.514)		(0.615)		(0.033)		(0.032)
	Summary Statistics for Reference Groups							
Sample	SBAE and Control Schools	SBAE schools only	SBAE and Control Schools	SBAE schools only	SBAE and Control Schools	SBAE schools only	SBAE and Control Schools	SBAE schools only
Reference group	Control Schools	SBAE schools without Video	Control Schools	SBAE schools without Video	Control Schools	SBAE schools without Video	Control Schools	SBAE schools without Video
Reference group mean		21.54	19.87	20.48	0.35	0.42	0.31	0.39
Reference group SD		8.90	9.87	9.70	0.48	0.50	0.46	0.49
Observations		650	1543	1132	1543	1132	1543	1132

Notes: This table reports evidence that video-treated parents are genuinely open to delegating farm tasks to students. Panel A shows that SBAE-treated parents and video-treated parents are persistently more optimistic about students' harvests than reference counterparts. The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. All regressions include school-grade section-gender fixed effects and robust standard errors are presented. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 13: Evidence that parents are genuinely open to delegating farm tasks *(continued)*

	Panel B: Correlation between parents' beliefs elicited after videos and responses to SBAE (midline)						
	Visited school farms (0/1)	# visits in 2021	Observed harvest on school farms (0/1)	Worked on school farms (0/1)	Met Club Advisor (0/1)	Correctly name Field Officer (0/1)	# PTA meetings
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Beliefs about tubers: students' - adults' (SD)	0.061***	0.136***	0.038***	0.055***	0.042***	0.027**	0.091*
	(0.017)	(0.031)	(0.013)	(0.013)	(0.016)	(0.011)	(0.051)
$PVideo_h$	0.055*	0.097*	0.010	0.030	0.012	0.028	0.110
	(0.032)	(0.058)	(0.025)	(0.024)	(0.030)	(0.021)	(0.097)
Rev_h	-0.040	0.002	0.004	0.009	0.004	0.011	0.119
	(0.032)	(0.058)	(0.025)	(0.024)	(0.030)	(0.021)	(0.097)
	Summary Statistics for Reference Groups						
Sample	Households with video	Households with video	Households with video	Households with video	Households with video	Households with video	Households with video
Reference group	Basic video without revelation	Basic video without revelation	Basic video without revelation	Basic video without revelation	Basic video without revelation	Basic video without revelation	Basic video without revelation
Reference group mean	0.43	0.61	0.17	0.18	0.35	0.11	1.01
Reference group SD	0.50	0.86	0.38	0.39	0.48	0.31	1.49
Observations	1001	1001	1001	1001	1001	1001	1001

Notes: This table reports evidence that video-treated parents are genuinely open to delegating farm tasks to students. Panel B shows that conditional on households' treatment status, parents' beliefs about students' harvests (relative to adults') strongly predict midline responses to SBAE. The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. All regressions include school-grade section-gender fixed effects and robust standard errors are presented. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 14: Midline use of subsistence practices on the designated `student parcel`, by parcel control rights

	Student-managed: # crops			Parent-managed: # crops			Other-managed: # crops		
	Treated-parent-controlled	Household-controlled	Externally-controlled	Treated-parent-controlled	Household-controlled	Externally-controlled	Treated-parent-controlled	Household-controlled	Externally-controlled
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$PVideo_h$	0.431**	0.493*	-0.166	-0.068	-0.184	-0.482	-0.108	-0.034	0.130
	(0.209)	(0.263)	(0.231)	(0.239)	(0.248)	(0.325)	(0.139)	(0.219)	(0.320)
Rev_h	-0.089	0.191	0.077	-0.181	-0.251	-0.637**	-0.045	-0.009	-0.019
	(0.203)	(0.256)	(0.225)	(0.232)	(0.241)	(0.315)	(0.135)	(0.212)	(0.310)
$PVideo_h \times Rev_h$	-0.019	-0.471	0.225	0.007	0.235	1.013**	-0.008	0.078	-0.452
	(0.293)	(0.369)	(0.324)	(0.335)	(0.348)	(0.455)	(0.194)	(0.307)	(0.448)
Summary: Intent-to-treat effects of the revelation nudge									
Rev Effect given Basic Video	-0.089	0.191	0.077	-0.181	-0.251	-0.637**	-0.045	-0.009	-0.019
	(0.203)	(0.256)	(0.225)	(0.232)	(0.241)	(0.315)	(0.135)	(0.212)	(0.310)
Rev Effect given Persuasive Video	-0.108	-0.280	0.301	-0.175	-0.016	0.376	-0.053	0.070	-0.471
	(0.203)	(0.256)	(0.224)	(0.232)	(0.241)	(0.315)	(0.134)	(0.212)	(0.310)
Pooled Rev Effect	-0.098	-0.045	0.189	-0.178	-0.134	-0.130	-0.049	0.031	-0.246
	(0.140)	(0.177)	(0.155)	(0.160)	(0.167)	(0.219)	(0.093)	(0.147)	(0.215)
Summary Statistics for Reference Groups									
Sample	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session
Reference group	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev	Basic Video without Rev
Reference group mean	0.42	0.94	0.35	0.62	0.73	1.23	0.16	0.39	0.89
Reference group SD	1.47	2.31	1.65	2.00	2.06	2.93	1.58	1.95	2.51
Observations	650	650	650	650	650	650	650	650	650

Notes: This table reports the use of subsistence practices on the designated `student parcel` during the rainy seasons of 2022 (midline; Panel A) and 2023 (endline; Panel B), with separate measures for `student parcel` looked after by students, treated parents, and other household members. The top portion of each panel reports the estimates under each specification; the second portion then reports the intent-to-treat effects of the revelation nudge under the basic video, the persuasive video, and in the pooled video sample. These effects are computed using linear combinations of estimates. All regressions include school-grade section-gender fixed effects, outcome-specific control variables, and robust standard errors clustered at the cluster level. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 15: Midline use of subsistence practices on the designated `student parcel`, by FFD treatment

	Rice (0/1)			Intercropping (0/1)			# crops			First principal component (SD)		
	Without FFD (1)	With FFD (2)	Diff due to: FFD (3)	Without FFD (4)	With FFD (5)	Diff due to: FFD (6)	Without FFD (7)	With FFD (8)	Diff due to: FFD (9)	Without FFD (10)	With FFD (11)	Diff due to: FFD (12)
FFD_s		0.160**			0.154**			0.816			0.409**	
		(0.065)			(0.063)			(0.529)			(0.160)	
$PVideo_h$	-0.043	-0.096	-0.054	-0.021	-0.085	-0.063	-0.371	0.185	0.556	-0.102	-0.174	-0.072
	(0.061)	(0.071)	(0.094)	(0.059)	(0.068)	(0.090)	(0.494)	(0.574)	(0.758)	(0.149)	(0.173)	(0.229)
Rev_h	-0.042	-0.170**	-0.128	-0.065	-0.150**	-0.084	-0.724	-1.103**	-0.379	-0.182	-0.441***	-0.259
	(0.061)	(0.069)	(0.092)	(0.059)	(0.066)	(0.089)	(0.493)	(0.557)	(0.746)	(0.149)	(0.168)	(0.226)
$PVideo_h \times Rev_h$	0.039	0.150	0.111	0.045	0.123	0.078	0.539	0.939	0.400	0.140	0.376	0.236
	(0.084)	(0.098)	(0.130)	(0.081)	(0.094)	(0.124)	(0.679)	(0.795)	(1.048)	(0.205)	(0.240)	(0.317)
Summary: Pooled FFD effects across video types and revelation conditions (no interactions)												
Pooled FFD		0.097***			0.099***			1.008***			0.302***	
		(0.033)			(0.031)			(0.263)			(0.079)	
Summary: Total effects under FFD												
Total: BVideo, NoRev, FFD		0.160**			0.154**			0.816			0.409**	
		(0.065)			(0.063)			(0.529)			(0.160)	
Total: BVideo, Rev, FFD		-0.010			0.004			-0.286			-0.032	
		(0.066)			(0.063)			(0.530)			(0.160)	
Total: PVideo, NoRev, FFD		0.064			0.069			1.002*			0.235	
		(0.068)			(0.065)			(0.548)			(0.166)	
Total: PVideo, Rev, FFD		0.044			0.043			0.838			0.170	
		(0.065)			(0.063)			(0.529)			(0.160)	
Summary Statistics for Reference Groups												
Sample	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session	Promo session
Reference group	Basic Video without Rev and FFD	Basic Video without Rev and FFD	Basic Video without Rev and FFD	Basic Video without Rev and FFD	Basic Video without Rev and FFD	Basic Video without Rev and FFD	Basic Video without Rev and FFD	Basic Video without Rev and FFD	Basic Video without Rev and FFD	Basic Video without Rev and FFD	Basic Video without Rev and FFD	Basic Video without Rev and FFD
Reference group mean	0.20	0.20	0.20	0.18	0.18	0.18	4.76	4.76	4.76	-0.05	-0.05	-0.05
Reference group SD	0.41	0.41	0.41	0.39	0.39	0.39	2.93	2.93	2.93	0.97	0.97	0.97
Observations	1001	1001	1001	1001	1001	1001	1001	1001	1001	1001	1001	1001

Notes: This table reports the use of subsistence practices on the designated `student parcel` during the rainy season of 2022 (midline), with separate estimates by whether the school is randomly assigned to host an annual Farmer Field Day (FFD). All regressions include section-gender fixed effects and robust standard errors are presented. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Figures to be included in the main text

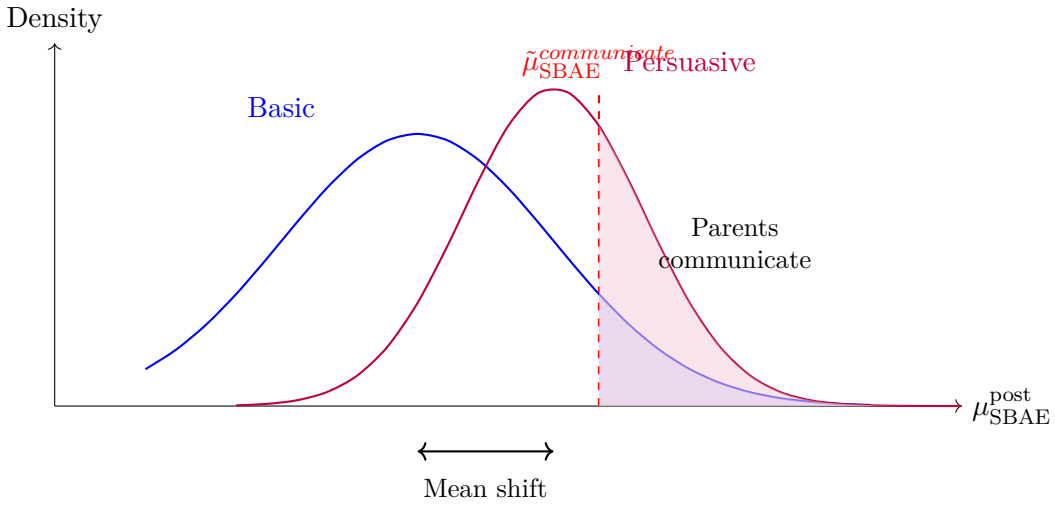


Figure 12: Video Main Effect on Delegation (Prediction 1).

Notes: The persuasive video shifts the distribution rightward (higher mean) and reduces variance, moving more parents above the communication threshold $\tilde{\mu}^{communicate}_{\text{SBAE}}$.

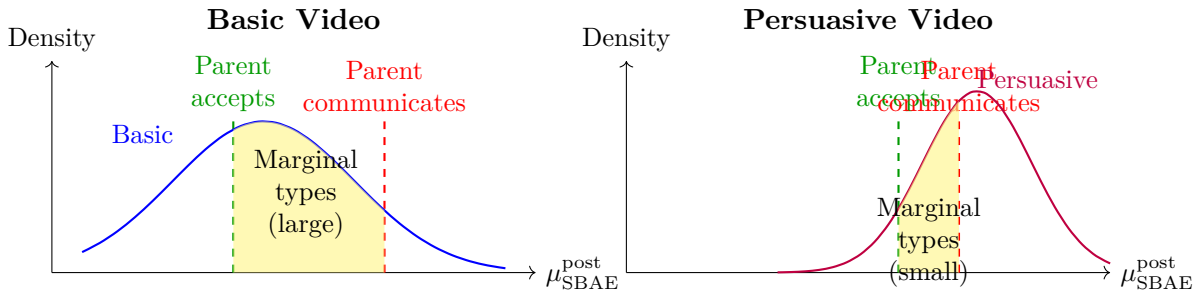


Figure 13: Nudge Effect on Delegation by Video Type (Prediction 2).

Notes: Marginal types are parents between the parent's acceptance threshold and parent's communication threshold. Basic video leaves large mass in this region; persuasive video already moves many above communication threshold, leaving fewer marginal types. Therefore: $\Delta^{Delegate}_{\text{Basic}} > \Delta^{Delegate}_{\text{Persuasive}}$ (substitutes).

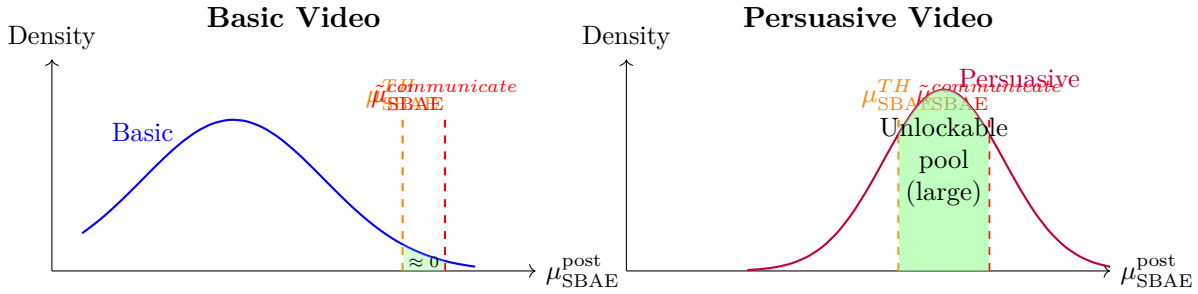


Figure 14: Nudge Effect on Timely Harvest by Video Type (Prediction 3).

Notes: The unlockable pool is parents between μ^{TH}_{SBAE} (would accept if asked) and $\tilde{\mu}^{communicate}_{\text{SBAE}}$ (won't initiate). Basic video leaves negligible mass here; persuasive video concentrates substantial mass in this region. Therefore: $\Delta^{TH}_{\text{Persuasive}} \gg \Delta^{TH}_{\text{Basic}} \approx 0$ (complements). **Both persuasive video AND nudge necessary.**

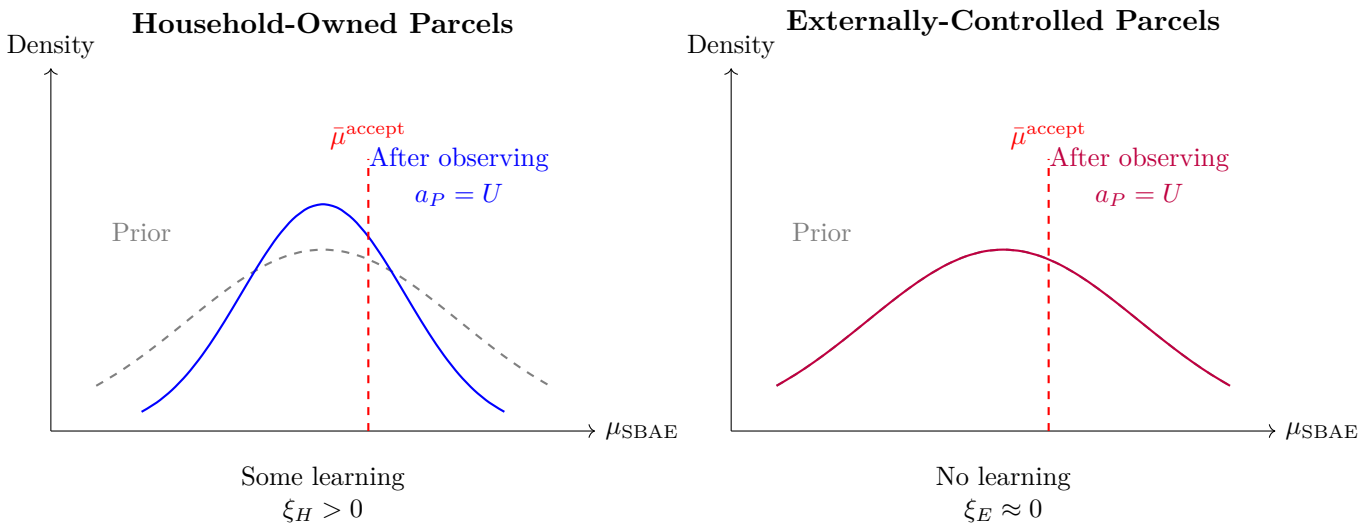


Figure 15: Differential Learning by Parcel Ownership Type.

Notes: Students can partially update beliefs about μ_{SBAE} from observing technology adoption on household-owned parcels ($\xi_H > 0$). For externally-controlled parcels with rigid traditional assignments, students learn essentially nothing ($\xi_E \approx 0$), maintaining prior beliefs even after observing adoption.

Appendix

Contents

Additional Tables A-1

Additional Figures A-5

Additional Tables

Table A1: External validity of sampled schools and students

	Sampled schools	Eligible schools	Targeted schools	Targeted counties
Panel A: Sampled schools vs. other schools in targeted counties				
# students	444.47	419.17	423.33	279.72
	(244.96)	(228.14)	(258.93)	(208.56)
# elementary students	182.48	174.76	170.46	126.43
	(118.07)	(112.08)	(120.34)	(115.91)
# junior high students	91.93	78.26	86.55	25.66
	(107.25)	(94.65)	(109.64)	(71.14)
# teachers	14.52	13.78	15.53	8.75
	(6.33)	(6.60)	(9.38)	(7.29)
Student-teacher ratio	31.83	32.48	29.34	39.29
	(13.02)	(15.29)	(14.14)	(25.00)
Functional PTA according to WASH (0/1)	0.84	0.78	0.74	
	(0.37)	(0.41)	(0.44)	
Water source according to WASH (0/1)	0.61	0.53	0.51	
	(0.49)	(0.50)	(0.50)	
# schools	197	304	447	1794

	Sampled students	All targeted students
Panel B: Sampled vs. all students in targeted grades		
% farming	100.00 (0.00)	98.65 (6.32)
% farming before 2021	99.38 (1.82)	94.96 (11.08)
% joined cooperative farming groups (kuu)	93.91 (18.39)	83.71 (18.89)
% joined kuu before 2021	82.12 (17.09)	77.14 (22.32)
% students sampled in each school	30.33 (17.33)	

Notes: This table compares the average characteristics of sampled schools and students to population-level statistics in Liberia. Panel A compares the characteristics of sampled schools to eligible schools (after screening by 4-H Liberia), targeted schools using the WASH database, and all schools listed in the EMIS database 2015-16 and in targeted counties. Panel B uses responses to the grade-level screening survey to compare the average characteristics of sampled students and all students in targeted grades. Standard errors are in parentheses.

Table A2: Balance in baseline characteristics

Baseline variable	Overall Mean	Reference Mean	$PVideo_h$	Rev_h	$PVideo_h \times Rev_h$
Panel A: Student characteristics					
Age	15.59	15.58	-0.026	-0.099	-0.033
	(2.09)	(2.06)	(0.152)	(0.147)	(0.215)
Grade	5.55	5.46	-0.027	-0.026	0.052
	(1.51)	(1.47)	(0.039)	(0.037)	(0.055)
Male (0/1)	0.51	0.51	0.020***	0.008	-0.019*
	(0.50)	(0.50)	(0.008)	(0.007)	(0.011)
Years of farming	7.85	7.75	-0.028	0.298	-0.107
	(3.28)	(3.20)	(0.321)	(0.311)	(0.455)
	F-test p-values		0.124	0.564	0.401
Panel B: Parent characteristics					
Age	42.04	42.57	-0.369	-0.601	-0.702
	(12.12)	(12.04)	(1.142)	(1.107)	(1.621)
Male (0/1)	0.46	0.48	0.013	-0.014	-0.040
	(0.50)	(0.50)	(0.050)	(0.048)	(0.071)
Parent or not (0/1)	0.63	0.63	0.002	0.030	-0.062
	(0.48)	(0.48)	(0.046)	(0.045)	(0.066)
Household head (0/1)	0.64	0.63	-0.043	-0.002	0.048
	(0.48)	(0.48)	(0.048)	(0.046)	(0.067)
Invited to surveys (0/1)	0.96	0.96	0.003	-0.016	0.017
	(0.20)	(0.20)	(0.019)	(0.019)	(0.028)
Farmer (0/1)	1.00	0.99	0.009	0.014**	-0.014*
	(0.06)	(0.11)	(0.006)	(0.006)	(0.008)
Read and write English (0/1)	0.68	0.66	0.005	0.006	0.050
	(0.47)	(0.47)	(0.045)	(0.044)	(0.064)
Finished elementary school (0/1)	0.63	0.64	0.012	-0.019	0.016
	(0.48)	(0.48)	(0.046)	(0.044)	(0.065)
Finished junior high school (0/1)	0.45	0.45	-0.045	-0.002	0.067
	(0.50)	(0.50)	(0.047)	(0.046)	(0.067)
	F-test p-values		0.672	0.398	0.546

Notes: This table shows a test of balance for the baseline characteristics of students (Panel A) and parents (Panel B) in the randomized experiment. Column 1 provides covariate means and standard deviations in the full sample. Column 2 provides covariate means and standard deviations in reference group of households receiving the basic video treatment and no revelation. Column 3 shows the marginal effects of the persuasive video treatment. Column 4 shows the marginal effects of the revelation treatment. Column 5 shows the interaction effects of the persuasive video and the revelation treatment. P-values from Wald tests of joint significance of treatment effects over all covariates are presented at the bottom of each panel.

Table A3: Balance in attrition rates across treatment arms

	Panel A: Balance in attrition		
	Midline Students Attrited (0/1)	Endline Students Attrited (0/1)	Endline Parents Attrited (0/1)
$PVideo_h$	-0.013	0.040	0.005
	(0.044)	(0.048)	(0.037)
Rev_h	-0.009	-0.020	-0.001
	(0.043)	(0.046)	(0.036)
$PVideo_h \times Rev_h$	-0.049	-0.042	-0.022
	(0.063)	(0.068)	(0.053)
Reference Mean	0.35	0.45	0.14
Overall Mean	0.33	0.45	0.14
Observations	1002	1002	1002
	Panel B: Reasons for attrition		
	Dropped out or switched schools	Dropped out or switched schools	Travelled
	32.0%	42.4%	5.0%
	— Confirmed dropped out	— Confirmed dropped out	Relocated
	16.4%	21.8%	6.0%
	— Confirmed switched schools	— Confirmed switched schools	Working
	15.2%	20.5%	0.8%
	Travelled	Travelled	Sick
	0.3%	0.8%	0.7%
	Moved to foreign countries	Moved to foreign countries	Moved to foreign countries
	0.1%	0.6%	0.4%
	Passed away	Passed away	Passed away
	0.1%	0.1%	1.1%
	Other reasons	Other reasons	Other reasons
	0.5%	1.2%	0.5%
	Panel C: Balance in major attrition reasons		
	Dropped out or switched schools (0/1)	Dropped out or switched schools (0/1)	Travelled or relocated (0/1)
$PVideo_h$	0.014	0.015	-0.009
	(0.044)	(0.047)	(0.033)
Rev_h	0.009	-0.027	0.020
	(0.043)	(0.046)	(0.032)
$PVideo_h \times Rev_h$	-0.078	-0.035	-0.048
	(0.062)	(0.067)	(0.046)

Notes: This table shows a test of balance for the attrition rates among students and parents in the midline and endline surveys. Panel A regresses a binary indicator of attrition on a the persuasive video treatment, the revelation treatment, and their interaction. Panel B shows the distributions of reasons for attrition in each survey. Panel C repeats the balance test in Panel A, but using only attrition due to major reasons. All regressions include school-section-gender fixed effects. The reference group is households who are assigned to the basic video treatment and no revelation. Robust standard errors are reported.

Table A4: Selection of students and parents into the endline sample

Baseline variable	Change in attrition probability in percentages (%)	E(Attritors) - E(Non-attritors) in standard deviations	F-test p-value: Differential selection by treatment arms
	Panel A: Student characteristics		
Age	1.541	0.074	0.657
	(1.095)	(0.053)	
Grade	10.422**	0.046**	0.123
	(4.295)	(0.019)	
Male (0/1)	10.248	0.005	0.312
	(21.657)	(0.011)	
Years of farming	0.177	0.025	0.381
	(0.518)	(0.074)	
	Panel B: Parent characteristics		
Age	-0.113	-0.103	0.850
	(0.117)	(0.107)	
Male (0/1)	5.004*	0.217*	0.990
	(2.613)	(0.113)	
Parent or not (0/1)	-2.724	-0.105	0.734
	(2.846)	(0.110)	
Household head (0/1)	1.627	0.068	0.798
	(2.693)	(0.112)	
Invited to surveys (0/1)	-4.360	-0.073	0.656
	(6.720)	(0.113)	
Read and write English (0/1)	-0.726	-0.028	0.236
	(2.861)	(0.109)	
Finished elementary school (0/1)	1.225	0.047	0.468
	(2.826)	(0.108)	
Finished junior high school (0/1)	-3.214	-0.127	0.229
	(2.749)	(0.108)	

Notes: This table summarizes how the endline samples of students (Panel A) and parents (Panel B) differ from the original samples. Column 1 regresses a binary indicator on each subject characteristic, controlling for treatment status. Column 2 regresses each subject characteristic on a binary indicator of attrition, controlling for treatment status. The marginal effects of attrition are then divided by the control group standard deviations. Column 3 regresses a binary indicator of attrition on all treatments, fully interacted with each subject characteristic. P-values from Wald test of joint significance of all treatment interactions are reported. All regressions include school-section-gender fixed effects. Robust standard errors are reported.

Additional Figures

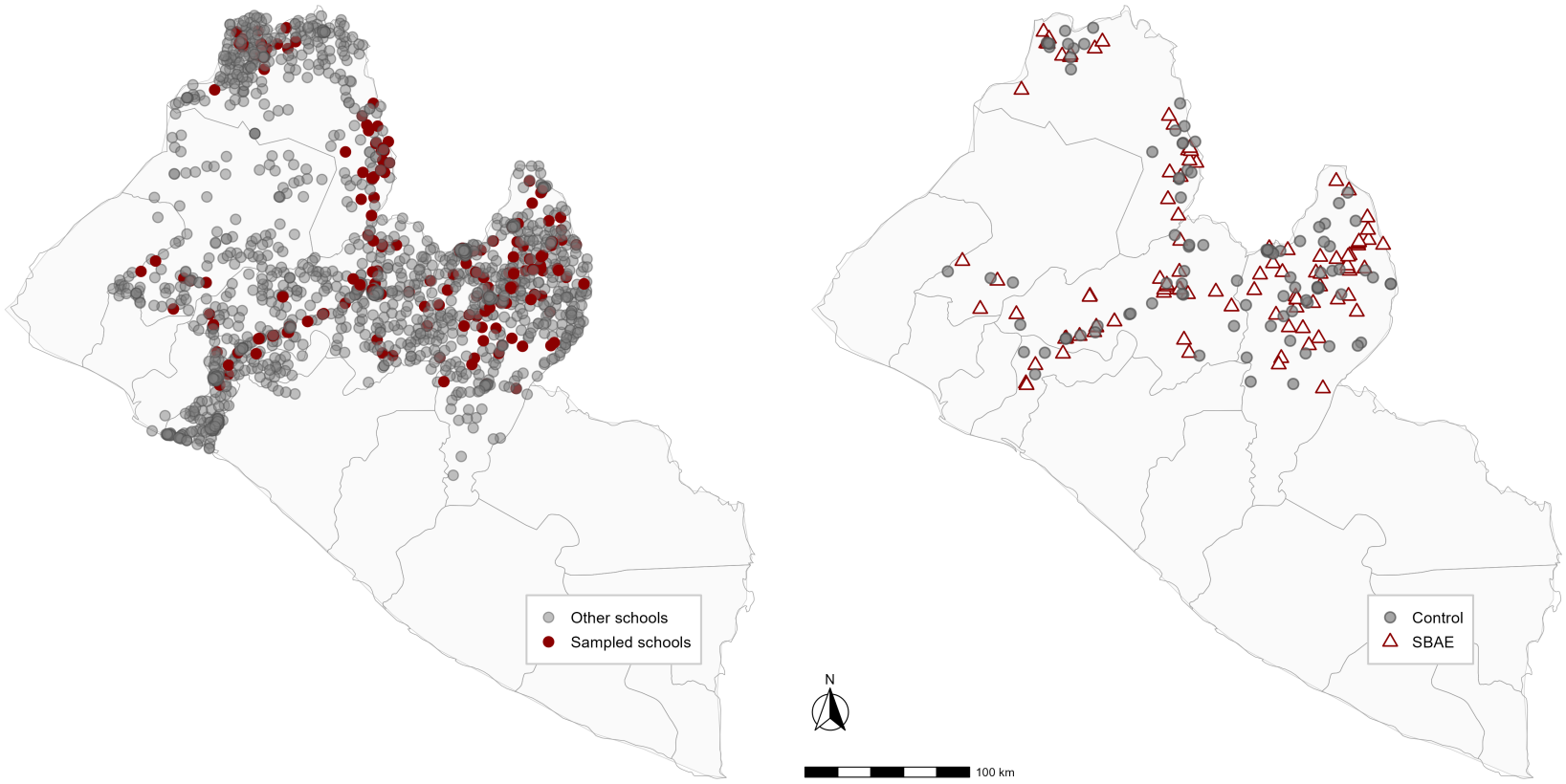


Figure A1: Geographical distribution of sampled schools